



Statistics 101 My Version

The Why and the What?

The background of the slide features several sets of thin, curved lines in light gray and white, creating a sense of motion and depth. These lines are primarily located on the left and right sides of the slide, framing the central content.

Our lives are run
by statistics

- **COVID**

- What are your chances of getting COVID?
- Should you be vaccinated?
- Once vaccinated, should you wear a mask?

- **Health care**

- Insurance
- Treatment

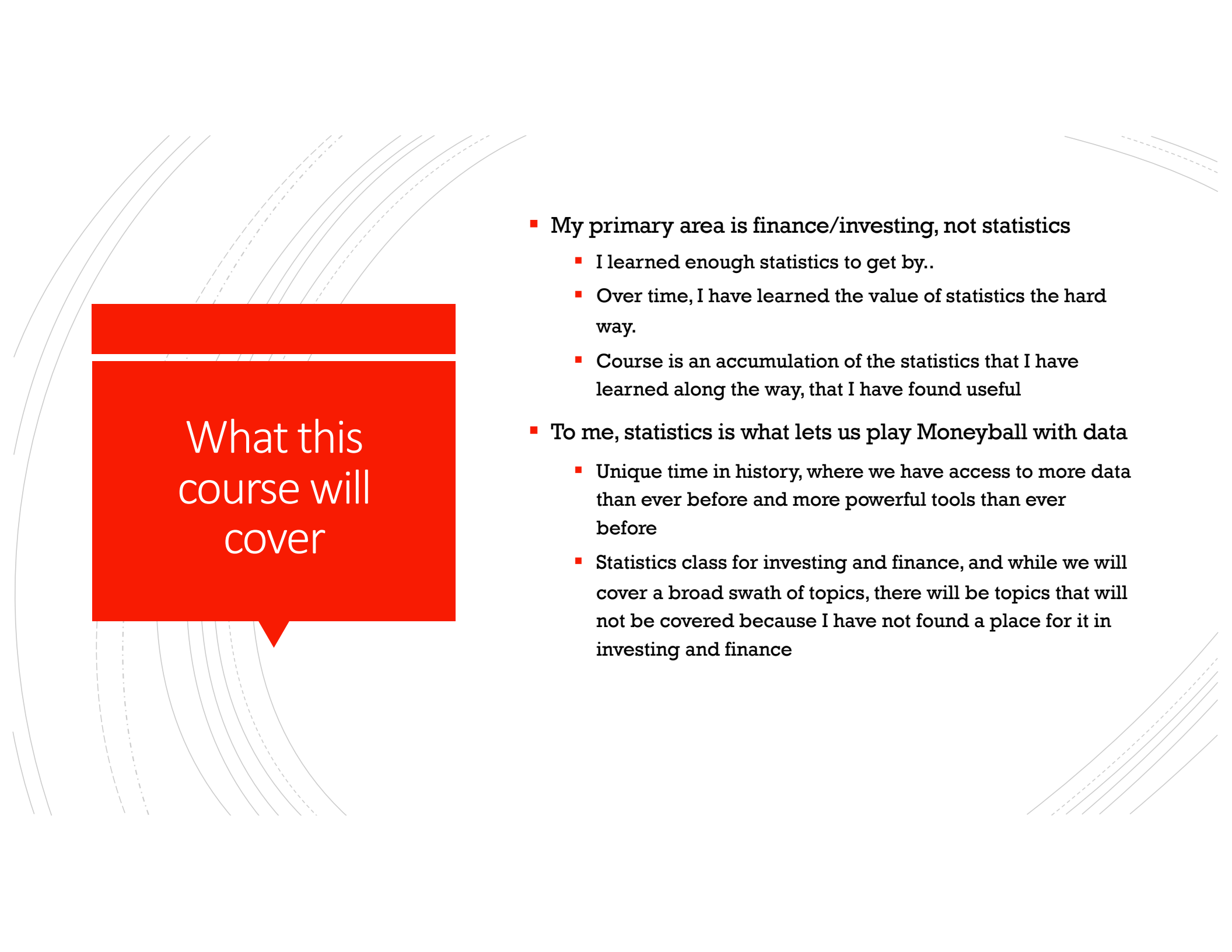
- **Politics**

- Polls
- Elections



And our grasp
of statistics is
fleeting...

- Intuition and gut feeling fail us
 - Anecdotal evidence
 - Drowning in data
- Treat estimates as facts...
 - Experts with agendas...
 - Policy driven by bad statistics..

The background of the slide features several thin, curved lines in a light gray color, some solid and some dashed, creating a sense of motion or a stylized wave pattern. On the left side, there is a large red speech bubble with a white outline, containing the text 'What this course will cover'.

What this course will cover

- My primary area is finance/investing, not statistics
 - I learned enough statistics to get by..
 - Over time, I have learned the value of statistics the hard way.
 - Course is an accumulation of the statistics that I have learned along the way, that I have found useful
- To me, statistics is what lets us play Moneyball with data
 - Unique time in history, where we have access to more data than ever before and more powerful tools than ever before
 - Statistics class for investing and finance, and while we will cover a broad swath of topics, there will be topics that will not be covered because I have not found a place for it in investing and finance



Who this course is for...

- Primarily investing and finance people, whose statistical knowledge is old/stale, and need a brush up. Perhaps, even for people who don't even remember taking a statistics class.
- It is not for
 - Researchers, whose statistical knowledge has to be far deeper
 - Those who want to play inside statistics games...

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Course Objectives

- Understand what tools you have in your toolbox to answer the questions you may face, when confronted with data, and which of those tools to use.
- Understand how to use that tool to answer your questions, and the limitations of the tool.
- Assess other people's use of statistical tools, to make sure that you are not being misled with data and to ask the right questions about the data usage.



SESSION 1: THE ESSENCE OF STATISTICS

Session 1
Moneyball

What is data?

- ❑ Data is a generic word that covers almost any type of information that you obtain.
- ❑ Data can come in different forms.
 - ▣ Quantitative versus Qualitative: Quantitative data is numerical. Qualitative data is not, though it can sometimes be converted into numerical form. (Example: Pain scale)
 - ▣ Discrete versus Continuous: Discrete data can take only certain values (integer or otherwise). Continuous data take any value, though it can have upper or lower bounds.
- ❑ While we have always relied on data for decision making, we have become much better at finding and chronicling the data, collecting and storing it, and making it accessible to a broader audience.

What is statistics?

- ❑ The Data Science: Statistics is the discipline that allows us to gather, analyze, depict and make sense of data.
- ❑ Data -> Information: Statistics is what helps us convert raw data, noisy and often contradictory, into information that can be used to make better decisions.
- ❑ Multifaceted: Defined as such, statistics includes
 - ❑ The collection and recording of data
 - ❑ The processing of that data
 - ❑ The analysis of the data
 - ❑ The depiction (presentation) of that data

Why do we need statistics?

1. As data access widens and becomes easier: We have access to more data, quantitative and qualitative, than we have ever had, because of innovations in how we collect data, how we store the data and how we access the data.
2. Data overload is a problem: That access to data is leading to information overload, where we find that the line between facts and opinion is blurred, leaving us more confused than we ever have been.
3. Leading to simplistic rules of thumb: As our brains start to shut down in the face of data overload, we seek out rules from simpler times, many of which have no basis in truth, but make use feel in control.
4. And mental short cuts: We also adopt short cuts, again with no rational basis, just because they have worked in the past.

Statistics drives practice, policy and laws...

- In almost every aspect of our lives, practice and policy is determined by statistics.
 - ▣ In fiscal policy, governments decide taxation and spending, based upon statistical assessments of their effects on the economy.
 - ▣ In health care, questions of what drugs should be approved and what treatment patients should get is based upon statistics (often in medical research)
 - ▣ In our personal lives, our choices of where to work, live and how to save/invest are at least loosely driven by statistics.
- Making good policy and personal decisions requires an understanding of statistics and data. If we misread the statistics or the statistics are unreliable, policy will as well.

Lying with statistics?

- Agenda-driven data: As access to data has increased, the misuse of that data has also gone up, especially when people have agendas and want to further them. These people mislead, without technically lying, as they selectively pick and choose which data they use, and how they present that data.
- Social media as magnifier: Bad data can take on a life of its own, especially with social media operating as a weapon to widen its reach.
- Caveat emptor: As people weaponize data and use selective and slanted statistics, based upon that data, we need to be able to protect ourselves from misinformation.
Understanding statistics allows us to:
 - Look for red flags that can be used to detect data manipulation
 - Asking the right questions to separate fact from fiction

Big Data and Data Analytics

- Big Data: Big data is a catch-all that references not just the quantity of data, but also that the data is about behavior that until a few years ago would have been considered private and in real time. (Think location data on your phone, as a simple example). For better or worse, big data has been facilitated by technology platforms, which collect the data, as we use them.
- Data Analytics: All statistics is data analytics, but in the context of big data, data analytics has expanded to cover
 - ▣ Very large datasets, creating sample sizes that we have never worked with in the past.
 - ▣ Combining quantitative with qualitative data to create composite results.

1. Data Collection and Sampling

- Data collection & storage: The very first exercises in data collection required interactions with subjects, paper documents and physical storage. As technology has evolved, data collection and storage has been digitized, and the risks have shifted.
- Population versus Sample: With large populations, you often have to sample that population, either cross sectionally, across time, or both.
- The Sampling Sins: While sampling, by itself, is part of statistics, there are two fundamental sins that you should try to avoid:
 - *Bias*: The sample has to be representative of the population. If the sampling method creates bias, the results from the sample cannot be extrapolated to the population.
 - *Noise*: Even if the sample is representative, the results that you obtain will have statistical error or noise that can muddy your conclusions.

2. Data Descriptive(Statistic)s

- When faced with sample data, data descriptives try to summarize the data with metrics. Those metrics can include
 - ▣ Measures of location: Measures of location try to identify the number or numbers around which the data is centered or is most likely to take.
 - ▣ Measures of dispersion: Measures of dispersion measure how much divergence there is on a data item, across a sample.
 - ▣ Measures of skewness: Measures of skewness look at whether how symmetric or asymmetric the data is around the central value.
- These data descriptives become short hand for characterizing the data, and the basis for analysis.

3. Data Distributions

- A data distribution is a visual description of the data item in a distribution. In a histogram, for instance, you break the data down into groupings and count the outcomes of each grouping.
- While you can convert any data into a histogram, there are sometimes advantages in replacing this histogram with one of the many standardized descriptions in statistics that fits it.
- These standardized distributions can be
 - Discrete or continuous: A discrete distribution puts the variable into discrete values (0 or 1, numerical scale, bond ratings). A continuous distribution allows the variable to take any value.
 - Symmetric or asymmetric: A symmetric distribution has a center and symmetry on the up and down sides.

4. Data Relationships

- In statistics, we can also examine how two or more variables are related to each other.
 - ▣ In some cases, we restrict ourselves to *chronicling whether that co-movement* is in the same direction, opposite directions and that there is no co-movement at all.
 - ▣ In others, we try to find *whether there is causation*, where one variable's movement is the cause of the other variable's movement.
 - ▣ Finally, if there is a link, we can use statistics *to try to predict a variable*, based upon observed values of another variable.
- As an example, consider the linkage between stock prices and interest rates.
 - ▣ We can measure whether interest rates and stock prices move together, in opposite directions and are unrelated.
 - ▣ We can also examine which direction the causation runs (do higher stock prices cause higher interest rates or vice versa)
 - ▣ And if there is causation or a link, we can see if we use changes in interest rates can be use to predict changes in stock prices or vice versa.

5. Probabilities

- Probabilities measure the likelihood of something happening and are at the heart of statistics and data analysis.
- The “something happening” can be a discrete event or a variable that comes from a continuous distribution.
 - Discrete event: That a company can go bankrupt or stay a going concern.
 - Continuous variable: The likelihood that a company will generate earnings that exceed \$1 billion.
- You can estimate the probability of an event (discrete and continuous) happening by
 - Looking at the past or historical data.
 - Assuming a statistical distribution for the data, and that you can estimate the parameters of that distribution.

6. And Probabilistic Tools

- Probit/Logit: A probit/logit measures the likelihood or probability of an event happening, based upon observable variables.
 - ▣ Example: Measuring the probability that a company will be acquired, given variables that you believe are correlated.
- Decision Trees: When you have a series of discrete events that are sequential, a decision tree allows you to compute the likelihood of events happening, conditional on events leading up to it.
 - ▣ Example: The process by which an FDA drug works its way through the drug approval pipeline, from research to commercial development.
- Monte Carlo Simulations: When you have multiple inputs that determine an output variable, a simulation allows you specify probabilistic distributions for each input variable and get a distribution for the output variable.
 - ▣ Example: Valuing a company, given its cash flows, growth and risk characteristics.



SESSION 2: SAMPLES AND POPULATIONS

Session 2
Bias and Noise

Populations and Sample

- ❑ Population versus Sample: A population includes the universe of all instances of an object or phenomenon that you are trying to study. A sample is a subset of that population that you collect data on, and use, to make judgments about the behavior of the population.
- ❑ Time Series versus Cross Section: The data that you are trying to study can be a phenomenon that you observe over time (time series data) or across different subjects at a point in time (cross sectional data).
 - ❑ Time Series Example: If stock returns over time is your population, stock returns from 1960-2021 is a sample.
 - ❑ Cross sectional Example: If all publicly traded companies is your population, looking at only US companies or companies with market caps that exceed \$10 million is a sample.

Why do we need sampling?

1. Practicality: If the population is too large to collect data on, and/or a subset (small or large) of the population is inaccessible, you have no choice but to sample the data.
2. Costs: Even if you could collect data on the entire population, the costs (in time and money) may outweigh the benefits of doing so.
3. Time trade off: Related to the second point is the question of how frequently you want to update the data. It is easier to update sampled data than data on the entire population.

Sampling Approaches

- Probability versus Non-probability Sampling: In a probability-based sample, the observations/subjects are picked at random. In a non-probability sample, the researcher hand picks the sample, based upon criteria that he or she picks.
- Variants of Random Sampling:
 - *Simple Random*: In a simple random sample, you pick your sample randomly across the entire population.
 - *Stratified Random*: In a stratified random sample, you first break your population down into groupings, and then randomly pick from within each of these groupings.
 - *Cluster Random*: In a cluster random sample, you break the population into groups, and then randomly select some of these groups and collect data on each group member.

Sampling Bias

- To the extent that you intend to extrapolate your findings on a sample to the entire population, you want to ensure that you don't have a biased sample.
- A biased sample is one that diverges from the population in its characteristics. That bias can arise for many reasons including:
 - Exclusion: Some parts of the population may not even make into the sampling universe.
 - Self-selection: Some parts of the population may be more easily accessible than other parts, given how you collect data.
 - Non-response: Some parts of the population may be less likely to respond to requests for data.
 - Survivorship: Success (or failure, sometimes) may make an observation more likely to be sampled.

And its consequences...

- If there is sampling bias, and you are or choose to be *unaware of that bias*, the extrapolations that you make from your sampling findings to the population will be biased.
- If there is sampling bias, and *you are aware of it*, you *can try to correct for it*, as you extrapolate your findings to the population.
- If *there is sampling bias*, and *you cannot correct for it*, you can narrow your findings to reflect the portion of the population that is represented by your sample.

Sampling Noise/ Error

- If a sample is unbiased, the results that you get from that sample can be extrapolated to the population, but with error (usually measured with a standard error on your forecast).
- That error is called sampling noise and will decrease as the sample size increases.
 - In a classic example, consider the odds of getting a head or a tail on an (unbiased) coin toss. Even though we know the population odds (50/50), you can get results that diverge on a small number of tosses. As the number of tosses increase, the sample numbers will also approach 50/50.
 - More generally, sampling noise is part and parcel of the process and all you can do is be transparent about it, and report it.

Independence + ID: Not just buzz words

- In almost any discussion of sampling and statistics, you will hear the words “*independence*” and “*identical distributions*” thrown in as pre-requisites or at least good qualities in a sample.
 - Independence: Events are independent when whether an event occurs or not is not determined by other events occurring.
 - Coin tosses are a classic example of independence
 - Are stock price changes independent?
 - Identical Distributions: Each event draws from the same probability distribution.
 - Coin tosses draw from the same distribution (50/50)
 - Do stock price changes draw from the same distribution?
- In finance, researchers often assume independence and identical distributions, in making assertions based upon samples, but the truth is that both characteristics are hard to find.

The Law of Large Numbers

- The Law of Large Numbers: As the number of observations in a sample increases, the sample average will approach the population average (or the true average), if the observations are independent(I) and identically distributed (ID).
- There are a few exceptions to the law of large numbers.
 - The first is if the sample observations are not independent or identically distributed.
 - The other is for distributions with fat tails (higher chance of extreme outcomes).
- The Weak Law of Large Numbers: Even when observations are not independent or identically distributed, the sample average will approach the population average, albeit slower and only if the variance is finite.

Sampling Error Theorems/Propositions

- Central Limit Theorem: Even if your population distribution is non-normal, the means of any samples that you pull from that distribution will follow a normal distribution, as long as you the observations are independent and identically distributed.
- Chebyshev's inequality : In a normal distribution, for instance, roughly two thirds of the values have to fall within one standard deviation of the mean and 95% within two standard deviations. Chebyshev's inequality is more general, stating that a minimum of just 75% of values must lie within two standard deviations of the mean and 88.89% within three standard deviations for a broad range of different probability distributions.



SESSION 2A: SAMPLES AND POPULATIONS IN FINANCE

Session 2
Bias and Noise

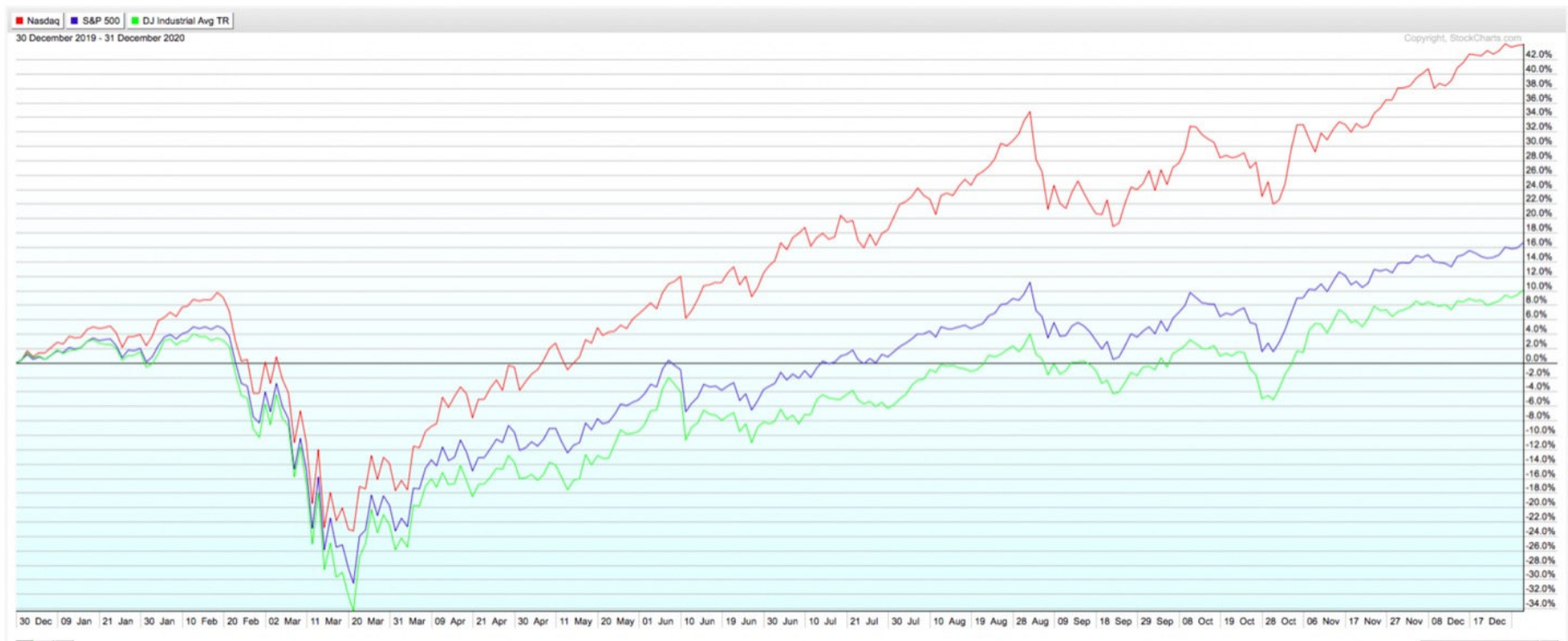
1. Market Indices

- ❑ In most stock markets, you have indices that track how well or badly stocks are doing, in the aggregate.
- ❑ While the way these indices are constructed and returns are computed vary, they all represent samples of stocks that are traded.
- ❑ The question of how well these indices do in measuring how well equities are doing collectively depends on
 - ❑ How many constituents there are in the index
 - ❑ The criteria that went into their selection
 - ❑ The weighting that is attached to the constituents
- ❑ Put simply, a market index is almost never a random sample, and that has to be kept in mind when reading or extrapolating from the index.

Equity Indices in the US

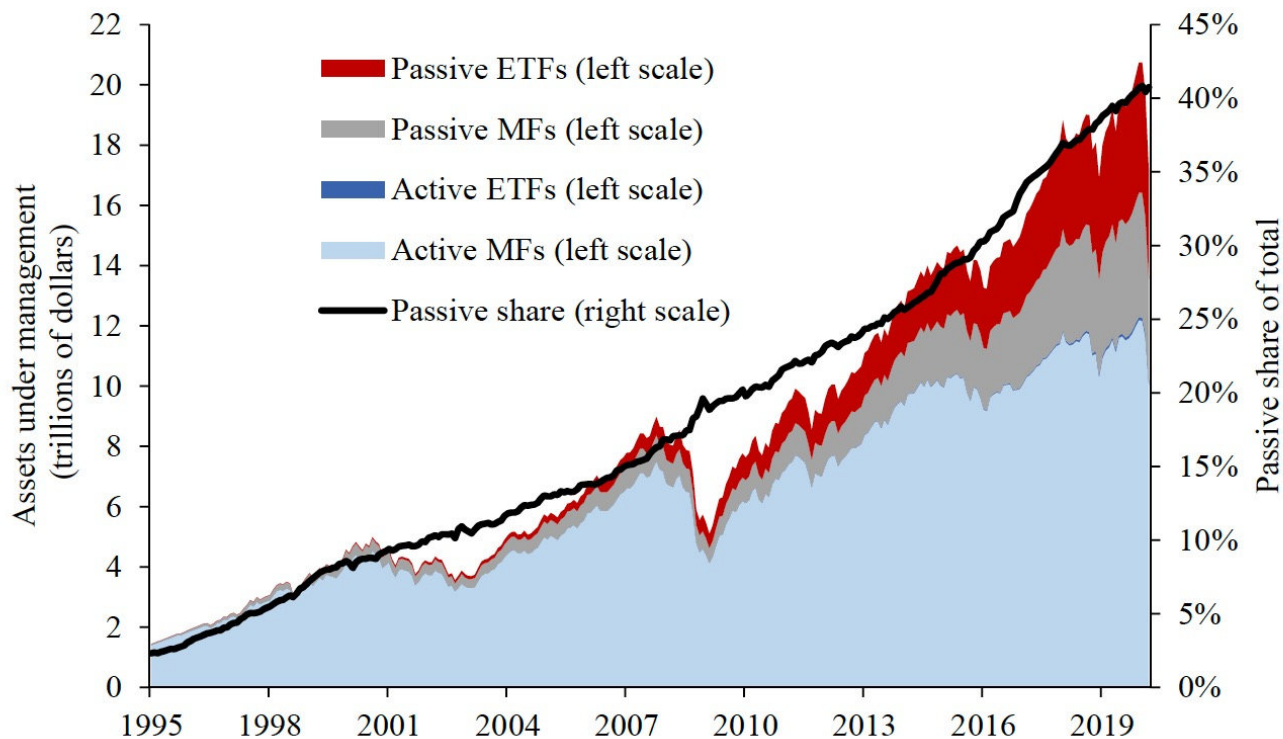
	DAILY					5 DAY		52 WEEK			
	HIGH	LOW	LAST	CHG	% CHG	CHG	% CHG	HIGH	LOW	% CHG	YTD % CHG
Dow Jones											
Industrial Average	35047.44	34790.62	34843.21	-273.19	-0.78	-87.72	-0.25	35192.11	26143.77	28.09	13.84
Transportation Average	14523.29	14266.14	14266.70	-319.31	-2.19	-93.10	-0.65	16170.05	10098.44	39.66	14.07
Utility Average	920.45	908.30	918.61	-2.08	-0.23	2.82	0.31	937.38	780.27	11.30	6.24
65 Composite	11498.47	11388.55	11407.20	-123.86	-1.07	-31.92	-0.28	11860.67	8824.17	28.37	12.85
Total Stock Market	45828.26	45651.94	45729.58	-158.45	-0.35	25.14	0.05	46043.06	32668.83	34.55	16.60
Barron's 400	1030.18	1024.57	1025.50	-4.68	-0.45	10.31	1.02	1030.18	687.57	41.87	20.34
S&P											
500 Index	4416.17	4400.23	4408.41	-14.74	-0.33	7.77	0.18	4429.97	3209.45	32.47	17.37
100 Index	2021.82	2014.06	2018.27	-6.06	-0.30	-5.32	-0.26	2038.79	1487.99	31.45	17.31
MidCap 400	2710.53	2684.20	2685.22	-23.66	-0.87	5.51	0.21	2780.08	1769.05	39.82	16.41
SmallCap 600	1339.28	1321.45	1322.68	-22.49	-1.67	-6.72	-0.51	1417.45	814.65	46.86	18.21
SuperComp 1500	1008.96	1005.20	1006.43	-4.21	-0.42	1.44	0.14	1010.64	732.07	33.22	17.33
Nasdaq Stock Market											
Composite	14812.37	14734.12	14791.93	30.63	0.21	29.34	0.20	14863.64	10519.49	34.49	14.77
Nasdaq 100	15117.38	15017.14	15096.34	34.91	0.23	78.24	0.52	15142.35	10677.85	35.69	17.13
Biotech	5366.94	5223.64	5314.63	65.09	1.24	126.36	2.44	5466.79	3962.38	23.11	11.67
Other U.S. Indexes											
NYSE Composite	16713.46	16610.20	16621.75	-91.71	-0.55	48.19	0.29	16761.24	12228.97	30.56	14.44
Russell 1000	2484.22	2471.72	2476.38	-7.55	-0.30	3.02	0.12	2490.27	1784.46	33.87	16.76
Russell 2000	2223.10	2195.37	2198.37	-25.21	-1.13	-26.59	-1.19	2360.17	1432.57	42.18	11.32
Russell 3000	2626.30	2612.86	2616.86	-9.37	-0.36	0.98	0.04	2635.01	1875.06	34.32	16.39
PHLX Gold/Silver	146.48	141.91	142.10	-1.37	-0.95	1.63	1.16	167.09	128.08	-11.82	-1.45
PHLX Oil Service	54.21	51.69	51.85	-3.38	-6.13	-4.55	-8.08	70.15	25.45	33.66	16.95
PHLX Semiconductor	3450.37	3405.71	3438.53	41.87	1.23	166.08	5.08	3450.37	2102.16	55.76	23.00

And they sometimes diverge .. As in 2020



2. Investable Indices: The Growth of Passive Investing

Figure 1: Total assets in active and passive MFs and ETFs and passive share of total



Source: Morningstar, Inc.

With variation across choices: Energy ETFs

Not Created Equal

A sampling of five energy-themed ETFs shows how different exchange-traded funds—and their returns—can be even when they have a very similar focus. Each fund has its own template for selecting companies within the sector, using criteria such as size, nationality and stock fundamentals. Each fund's performance and expense ratio differs widely, too.

Sources: Morningstar; largest holdings as of Oct. 15 from fund websites
The Wall Street Journal

Energy Select Sector SPDR (XLE)

Holds only large-cap U.S. companies in the S&P Energy Select Sector Index.

NO. OF STOCKS
45

LARGEST HOLDING % OF ASSETS
Exxon Mobil, 19%

AVERAGE MARKET CAP
\$60 billion

EXPENSE RATIO
0.18%

RECENT ASSETS
\$7.7 billion

AVG. DAILY TRADING VOLUME
15,000,000

YTD PERFORMANCE
(THROUGH 10/15)
6.85%

Vanguard Energy (VDE)

Invests in large, mid and small-cap stocks.

NO. OF STOCKS
171

LARGEST HOLDING % OF ASSETS
Exxon Mobil, 23%

AVERAGE MARKET CAP
\$45 billion

EXPENSE RATIO
0.19%

RECENT ASSETS
\$2.2 billion

AVG. DAILY TRADING VOLUME
128,815

YTD PERFORMANCE
(THROUGH 10/15)
5.26%

iShares S&P Global Energy (IXC)

Tracks the S&P 1200 Energy Sector Index. Only 50% of stocks are from the U.S.

NO. OF STOCKS
92

LARGEST HOLDING % OF ASSETS
Exxon Mobil, 16%

AVERAGE MARKET CAP
\$74.7 billion

EXPENSE RATIO
0.48%

RECENT ASSETS
\$1.1 billion

AVG. DAILY TRADING VOLUME
195,523

YTD PERFORMANCE
(THROUGH 10/15)
4.37%

iShares Dow Jones U.S. Oil Equipment Index (IEZ)

Focuses on oil equipment and services companies.

NO. OF STOCKS
47

LARGEST HOLDING % OF ASSETS
Schlumberger Ltd., 19%

AVERAGE MARKET CAP
\$13.4 billion

EXPENSE RATIO
0.47%

RECENT ASSETS
\$354 million

AVG. DAILY TRADING VOLUME
185,663

YTD PERFORMANCE
(THROUGH 10/15)
0.04%

First Trust Energy AlphaDEX Fund (FXN)

Ranks stocks based on fundamentals like price appreciation, sales growth and cash flow.

NO. OF STOCKS
52

LARGEST HOLDING % OF ASSETS
Chesapeake Energy, 3.9%

AVERAGE MARKET CAP
\$9.1 billion

EXPENSE RATIO
0.70%

RECENT ASSETS
\$139 million

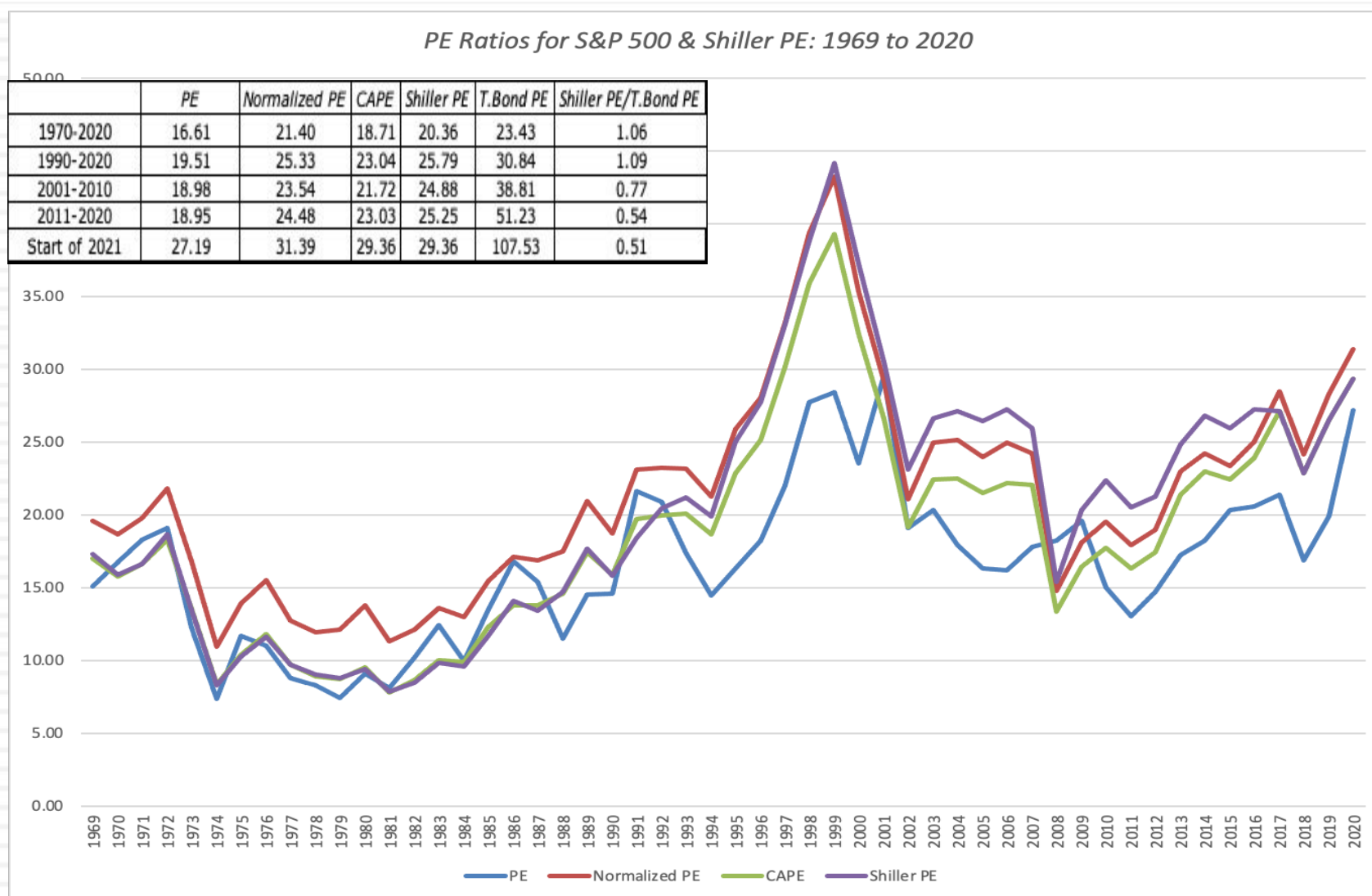
AVG. DAILY TRADING VOLUME
66,911

YTD PERFORMANCE
(THROUGH 10/15)
2.39%

3. Time Sampling

- In time sampling, you look at a sub-period of time to make judgments about investments. You do so because:
 - ▣ Data is unavailable or unreliable for other periods
 - ▣ The market has changed structurally, and you want to look at a period that you believe is more representative of the current market
 - ▣ The hypothesis that you are testing is about a specific period of time.

PE Ratios: 1960 - 2020

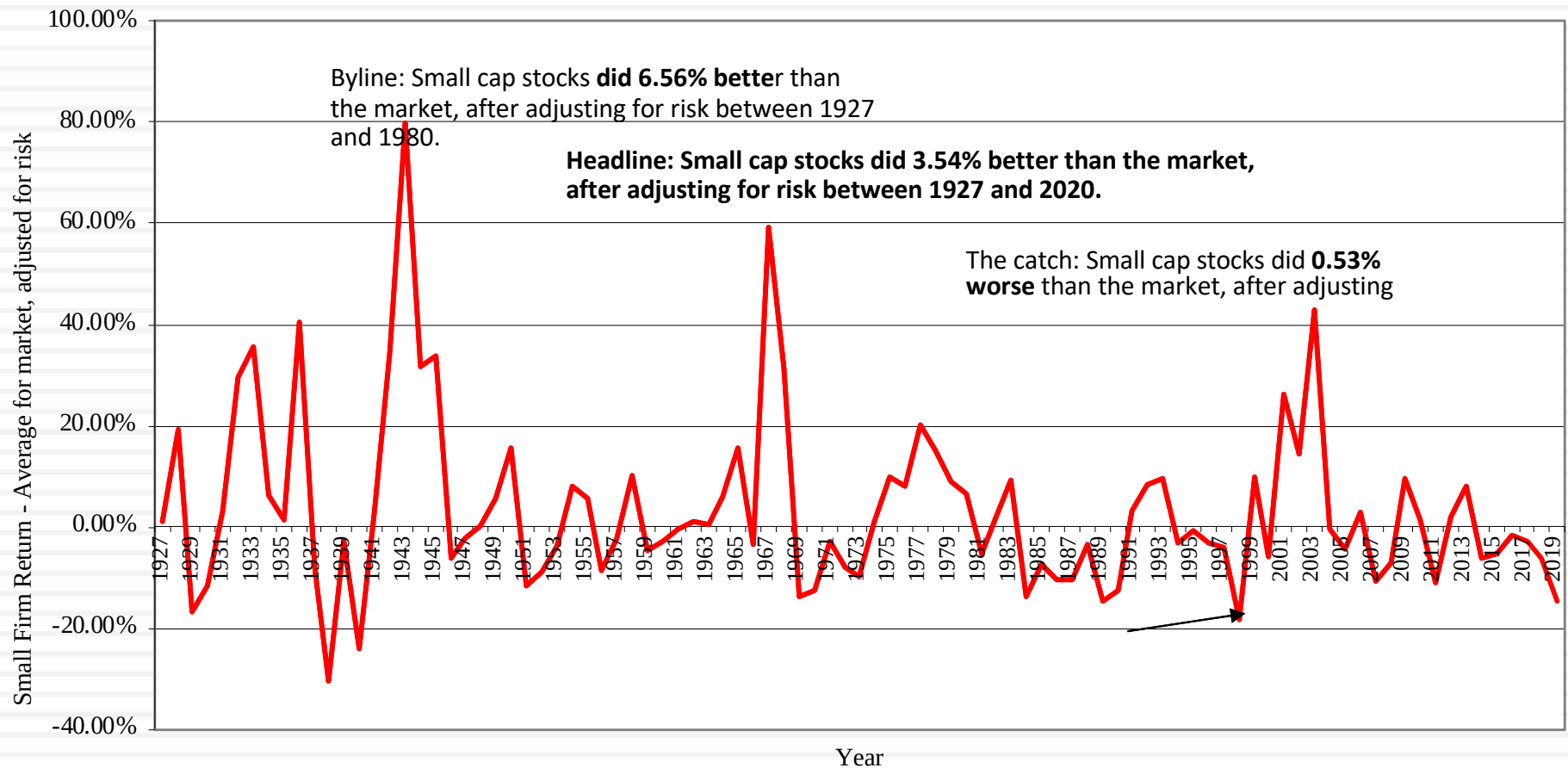


Bias in time sampling

- While there is clearly an advantage to looking at long periods of time to make judgments about investment performance, there is always the danger that the performance can change during the time period studied.
- In 1981, Rolf Banz found that small cap stocks earned significantly higher returns than expected, after adjusting for risk and market performance.
- That finding, formalized as the “small cap premium” has become part of finance lore and practice, with many analysts adding this premium on to get the expected return on small cap stocks.

The Small Cap Premium: Changing numbers over time

Small Firm Premium over time- 1927 -2020



4. Testing Investment Strategies

Sampling bias:

What was the universe of stocks? What was the basis for PE classes? What happened to negative earnings companies?

Value vs Growth: US Stocks, by decade

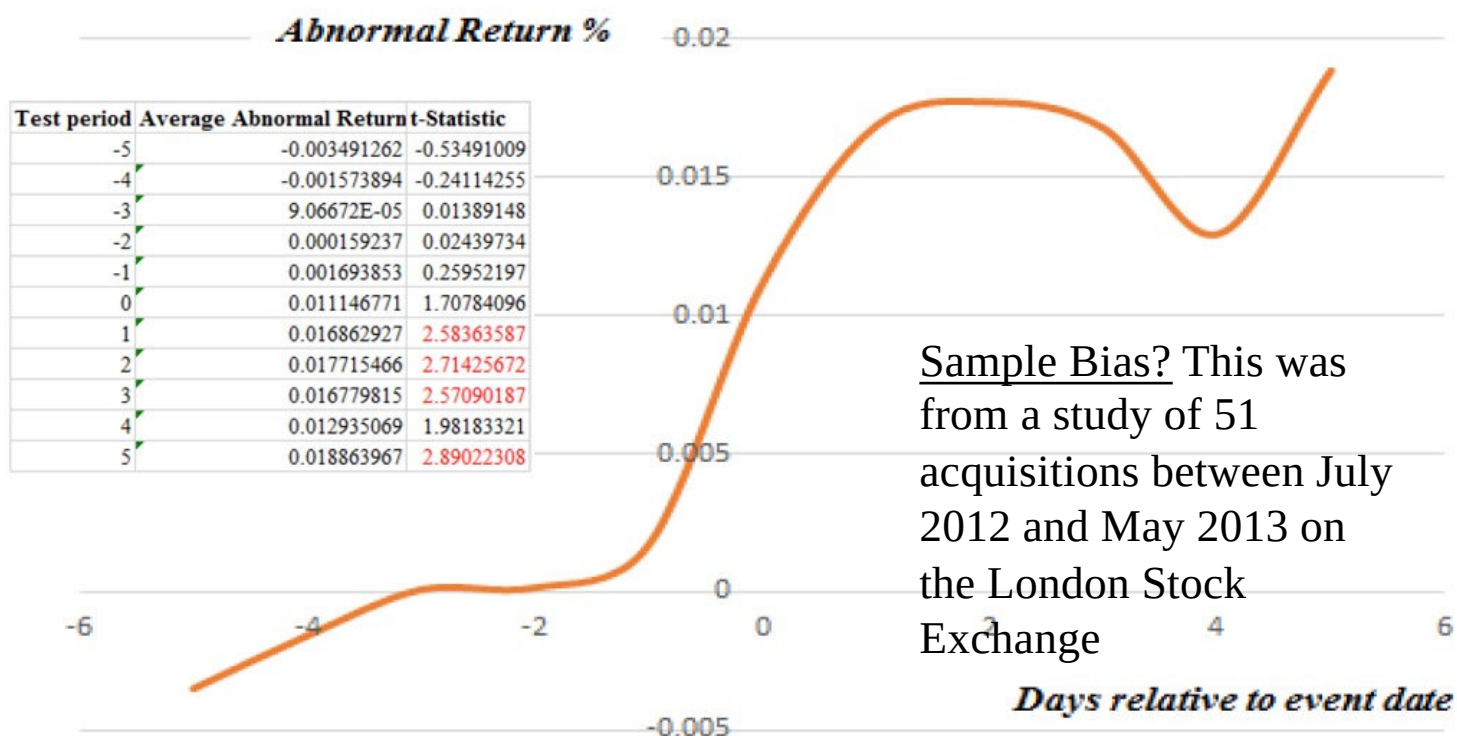
	<i>Lowest PBV</i>	<i>Highest PBV</i>	<i>Difference</i>	<i>Lowest PE</i>	<i>Highest PE</i>	<i>Difference</i>
1930-39	6.04%	4.27%	1.77%	NA	NA	NA
1940-49	22.96%	7.43%	15.53%	NA	NA	NA
1950-59	25.06%	20.92%	4.14%	34.33%	19.16%	15.17%
1960-69	13.23%	9.57%	3.66%	15.27%	9.79%	5.48%
1970-79	17.05%	3.89%	13.16%	14.83%	2.28%	12.54%
1980-89	24.48%	12.94%	11.54%	18.38%	14.46%	3.92%
1990-99	20.17%	21.88%	-1.71%	21.61%	22.03%	-0.41%
2000-09	8.59%	-0.49%	9.08%	13.84%	0.61%	13.23%
2010-19	11.27%	16.67%	-5.39%	11.35%	17.09%	-5.75%

Survivor bias: Did you start with all listed companies at the start of each period? What happened to the companies that were delisted?

Timing bias: Do return patterns change over time? Does picking a different time period change your result?

5. Event Studies

Graph showing the Average Abnormal Return around an Acquisition Announcement



Timing questions: In an event study, you need to identify the date of the event precisely. If you are wrong on that judgment, your study results will be skewed.



SESSION 3: DATA DESCRIPTIVES

Session 3
Means and Moments

Data Descriptives: An Overview

- ❑ Data descriptives are designed to capture the features of data in metrics that can be used to summarize the data and make comparisons across time and samples.
- ❑ Broadly speaking, there are four groups of descriptive statistics:
 - ❑ *Measures of centrality* are meant to capture the central value of a variable, across a sample.
 - ❑ *Measures of dispersion* try to measure the spread of data around the central value, with greater dispersion referencing a larger spread.
 - ❑ *Measures of symmetry* capture the skew in the spread of the data around the central value
 - ❑ *Measures of extremes* measure the likelihood of extreme values (the fatness of distributional tails)

Measures of Centrality

1. Mean: Perhaps, the most common measure of centrality in a data is the average. It is computed by aggregating the numerical values across all observations and dividing by the number of observations:
2. Median: The median literally represents the mid-point in the data, where half of all of the observations are higher than it and half are lower.
3. Mode: When the data take on different values, the mode is the value that you observe most frequently across observations.

Comparing Central Tendency Measures

1. The average matters: The average is the most widely used measure of central tendency because other statistical measures are built around it; the standard deviation, for instance, is computed around the average.
2. But the median may be more meaningful: When the data is symmetric, i.e., outliers lie on both sides of the average in rough equivalence, the mean and the median will converge. When the outliers are more in one direction than in the other, the average will be skewed by the outliers, and the median may be more meaningful.
3. And the mode less so: The mode is the least used measure of central tendency, at least in finance and investing, because it is designed more for discrete or categorical data, than it is for continuous data.

Measures of Dispersion

- Range and variants: With the range, you look at the difference between the highest and lowest values for a variable, across a sample. In variants, you can look at the difference between the first and the third quartile of the data (interquartile range) or between the first and the ninth decile of the data.
- Standard deviation/Variance: With the standard deviation, you estimate the difference between each value of a variable and its mean and arrive at a measure of dispersion.
- Coefficient of variation: With the coefficient of variation, you divide the standard deviation of a data series by its mean, to provide a measure of comparison with data series of different levels.

The Range

- The range is the difference between the maximum and minimum values of a data variable in a sample.
 - ▣ Range = Maximum Value – Minimum Value
 - ▣ This range can be standardized by dividing by the average of the two values.
 - ▣ Standardized Range = $(\text{Max} - \text{Min}) / [(\text{Max} + \text{Min})/2]$
- It remains the simplest measure of dispersion for a data series, and the range requires no explicit assumptions about how the data is distributed.
- However, by using only the highest and lowest values, and ignoring the intermediate numbers, it wastes data and can give misleading measures of dispersion.
- In general, though, breaking data down into deciles or quartiles will give you useful information about dispersion.

Standard Deviation/ Variance & Coefficient of Variation

- The standard deviation is a measure of dispersion which uses all of the observations, computes the difference between each one and the mean, and summing up the squared differences:
 - Variance = $\frac{\sum_{j=1}^n (X_j - \mu)^2}{n}$ (where μ is the average across the n observations, and X_j is the value that of the j^{th} observation); The sum of the squared differences is divided by n, if your data comprises the entire population, or n-1, if it is a sample.
 - The standard deviation is the square root of the variance.
- When there is more divergence from the mean, the standard deviation will be higher, but it will be in the same units as the base data. Thus, if the base data is in dollars, the standard deviation will be in dollars, and if it is in percent, it will be in percent.
- Since standard deviations cannot be compared across two samples with different units or levels, you can compute a standardized version of the measure:
 - Coefficient of Variation = Std Deviation in Value/ Average Value

Standard Deviation and Standard Error

- The standard deviation measures the amount of variability, or dispersion, from the individual data values to the mean
- The standard error of the mean measures how far the sample mean (average) of the data is likely to be from the true population mean. It is computed as follows:
 - ▣
$$\text{Standard Error} = \frac{\text{Standard Deviation}}{\sqrt{\text{Number of observations in sample}}}$$
- As sample size increases, there will be no discernible effect on the former, but the latter will always decrease.
- When you are extrapolating from sample findings to the population, the standard errors become useful because they can be used to provide ranges for estimates. Thus, if your average is μ , and your standard error is SE, drawing on the central limit theorem, you can estimate the population mean:
 - ▣ With 67% confidence: $\mu \pm SE$
 - ▣ With 95% confidence: $\mu \pm 2 * SE$

Measures of Asymmetry

- When the data is symmetric, the deviations from the mean fall equally or roughly equally on either side of the mean.
- When the data is asymmetric, deviations on one side of the mean are much more pronounced than deviations on the other side. This deviation is measured with skewness.
 - If the deviations are more pronounced/extreme for the observations that have values higher than the average, the distribution is positively skewed.
 - If the deviations are more pronounced/extreme for the observations that have values lower than the average, the distribution is negatively skewed.
- When data is asymmetric, the average will be skewed in the same direction as the asymmetry, and in some cases the skew can be large enough to make it unrepresentative of the sample.

Boundedness, and consequences

- In some cases, data can also be bounded, where there are limits on how low a value can be (lower bound) or or high (upper bound). Those limits can come from either how the data variable is defined, or from how it is recorded.
- In others, data is unbounded, where it can take values, albeit very, very rarely, that approach infinity (plus or minus).
- When data is bounded on only one side and unbounded on the other, asymmetry almost always follows.

Measures of extreme values

- You can measure of how much, and how frequently, data takes extreme values, relative to its central value. That measure is called kurtosis.
- While variance and kurtosis are both affected by the presence (or absence) of extreme values, they measure different phenomenon.
 - You can have high variance and low kurtosis, low variance and high kurtosis or high variance and high kurtosis.
 - Distributions that have more frequent occurrences of extreme values are referred to as having fat tails or *leptokurtic*. Distributions that have less frequent occurrences of extreme values are referred to as *platykurtic*.

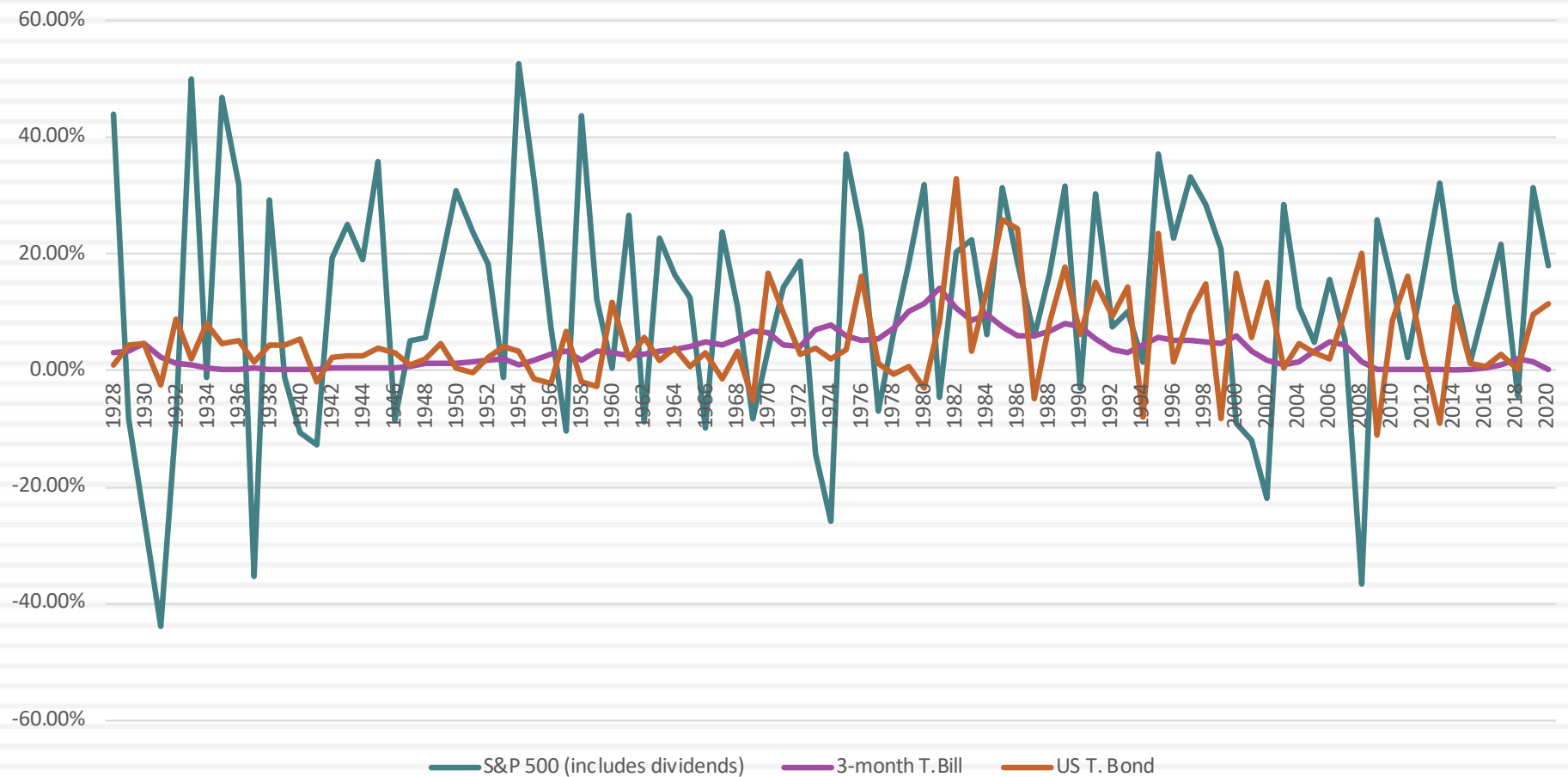


SESSION 3A: DATA DESCRIPTIVES IN FINANCE

Session 3
Means and Moments

1. Financial Asset Class Returns: Historical Data

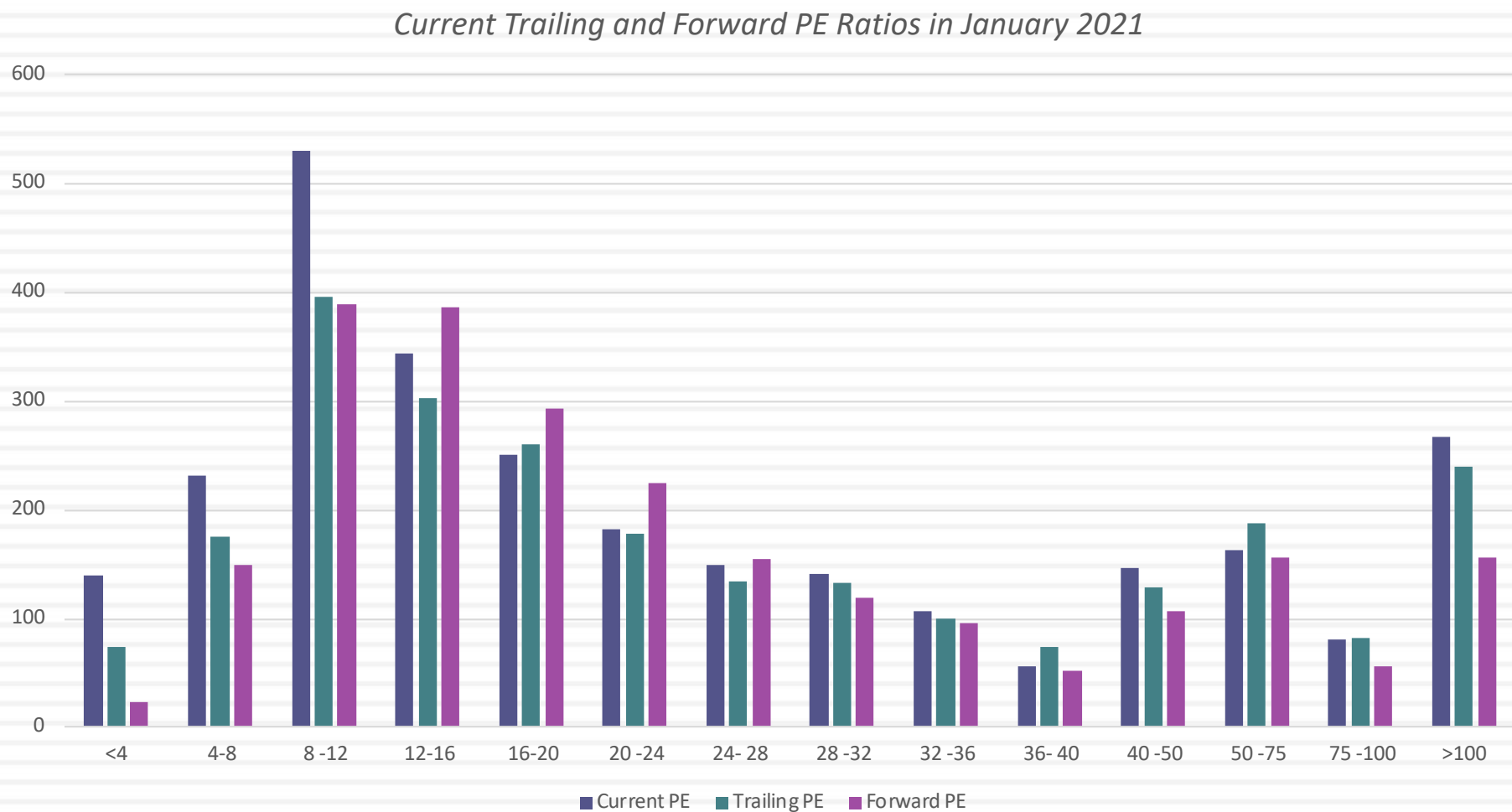
US Stocks, T.Bond and T.Bill Returns: 1928-2020



Data Descriptives: Financial Asset Class Returns from 1928 to 2020

	Stocks	T. Bills	T. Bonds
Mean	11.64%	3.36%	5.21%
Standard Error	2.02%	0.32%	0.79%
Median	14.22%	3.00%	3.29%
Standard Deviation	19.49%	3.15%	7.66%
Kurtosis	3.005	3.7944	4.5196
Skewness	-0.4196	1.0079	0.9653
Minimum	-43.84%	0.00%	-11.12%
Maximum	52.56%	14.03%	32.82%
Range	96.40%	14.03%	43.94%
25th percentile	-1.19%	0.93%	0.92%
75th percentile	25.06%	5.06%	8.79%
Inter-quartile Range	26.25%	4.13%	7.87%

2. PE Ratios: Cross Sectional Data

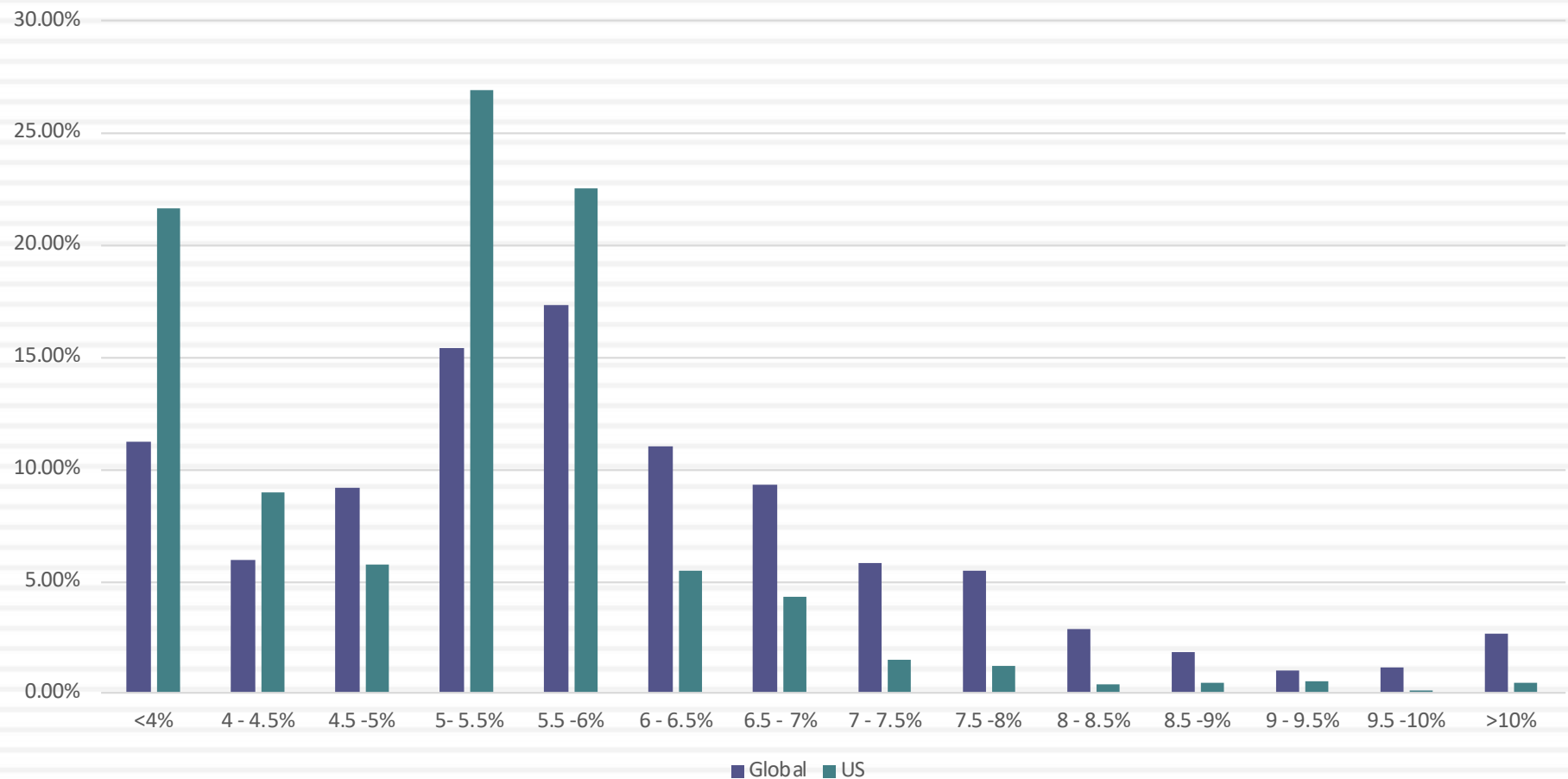


Data Descriptives

	Current PE	Trailing PE	Forward PE
Total Number of firms	7584	7584	7584
Firms with PE	2780	2481	2354
Average	109.79	103.25	79.74
Median	18.15	20.30	18.89
10th Percentile	6.95	7.68	8.96
First Quartile	10.41	11.50	12.36
Third Quartile	37.26	40.79	33.20
90th Percentile	95.44	96.80	69.40
Maximum	36157.14	25020.00	42390.00

3. Costs of Capital: Global companies in 2021

Cost of Capital in US\$ in January 2021



Statistics, by region

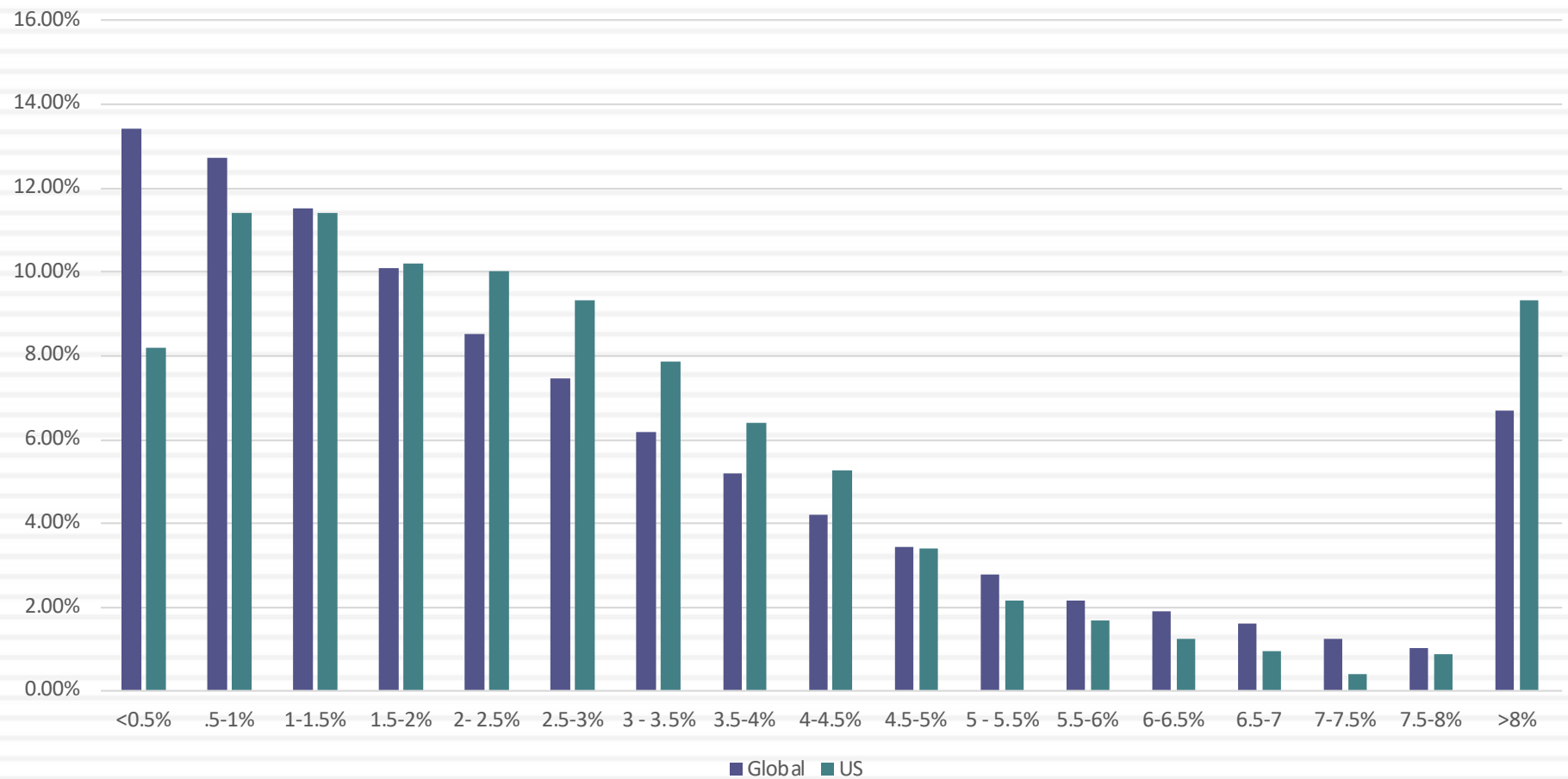
Sub Group	# firms	Average	Median	Minimum	Maximum
Africa and Middle East	2,245	6.69%	6.27%	1.56%	24.71%
Australia & NZ	1,766	4.91%	5.00%	1.71%	9.10%
Canada	2,828	5.11%	5.02%	1.71%	11.39%
China	6,810	5.87%	5.95%	1.80%	13.28%
EU & Environs	5,653	5.92%	5.68%	1.71%	21.17%
Eastern Europe & Russia	544	6.49%	6.44%	1.96%	13.84%
India	3,758	6.58%	6.97%	2.06%	16.03%
Japan	3,893	5.70%	5.83%	1.82%	10.28%
Latin America & Caribbean	1,082	7.43%	6.66%	1.82%	32.87%
Small Asia	9,185	6.61%	6.34%	1.71%	16.10%
UK	1,233	5.63%	5.87%	1.80%	18.72%
United States	7,582	5.08%	5.30%	1.71%	16.98%
All firms (Global)	46,580	5.93%	5.78%	1.56%	32.87%

Distributional Details

	<i>All in US \$ for comparability</i>				
<i>Sub Group</i>	1st Decile	1st Quartile	Median	3rd Quartile	9th Decile
Africa and Middle East	5.33%	6.08%	7.84%	9.49%	11.57%
Australia & NZ	4.88%	6.05%	7.32%	7.82%	8.38%
Canada	5.07%	7.00%	7.59%	7.92%	8.59%
China	5.52%	6.17%	7.50%	8.69%	9.43%
EU & Environs	4.99%	6.02%	7.48%	8.76%	10.01%
Eastern Europe & Russia	5.67%	6.35%	7.92%	9.48%	10.72%
India	6.17%	6.68%	7.71%	9.01%	10.15%
Japan	5.34%	5.85%	7.04%	8.38%	9.18%
Latin America & Caribbean	5.49%	6.35%	8.00%	10.02%	12.68%
Small Asia	5.69%	6.57%	8.03%	9.51%	10.69%
UK	5.05%	5.92%	7.47%	8.55%	9.09%
United States	4.80%	6.06%	7.40%	8.27%	9.15%
Global	5.32%	6.27%	7.58%	8.71%	9.88%

4. Dividend Yields in 2021

Dividend Yields among Dividend Paying Companies in January 2021



Descriptive Statistics

<i>Sub Group</i>	<i># firms</i>	<i>All companies</i>			<i>Only dividend paying</i>	
		<i>Average</i>	<i>Median</i>	<i>% of companies with zero dividends</i>	<i>Average</i>	<i>Median</i>
Africa and Middle East	2,207	2.83%	0.01%	49.21%	5.58%	3.89%
Australia & NZ	1,713	0.87%	0.00%	71.80%	3.09%	2.67%
Canada	2,752	0.48%	0.00%	87.06%	3.72%	3.03%
China	6,785	2.30%	0.72%	27.99%	3.19%	1.44%
EU & Environs	5,577	1.56%	0.00%	63.33%	4.26%	2.06%
Eastern Europe & Russia	506	2.30%	0.00%	56.52%	5.29%	4.12%
India	3,737	0.50%	0.00%	69.15%	1.61%	0.86%
Japan	3,887	1.62%	1.42%	29.20%	2.29%	2.08%
Latin America & Caribbean	1,068	3.10%	0.57%	41.20%	5.27%	2.50%
Small Asia	9,095	1.81%	0.25%	47.49%	3.45%	2.63%
UK	1,228	1.50%	0.00%	50.65%	3.03%	1.91%
United States	7,420	1.04%	0.00%	74.08%	4.00%	2.43%
Global	45,976	1.57%	0.00%	54.43%	3.44%	2.12%



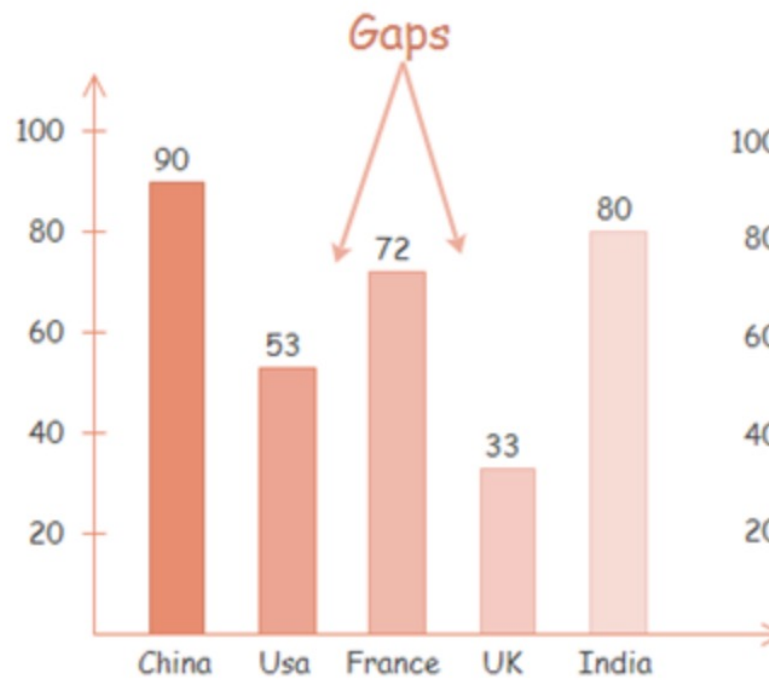
SESSION 4: DATA DISTRIBUTIONS

Session 4
Accounting & Statistics

Data Distributions: Why they matter!

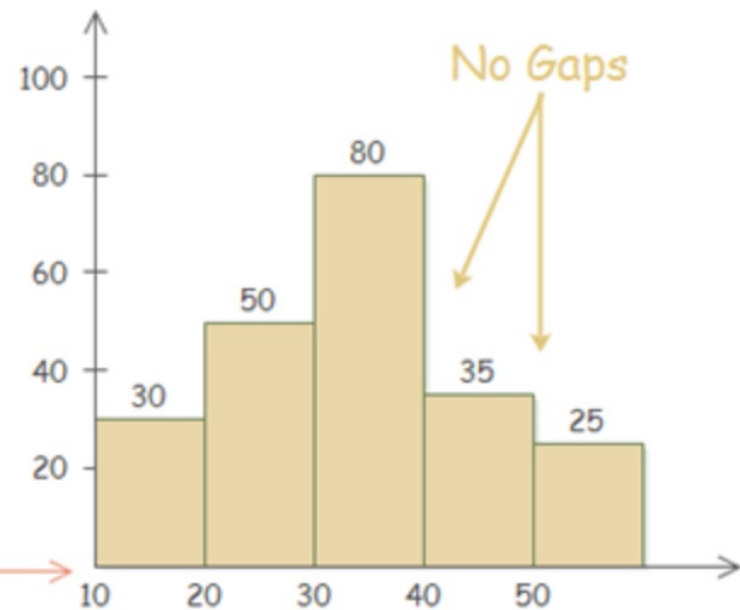
- While data descriptive statistics provide you with useful information about location, dispersion and symmetry in your data, data distributions provide the same information in fuller and usually visual form.
 - ▣ In its simplest and least complicated form, data (in either discrete or continuous form) can be converted into a histogram or a bar chart.
 - ▣ That histogram/chart provides the same information that data descriptive statistics provide
- When the distributions approach or resemble classic statistical distributions, you gain the benefit of being able to draw on their established distributional properties to extrapolate from the data.

Histogram: Visualizing Data



← Categories →

Bar Chart



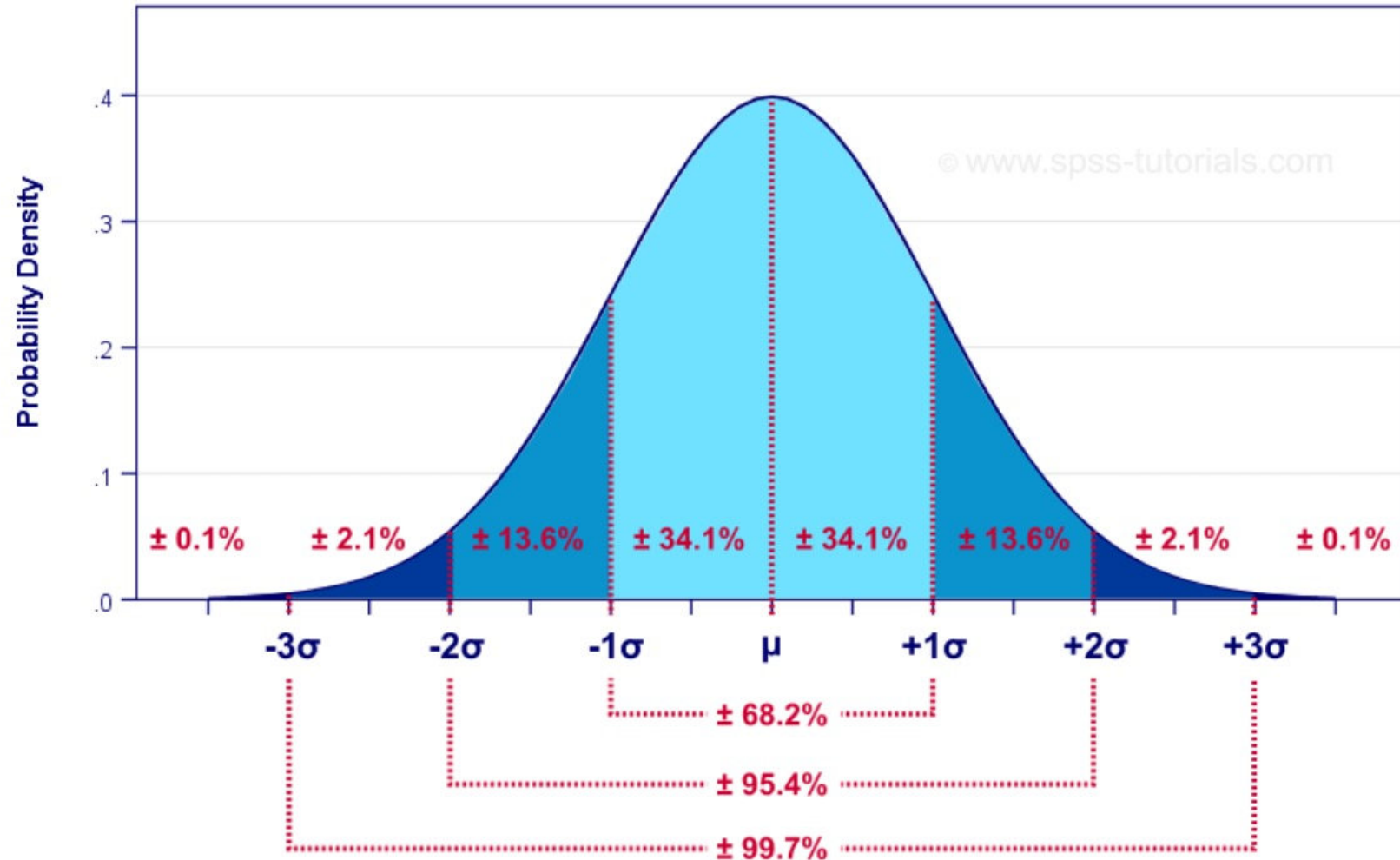
← Number Ranges →

Histogram

Statistical Distributions

- While histograms summarize data in visual form, there is an advantage to replacing them with one of many standardized statistical distributions (normal, exponential, uniform etc.).
 - ▣ You have to find a distribution that best fits your data in terms of symmetry, skewness and tails.
 - ▣ The benefit of using a standardized distribution is that you can draw on their established distributional properties to extrapolate from the data or to make probabilistic statements relating to the data.

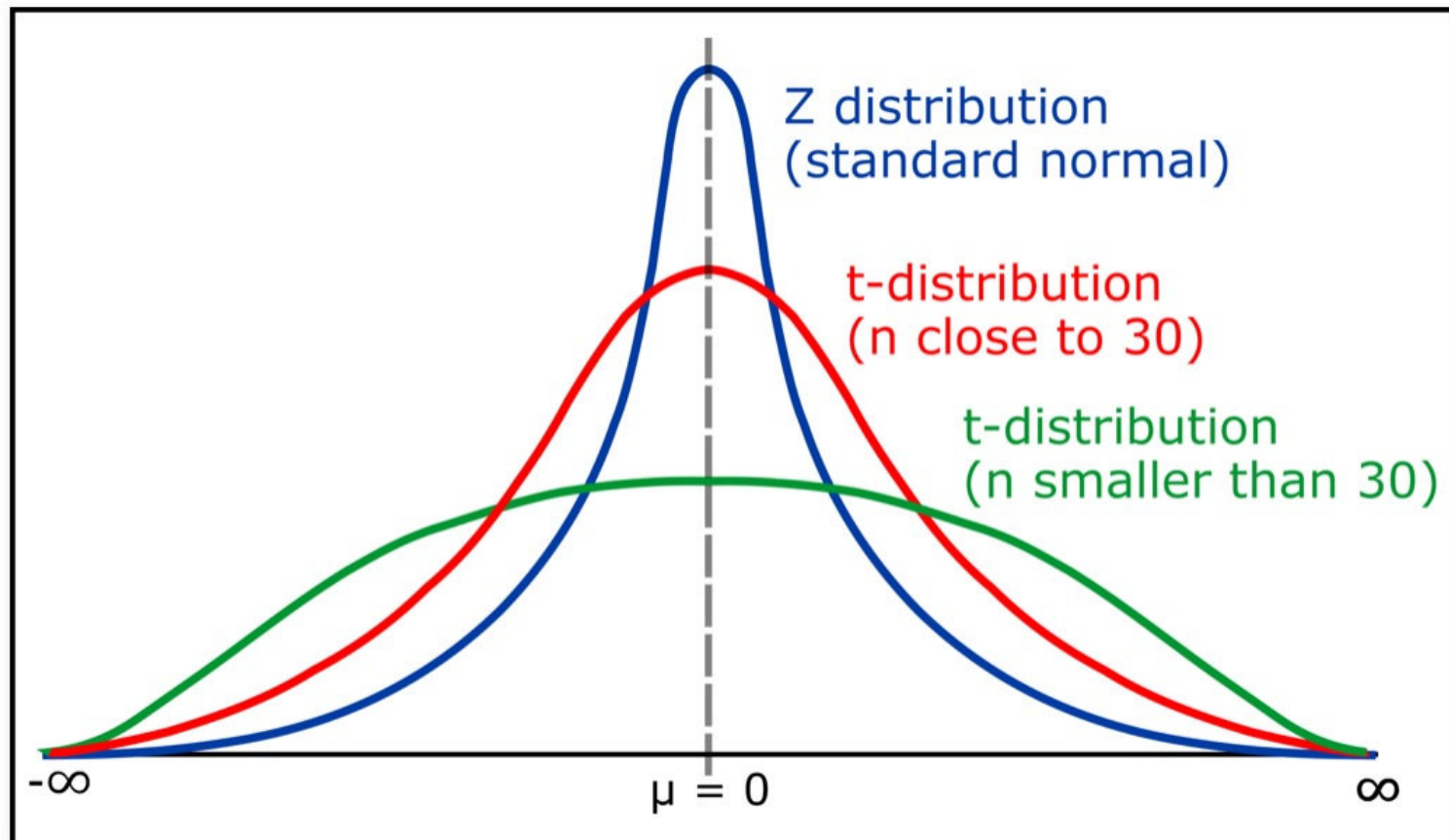
The Normal Distribution



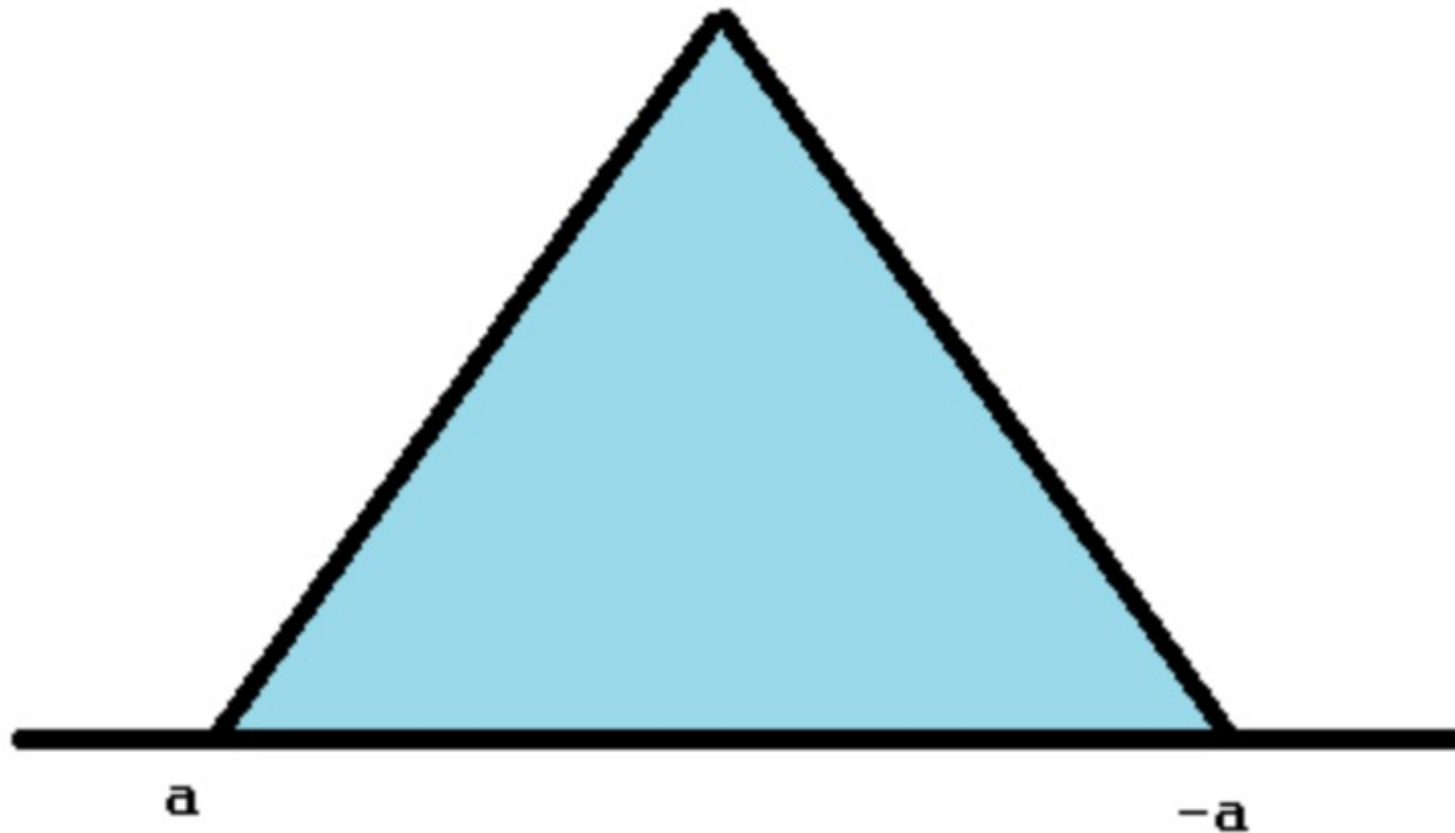
And its properties...

- Dispersion: The standard deviation in a normal distribution measures the spread around the mean, but with links to probabilities of a number occurring in the sample falling within or out of that spread.
- Skewness: A normal distribution is symmetric and has no skewness.
- Kurtosis: A variable that is normally distributed can take on values from minus infinity to plus infinity, but the likelihood of extreme values is constrained. The kurtosis for a normal distribution is three, which becomes the standard against which other distributions are measured.

A Close Relative: The t distribution



A more distant relative: The (Symmetric) Triangular Distribution

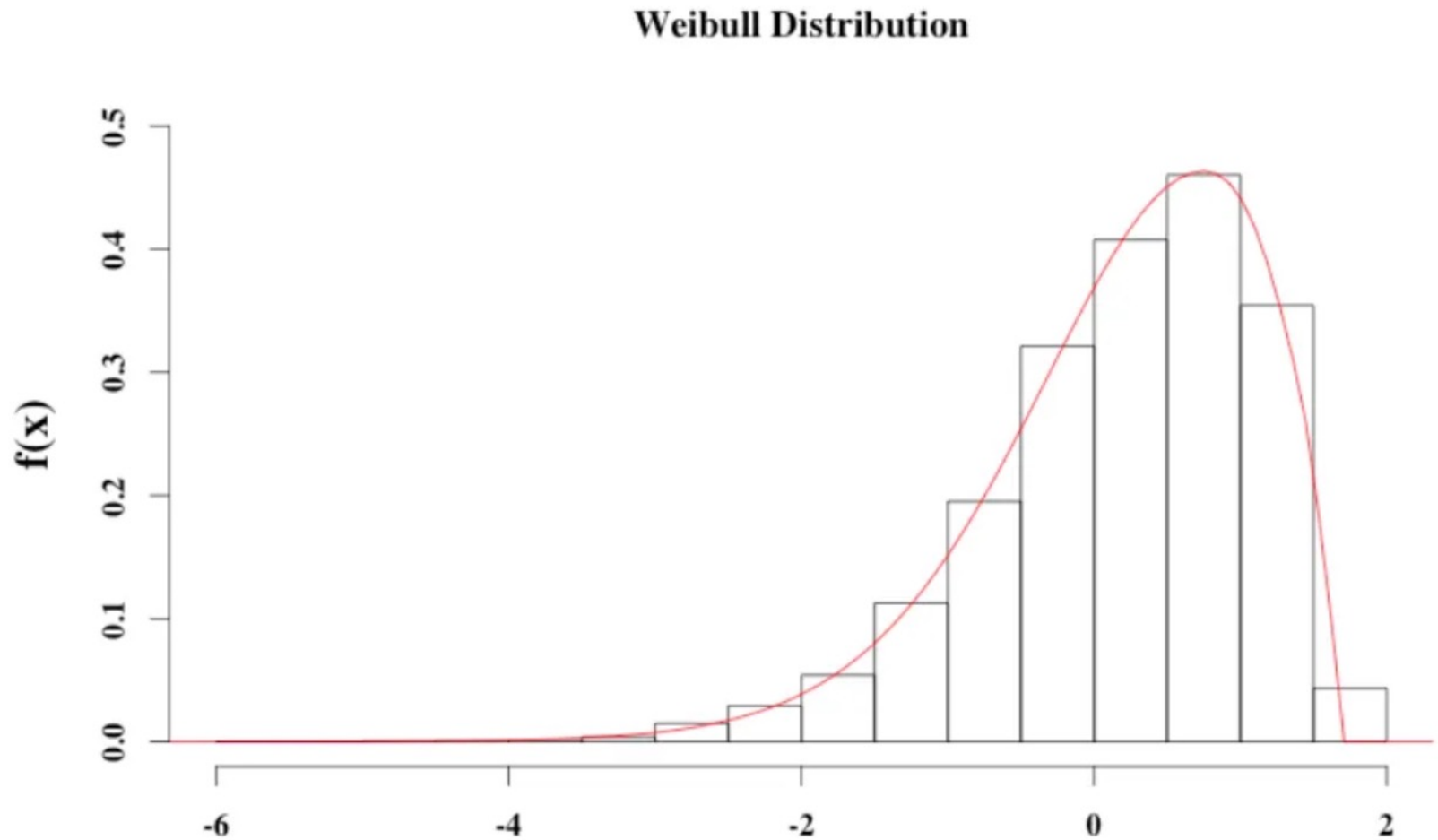


A Uniform Distribution

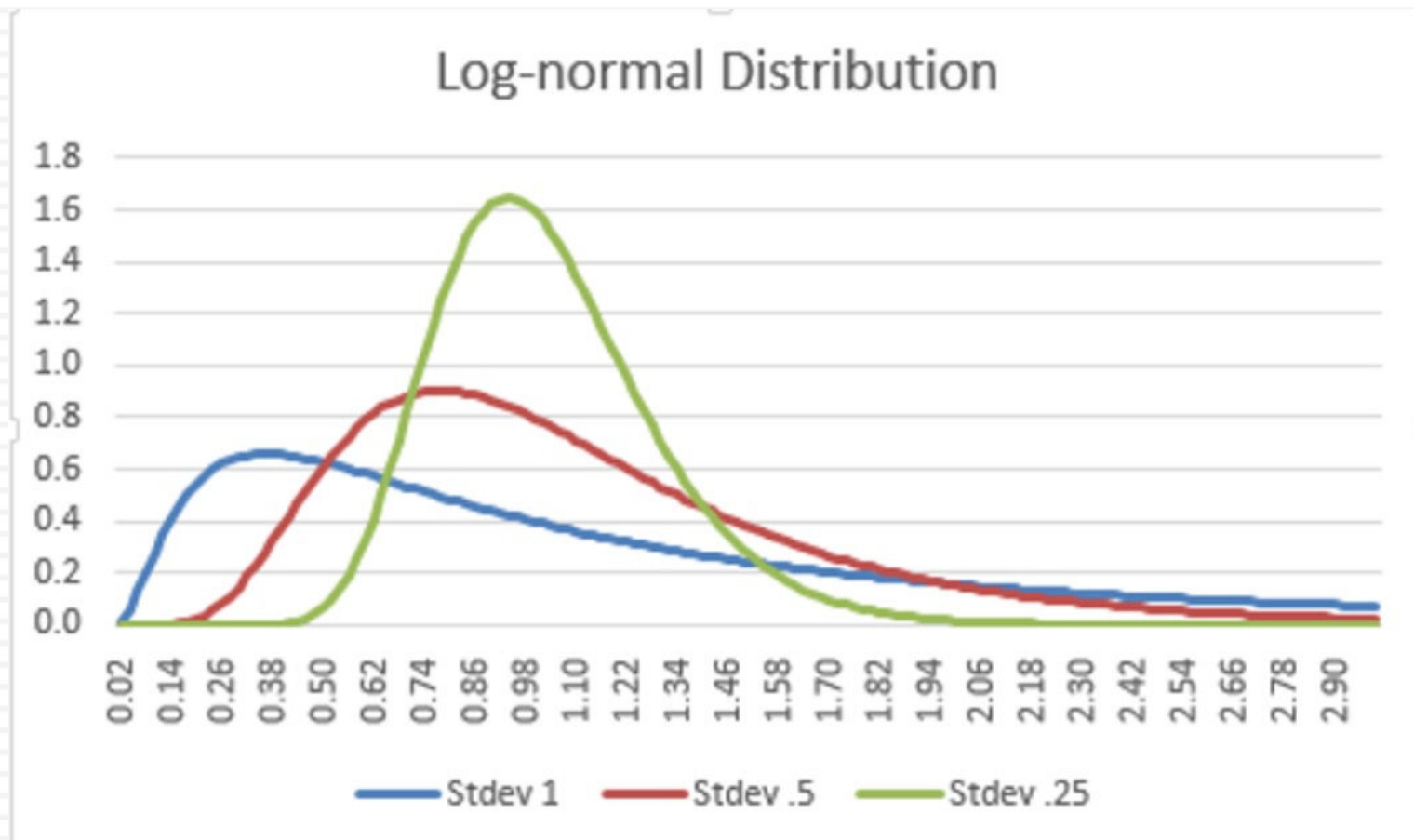


$$\text{mean: } \mu = E(X) = \frac{b+a}{2}$$

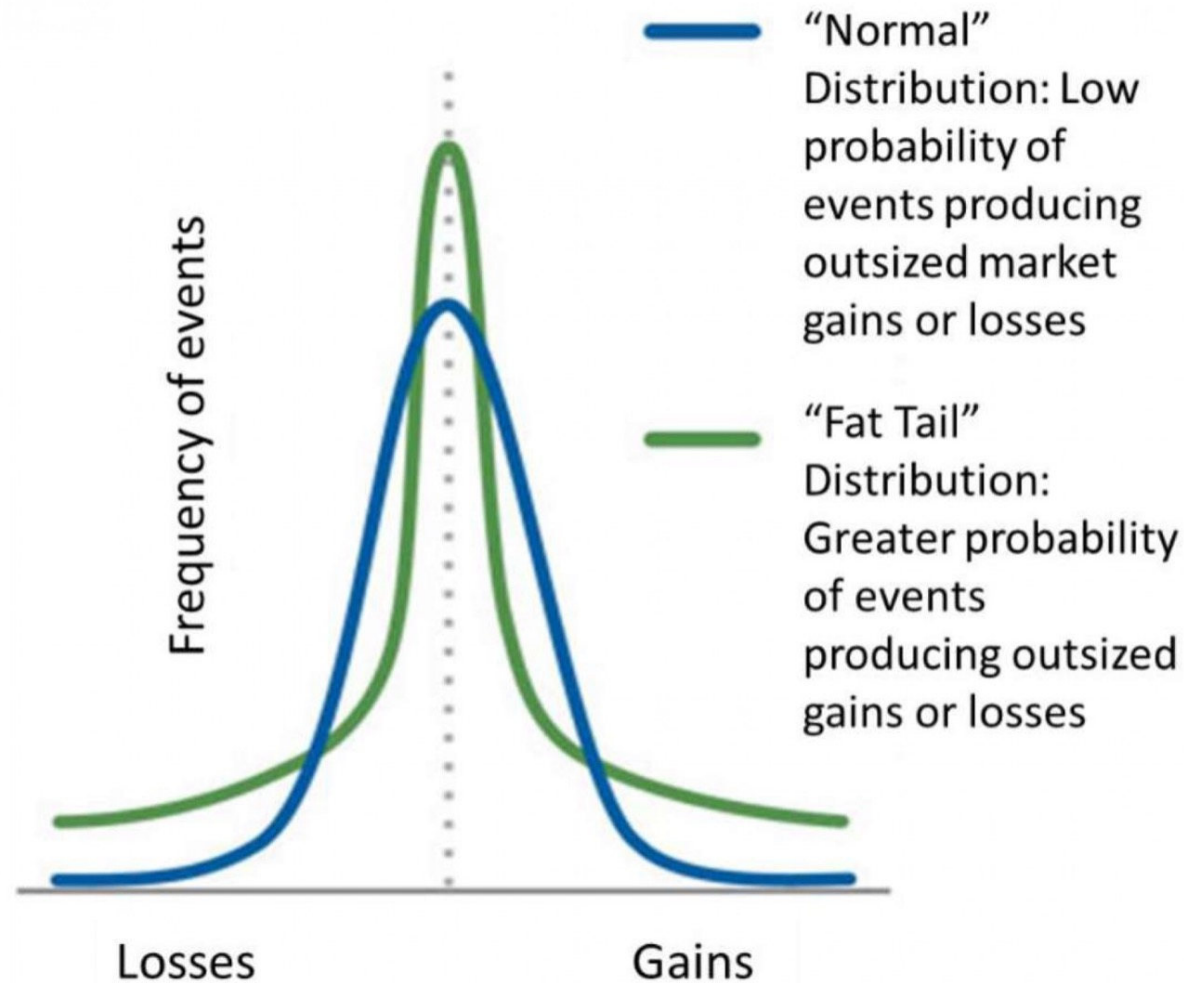
Negative Skew: Minimum Extreme Value



Positive Skew: Log Normal Distribution

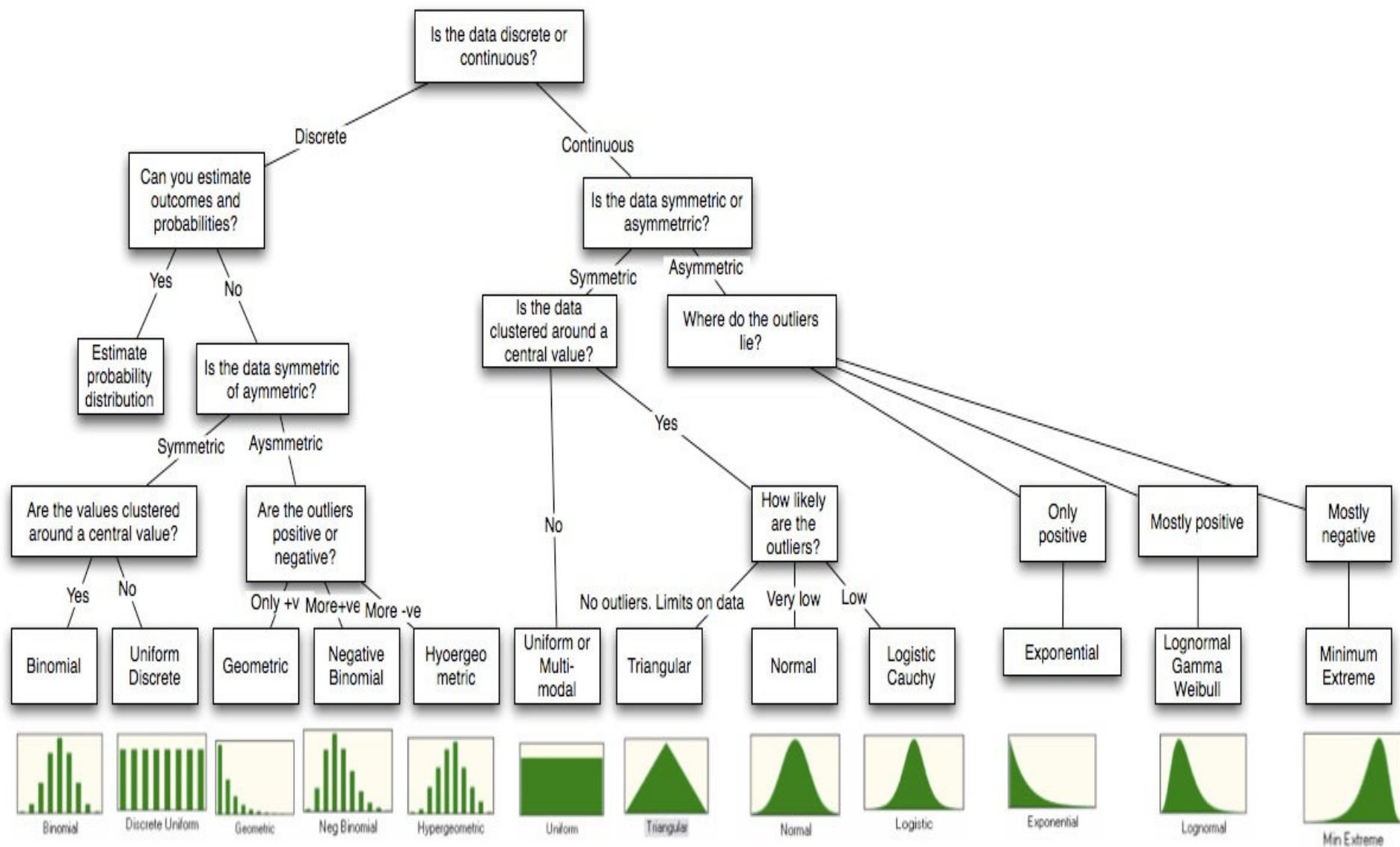


Thin tails and Fat tails...



Measured with kurtosis...

- Kurtosis is a measure of the combined weights of the tails, relative to the rest of the distribution.
- Most often, **kurtosis** is measured against the **normal distribution**. Pearson's kurtosis is the excess kurtosis over three
 - ▣ If the Pearson **kurtosis** is close to 0, then a **normal distribution** is often assumed. These are called mesokurtic **distributions**.
 - ▣ If the Pearson **kurtosis** is less than 0, then the **distribution** has thin tails and is called a platykurtic **distribution**. (Uniform distribution is a good example)
 - ▣ If the Pearson **kurtosis** is greater than 0, then the **distribution** has fat tails and is called a leptokurtic **distribution**.





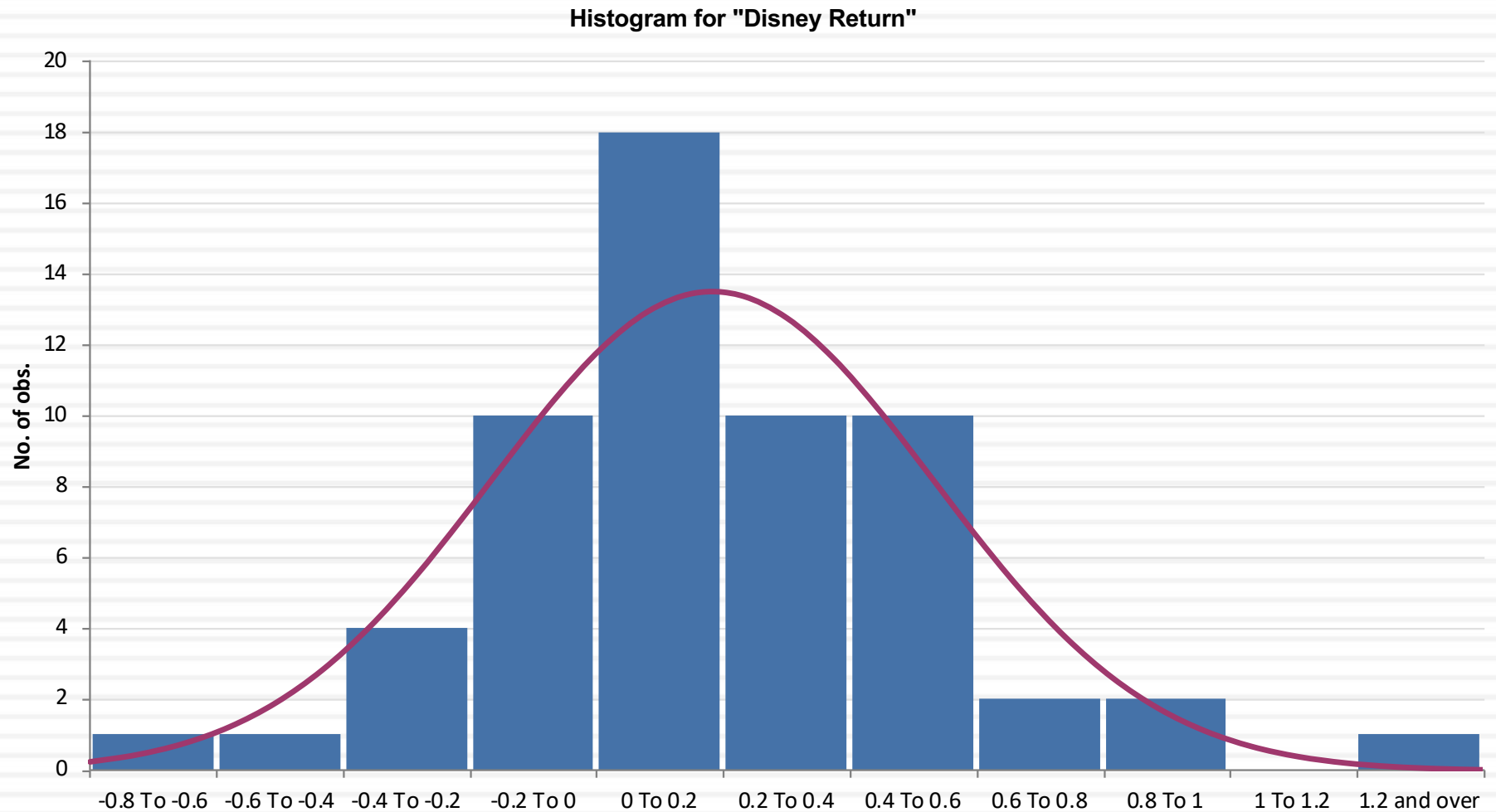
SESSION 4A: DATA DISTRIBUTIONS IN FINANCE

Session 4
Accounting & Statistics

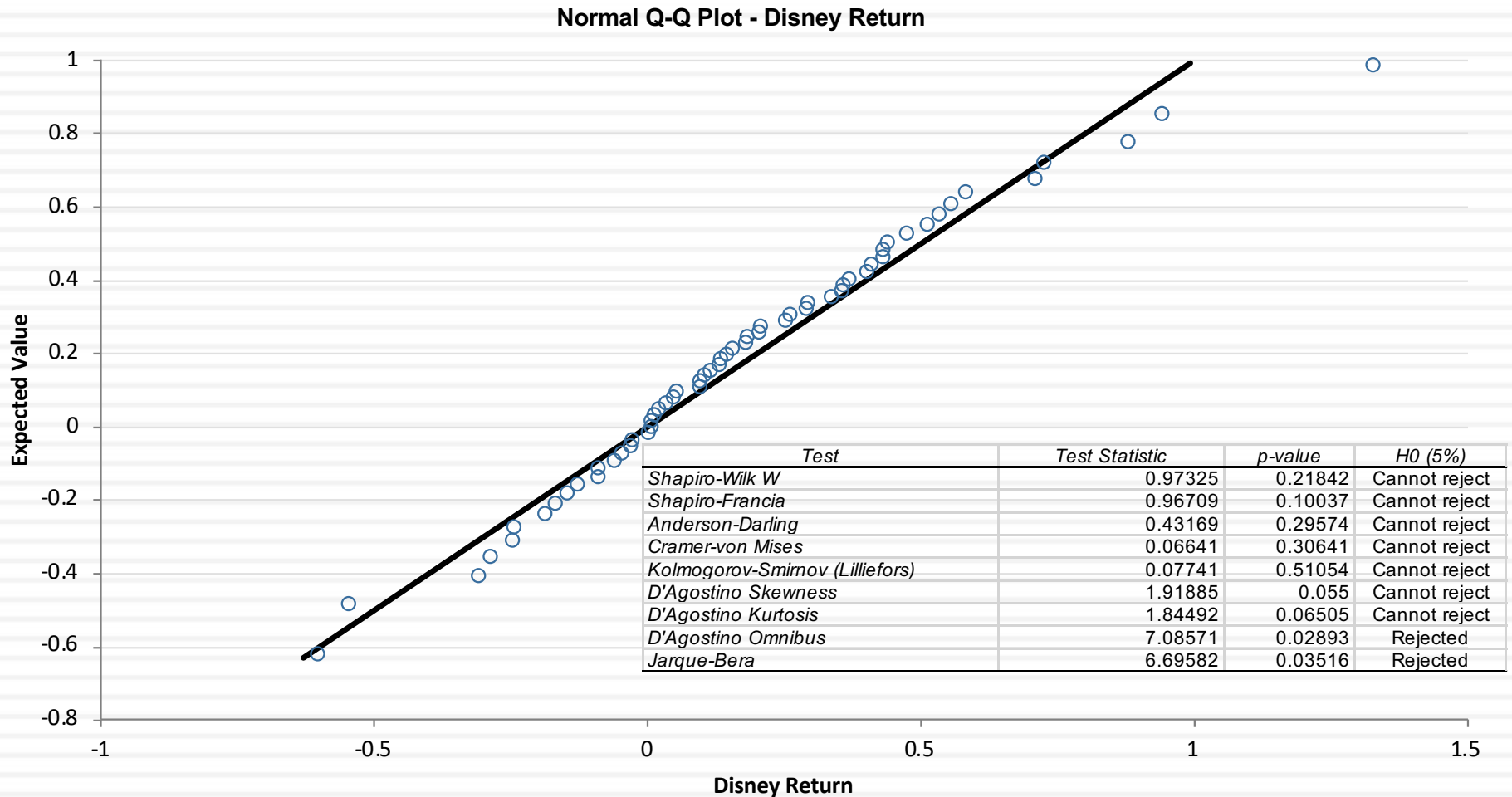
The Normal Distribution in Finance

- One of the critiques of finance is its dependence on the normal distribution in both developing theory and in empirical tests.
- The reason for that dependence is two-fold:
 - ▣ The normal distribution has properties that make it easy to work with, both in theory and in practice.
 - ▣ In financial markets, we have had access to large amounts of data (we had big data, before the rest of the world even discovered it), and with it came the temptation to use the normal distribution on almost every aspect of finance.
- The normal distribution assumption comes into play especially when it comes to returns that you can earn on investments (and stock prices).

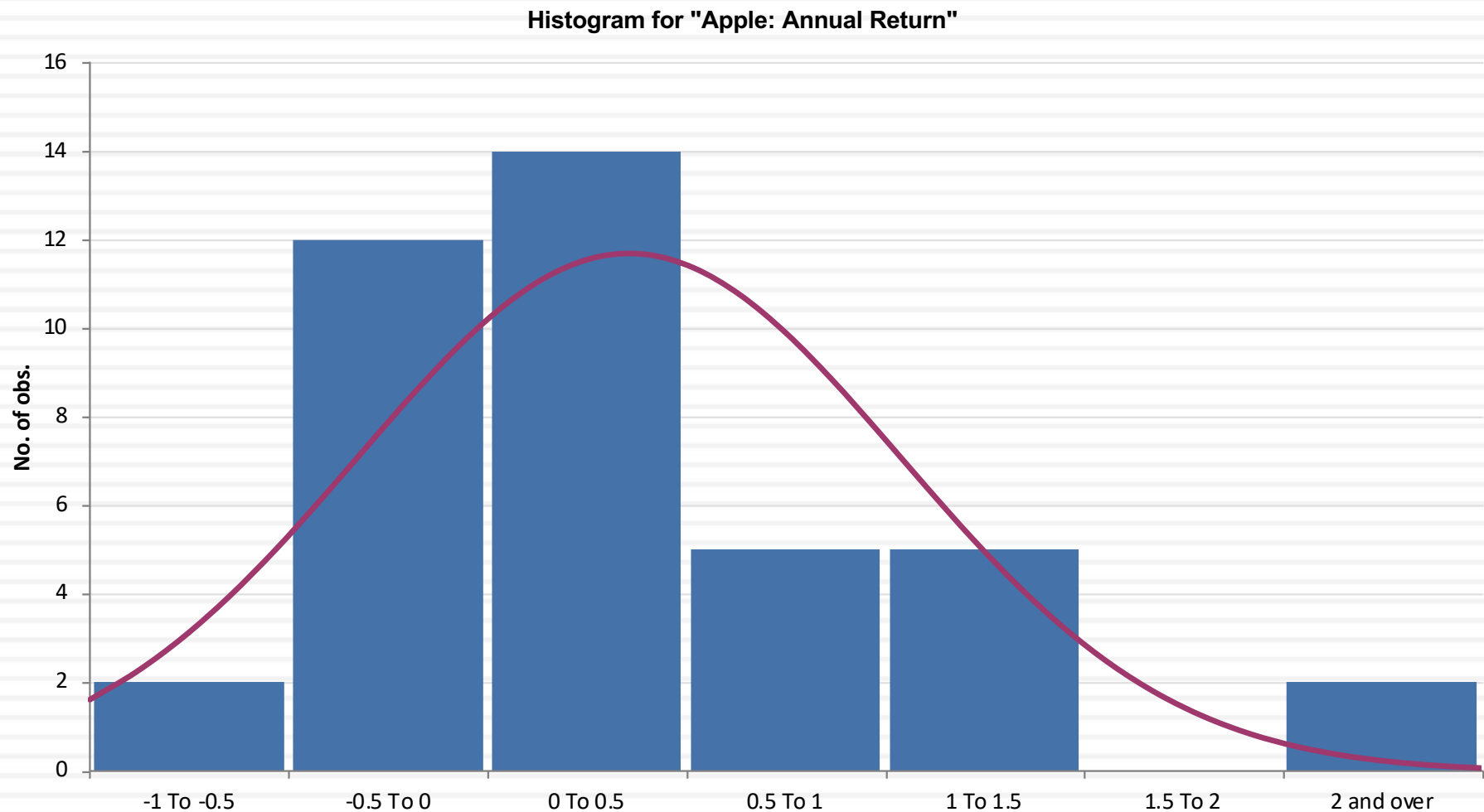
An Individual Stock: Annual Returns on Disney from 1962 - 2021



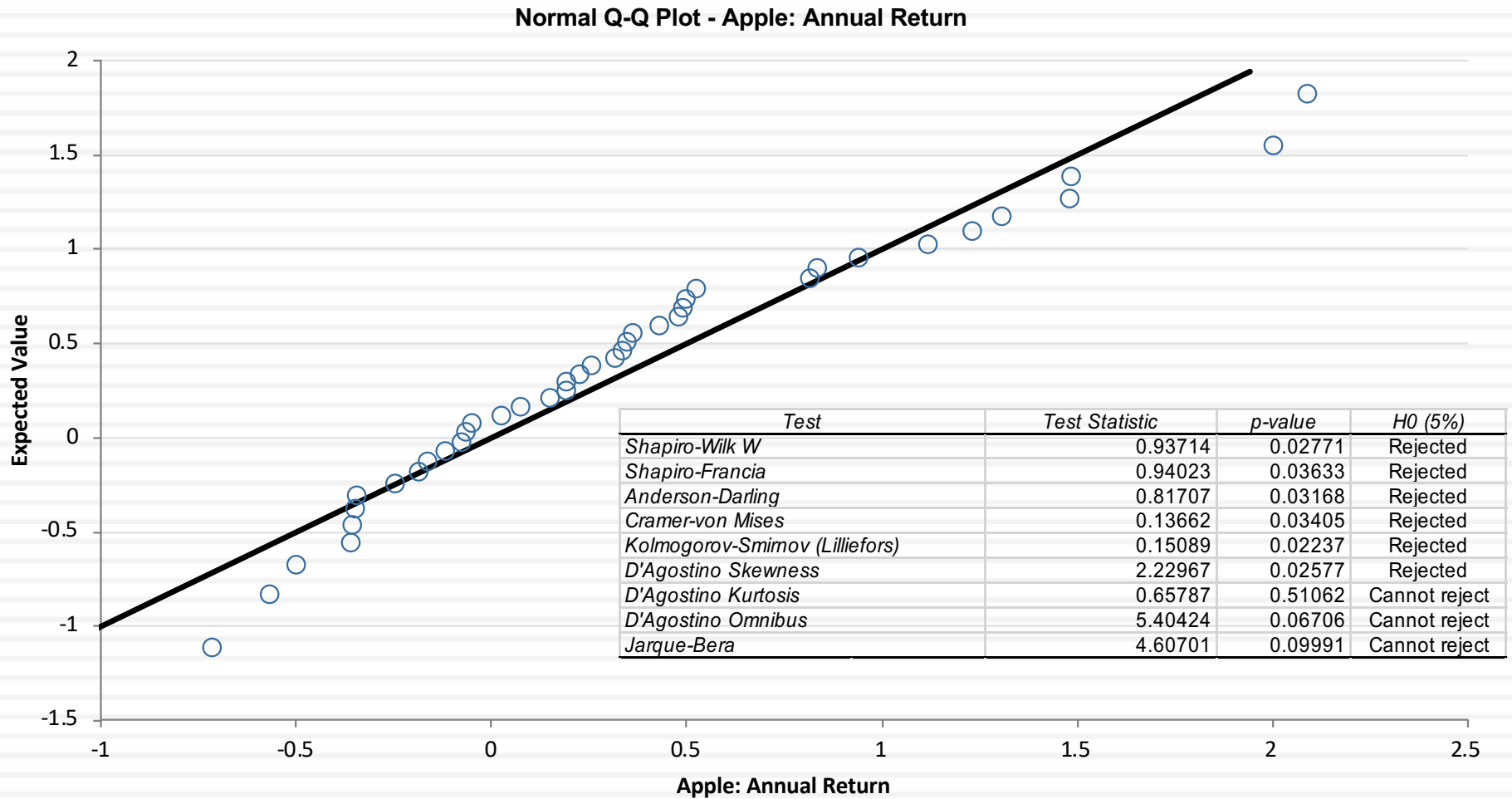
Testing for Normality



Another Individual Stock: Annual Returns on Apple from 1981 to 2020

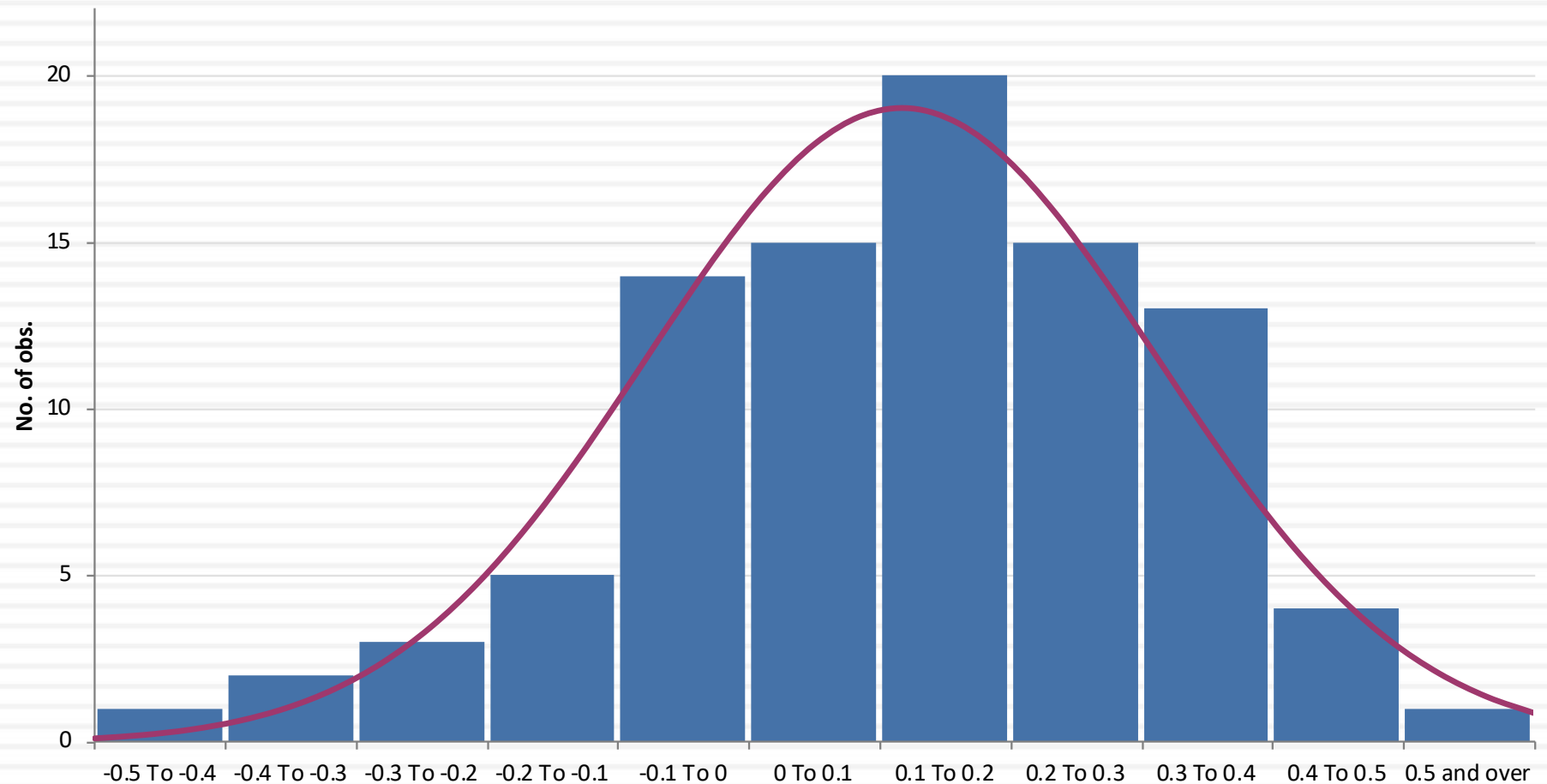


Testing for Normality



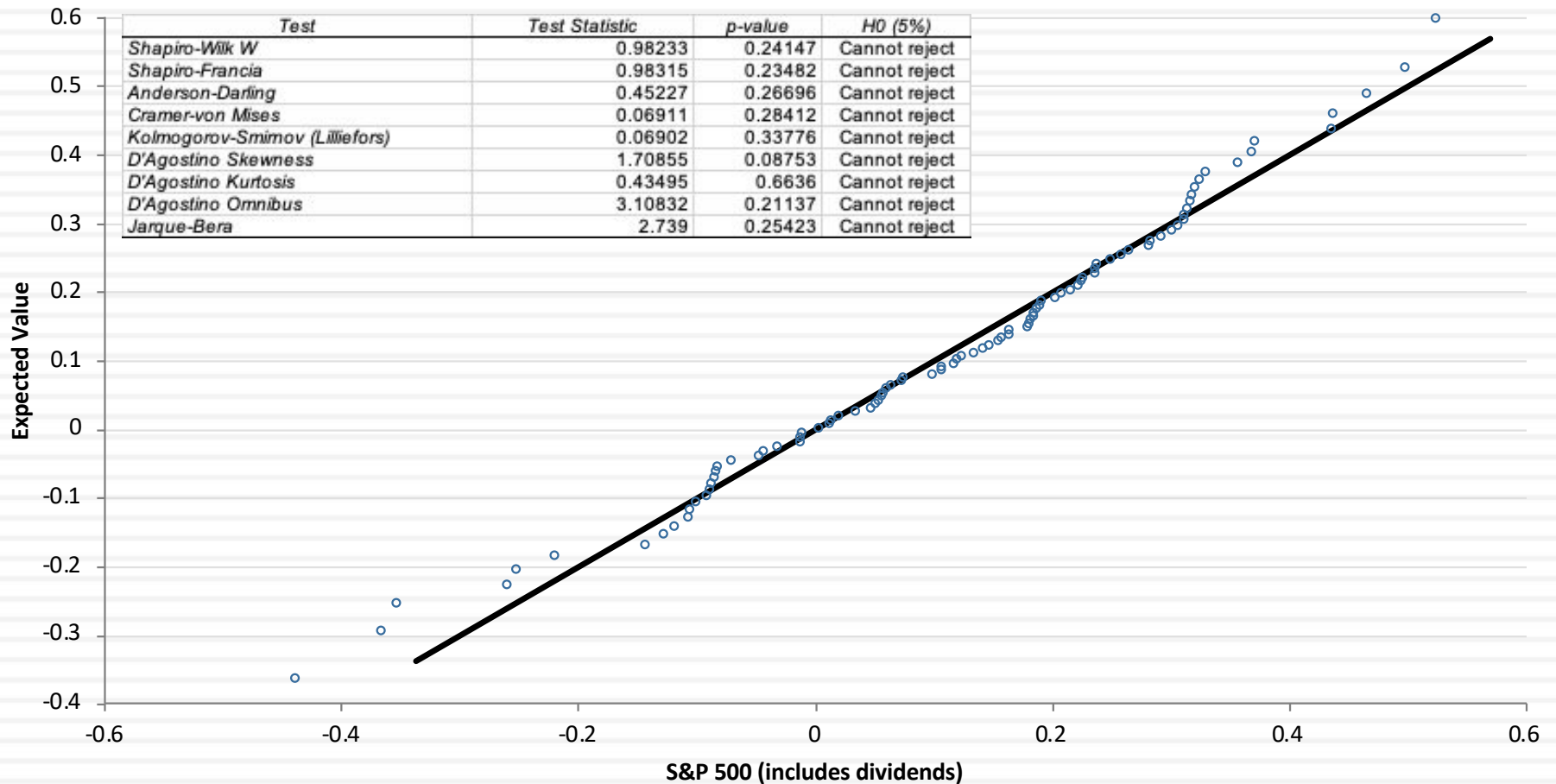
Returns on S&P 500: Annual Returns from 1928 to 2020

Histogram for "S&P 500 (includes dividends)"

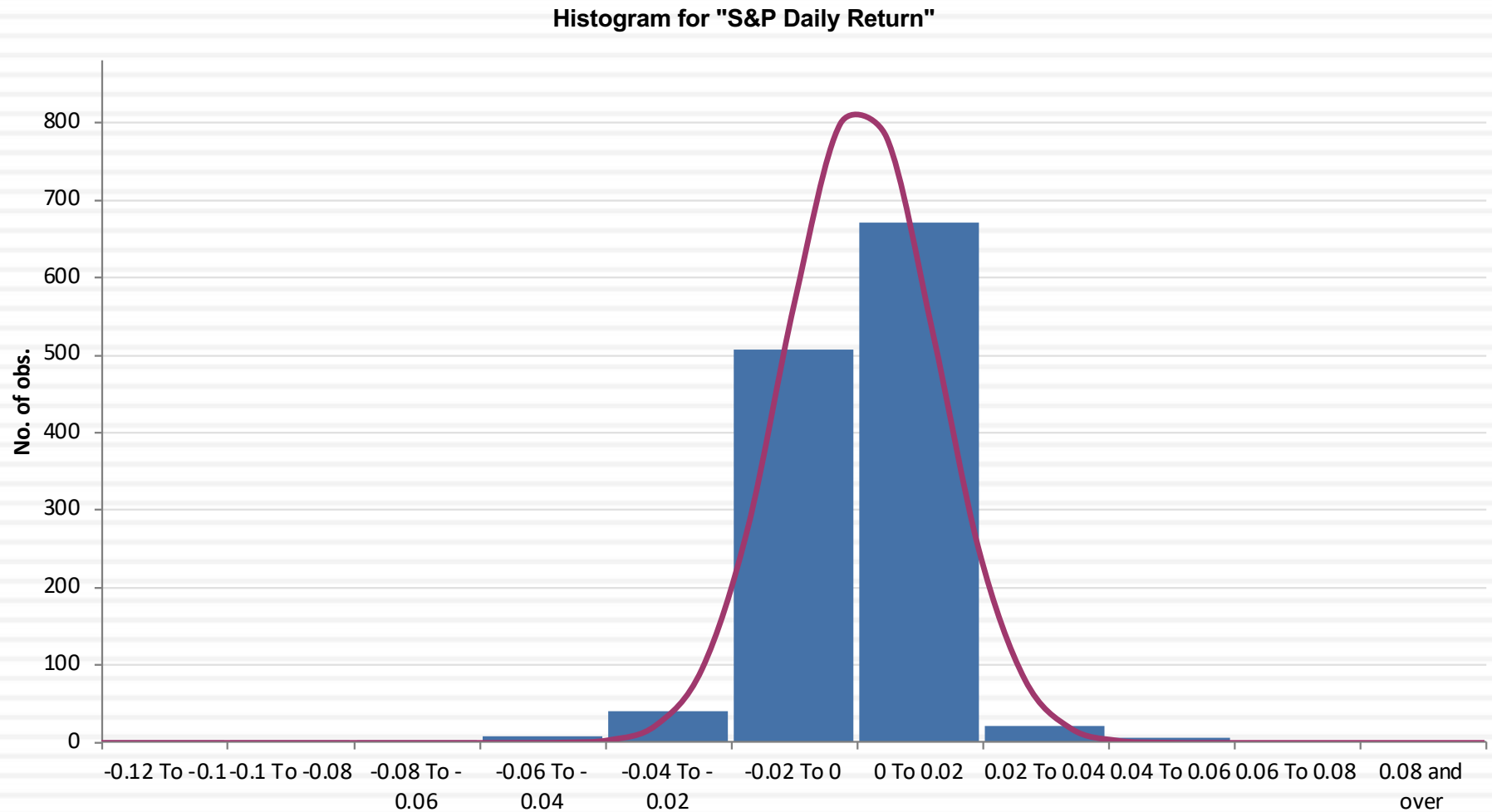


Testing for Normality

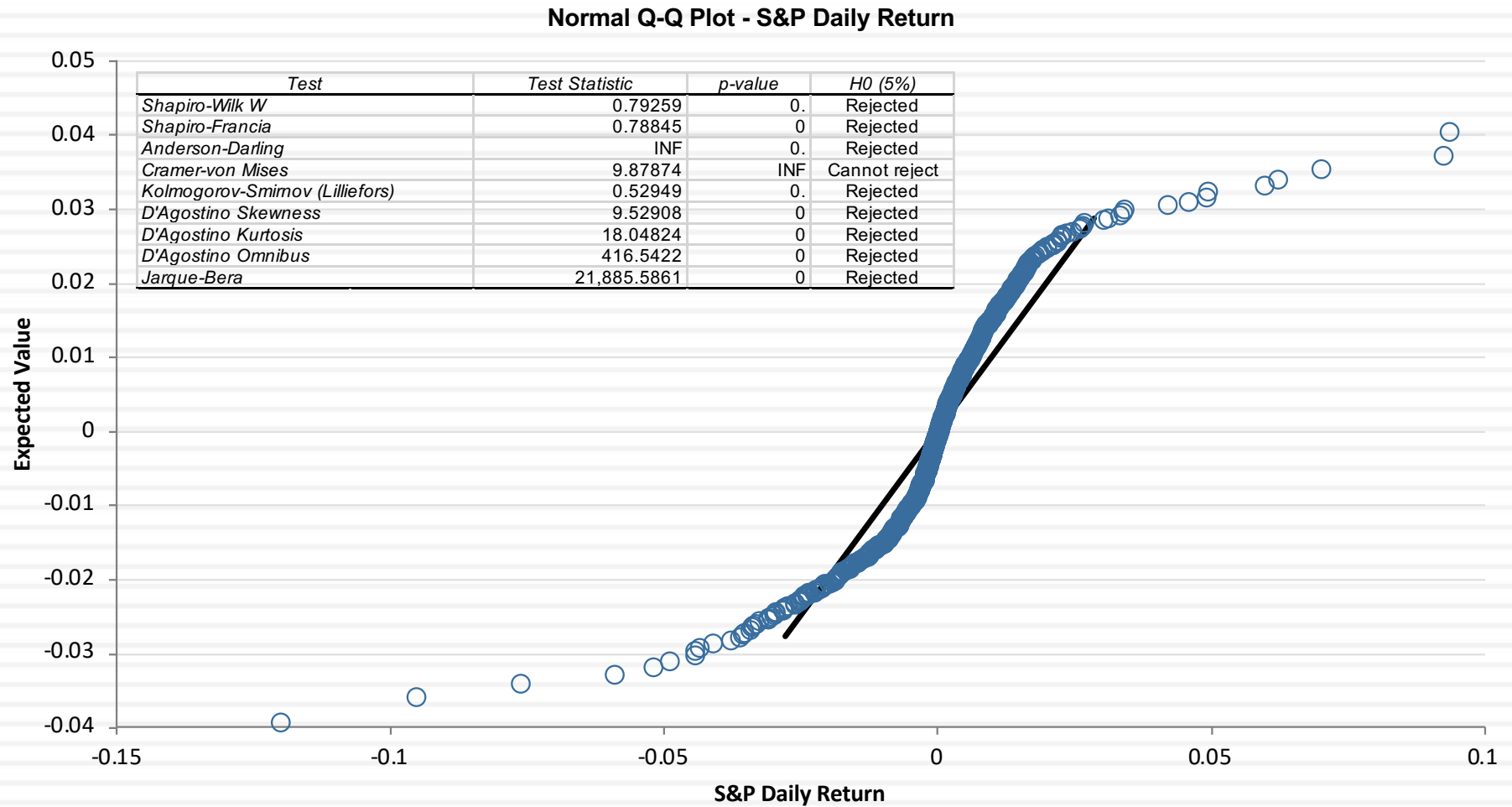
Normal Q-Q Plot - S&P 500 (includes dividends)



Returns on the S&P 500: Daily from 2016 to 2020



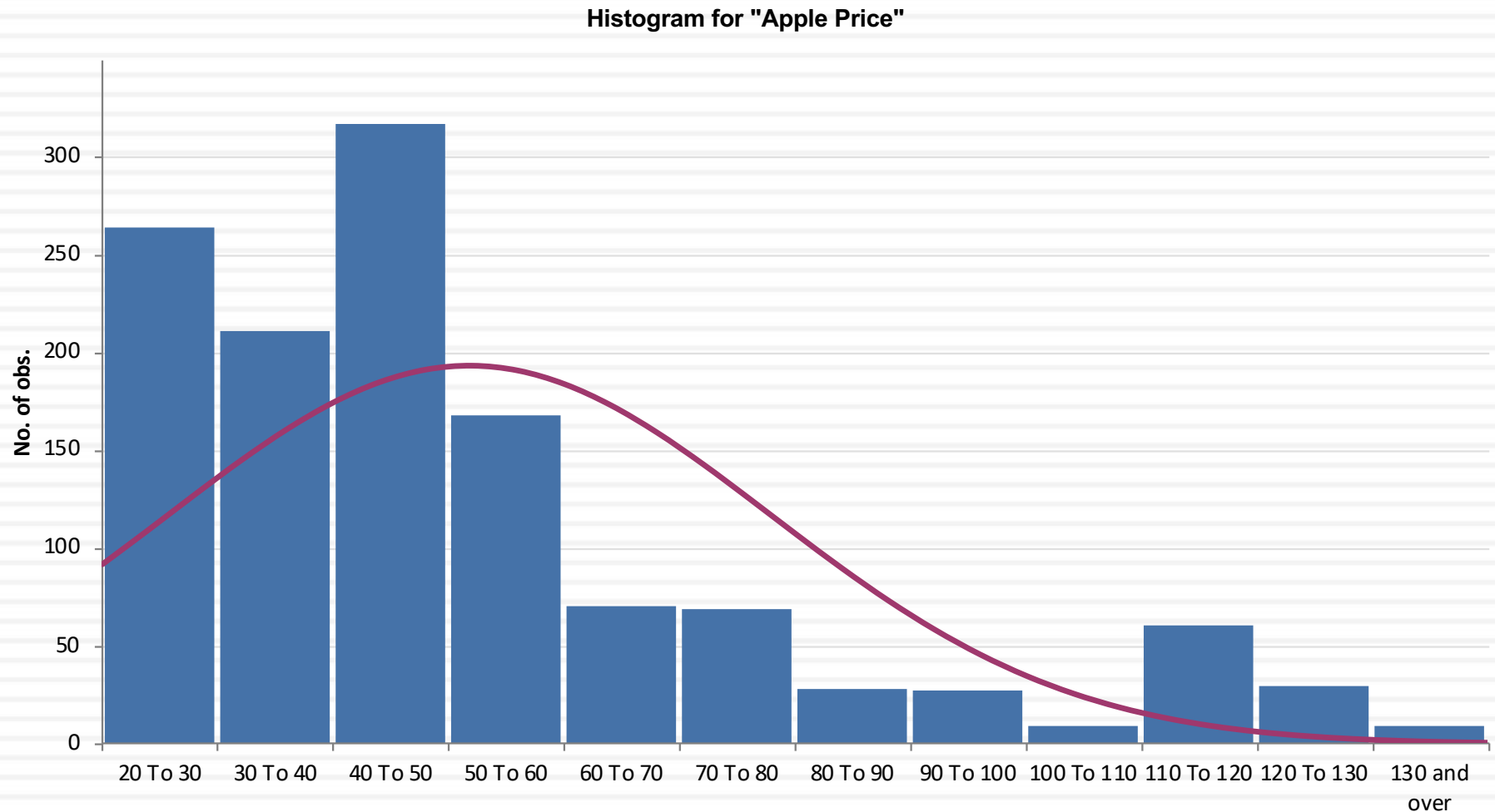
Testing for Normality



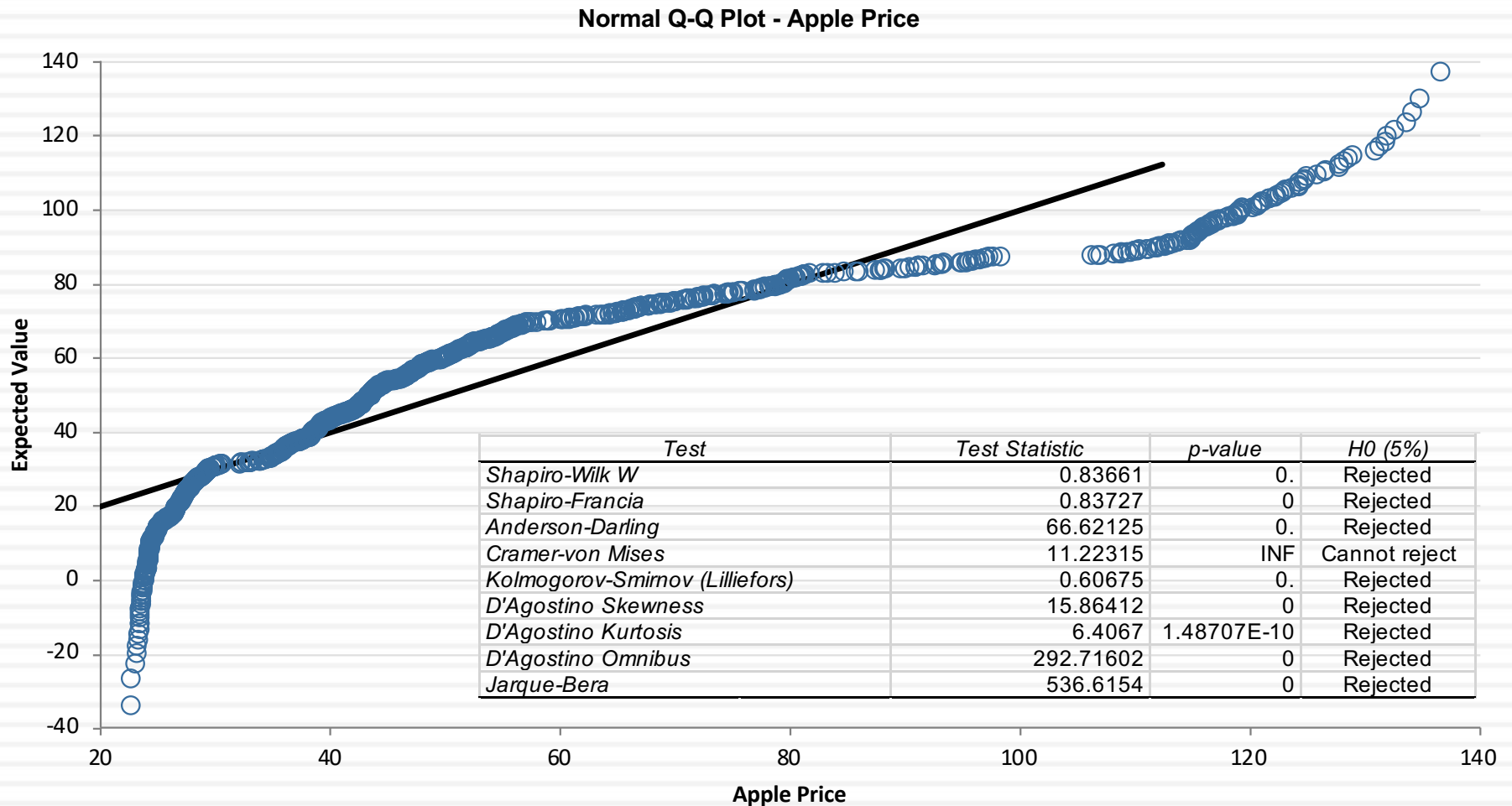
The perils of assuming normality, when the data is not...

- If returns follow a normal distribution, it is extremely unlikely that you will see returns that deviate more than two or three standard deviations from the average.
- Stock returns, especially for short term (daily, weekly) and on individual stocks (as opposed to portfolios) have fatter tails than you would expect to see in a normal distribution.
 - Simply put, you are going to see returns that can fall four, five or even six standard errors from the mean.
 - If your investment and risk management strategies are based upon normality, you will find yourself surprised (with catastrophic consequences), when this happens.

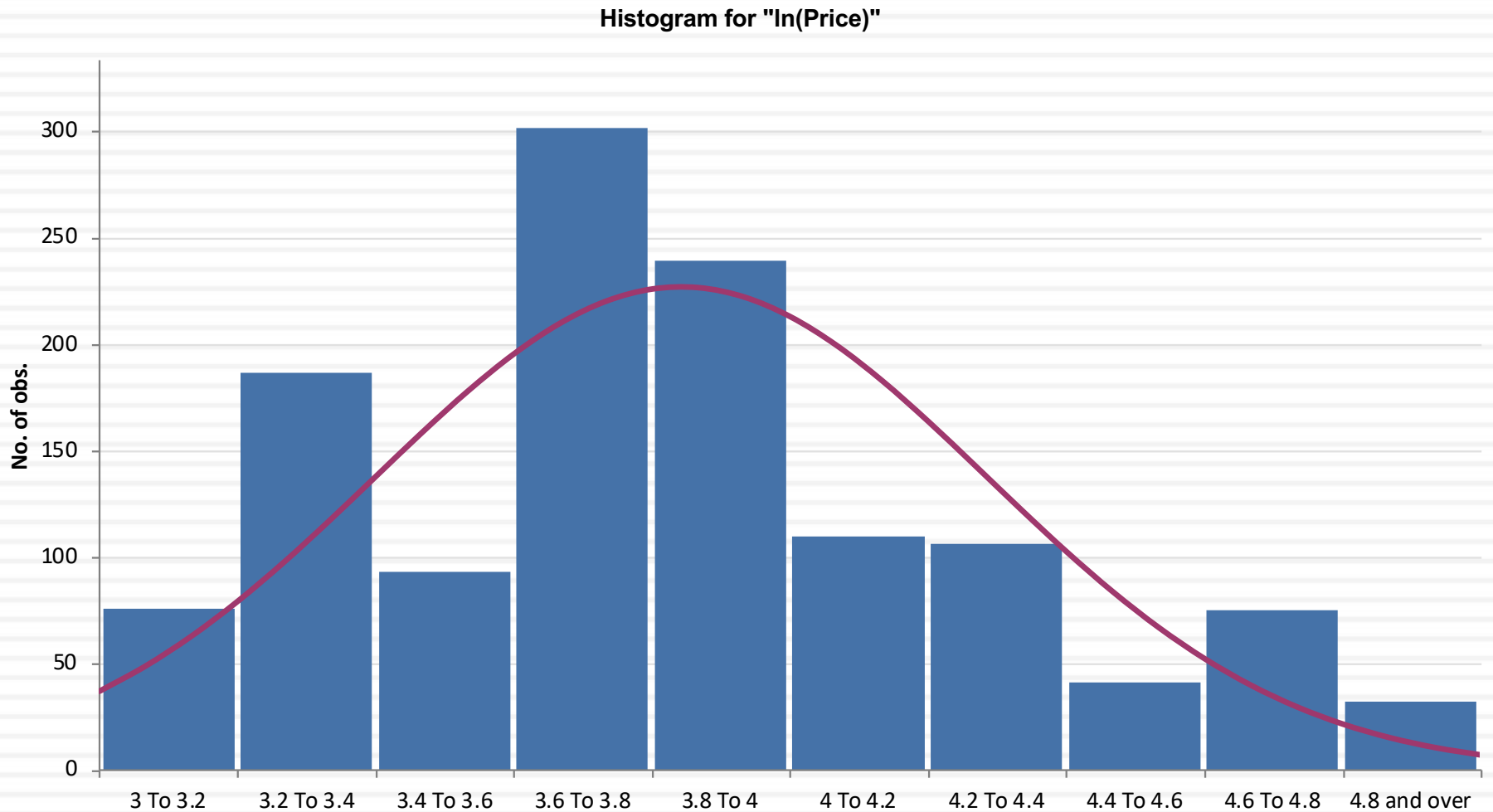
Stock prices are definitely not normally distributed... Apple Daily Stock Prices



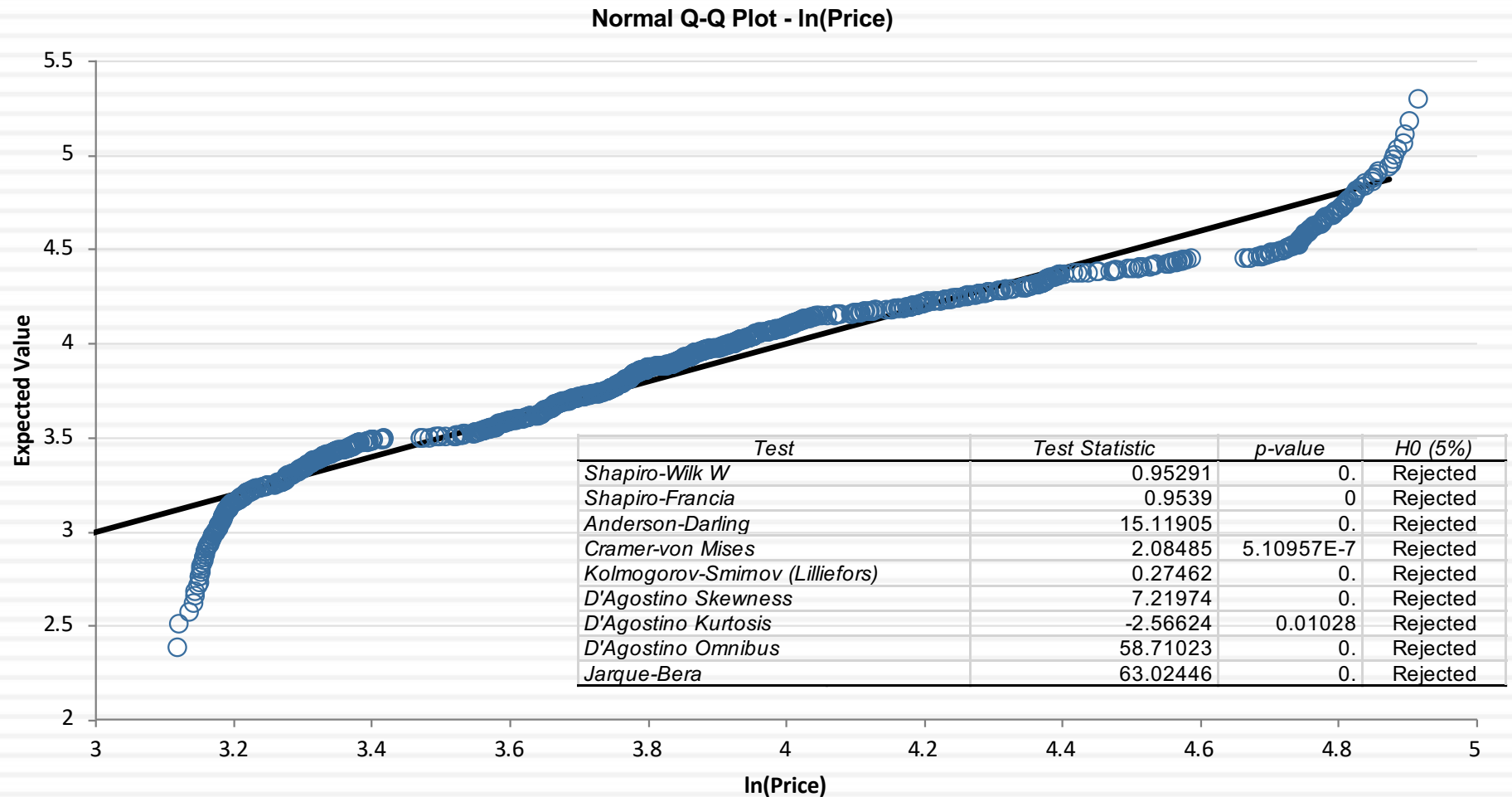
Normality Test



A Transformation: $\ln(\text{Apple})$ Daily



And normality tests...





SESSION 5: DATA RELATIONSHIPS

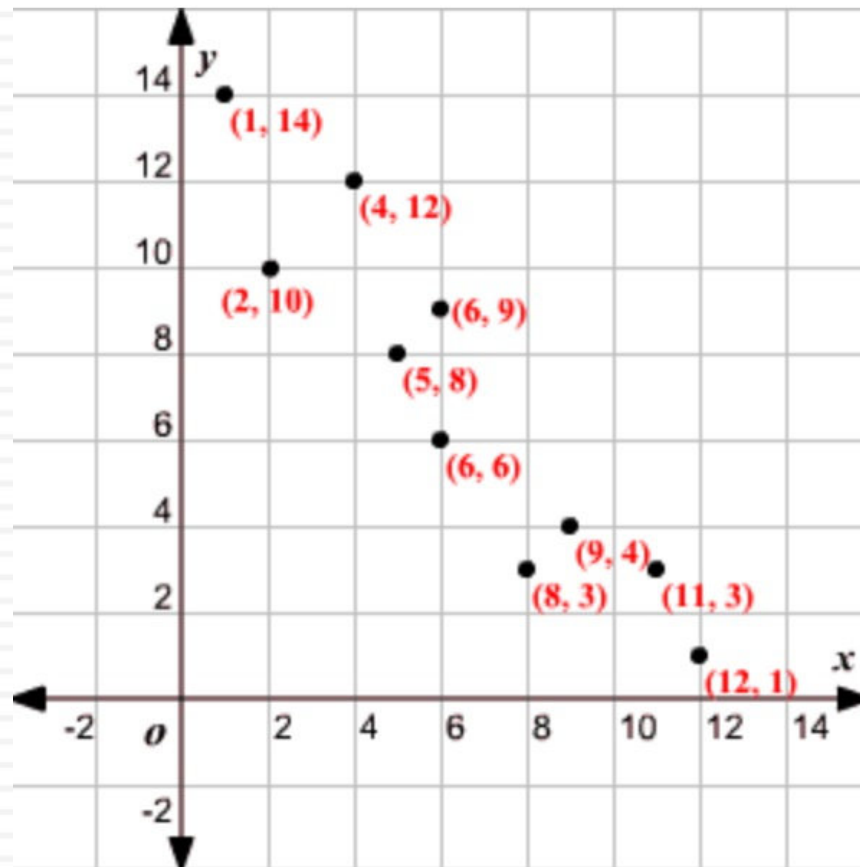
Session 5

Correlations, Covariances and Regressions

Exploring data linkages/relationships

- When you have two or more data series, you can check for linkages between the data, i.e., whether the data move together (positive co-movement), move inversely (negative co-movement) or are unrelated (no co-movement).
- If there is a linkage, you can explore further to see if
 - ▣ Time lags and leads: Changes in one data variable lead changes in the other
 - ▣ Correlation vs Causation: Changes in one data variable are "causing" changes in the other
 - ▣ Prediction: You can predict one variable, using the other variable.

A Scatter Plot



Correlation Coefficients

- The *correlation coefficient* measures how two variables move together. The most widely used one is Pearson's correlation coefficient:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

- The correlation coefficient lies *between -1 and +1*, with positive (negative) values indicating that the variables move together (inversely).
 - ▣ A correlation of zero indicates no co-movement.
 - ▣ A correlation coefficient of 1 (plus or minus) indicates perfect co-movement

And Covariances..

- The covariance, like correlation, is a measure of how two variables move together.

$$cov_{x,y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N - 1}$$

- Like the correlation, a positive covariance indicates that two variables move together, a zero covariance that there is no relationship between the two variables, and a negative covariance an indication that they move in opposite directions.
- Unlike the correlation, the covariance is not bounded between 1 and -1, and will reflect the variable values.

Parting propositions..

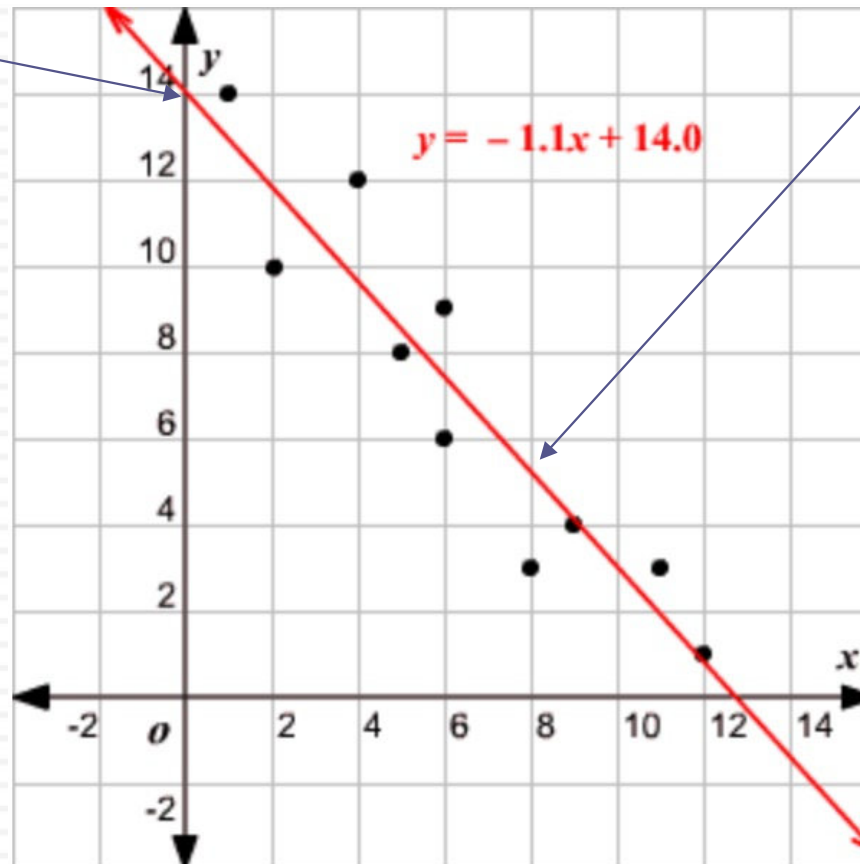
- Correlations can be spurious: If two variables are correlated, we are often tempted to create elaborate explanations for why. In many cases, that correlation can be spurious, with the two variables that are correlated both being driven by a third and often unseen variable.
- Correlation $\neq$ Causation: Correlation is a statistical measure of how two variables move together. To get from correlation to causation, you need to do more work (including a hypothesis of why, and an experiment designed to show that causation).
- Past versus Future: The data that we use to estimate correlation come from the past, and past correlation is not always a predictor of future correlation.

A Best Fit Line

The Intercept: In the scatter plot, the intercept is where the best-fit line crosses the Y axis. In simple terms, it is the value that the Y variable has when the X variable is zero.

The R squared: This measures the how well the line fits the data. If the line is a perfect fit (every point is on it), the R squared will be one (as will the correlation).

The Best Fit Line: In an ordinary-least-squares (OLS) regression, the best fit line is the one that minimizes the squared distances from the line.

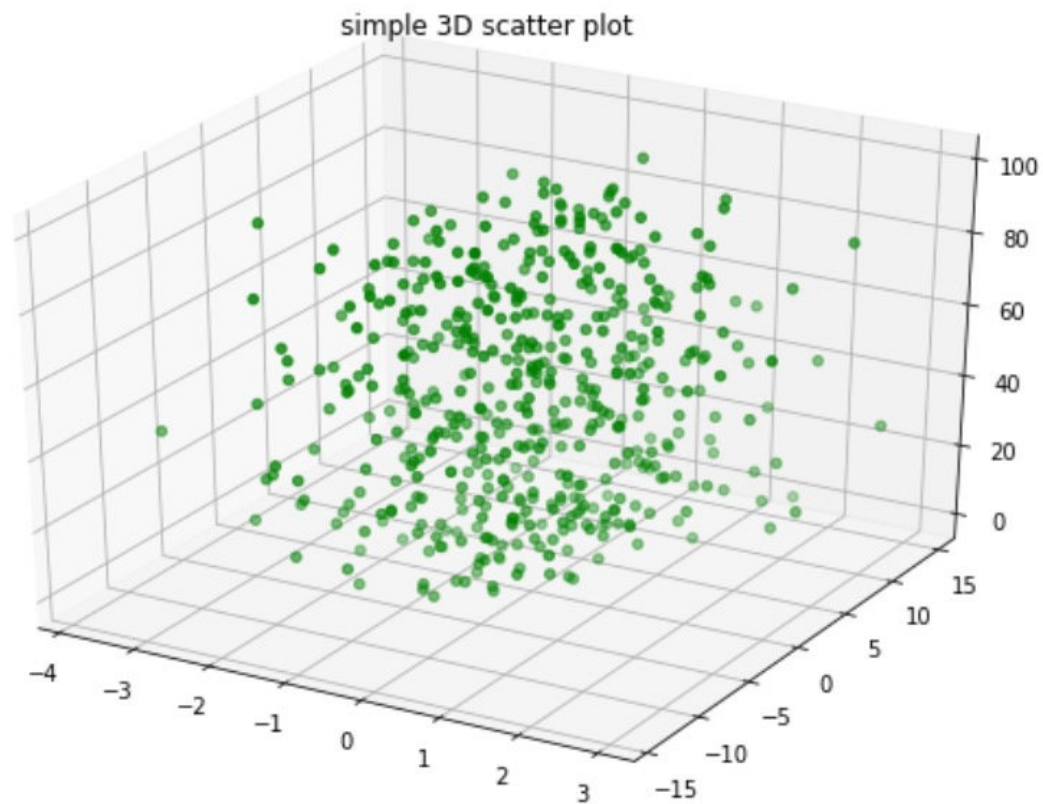


The Slope: The slope of the best fit line is also a measure of how changes in the X variable show up as changes in the Y variable. When there is no (positive, negative) relationship between the variables, the slope will be zero (positive, negative).

More than two variables?

- While a simple regression has one dependent variable and one independent variable, there are times where you may want to look at more than one independent variable in trying to explain a dependent variable.
- You can extend the tools that you use for two variables to multiple variables, by:
 - ▣ *Three dimensional plots*, if you have two independent variables, or multiple scatter plots, if you have more than two.
 - ▣ *Correlation matrices* that look at correlations between all possible pairs of variables.
 - ▣ *Multiple regressions*, where you have one dependent variable and many independent variables.

A 3-dimensional Scatter Plot...



Independent Variables Plus

- Dummy Variables: A dummy variable takes on a value of either zero or one and can be used to capture the effect of grouping in a multiple regression.
 - ▣ Discrete Variable: If the independent variable is discrete, you may have no choice but to use a dummy variable to capture its effects. *(If you have both private and publicly traded companies in a group that you are studying, and you believe that they behave differently, you could break them down on that basis).*
 - ▣ Continuous Variable: Even if a variable is continuous, there may be an advantage to converting it into a discrete dummy variable. *(Default risk is a continuous variable, but you could break companies down into investment and non-investment grade, based upon ratings.)*
- Joint Effects: If you believe that your dependent variable is affected by interactions between independent variables, you can capture the cross effects by looking at products of the independent variables. *Thus, if you are regressing PE ratios against market cap and expected growth rates, you can do the following:*

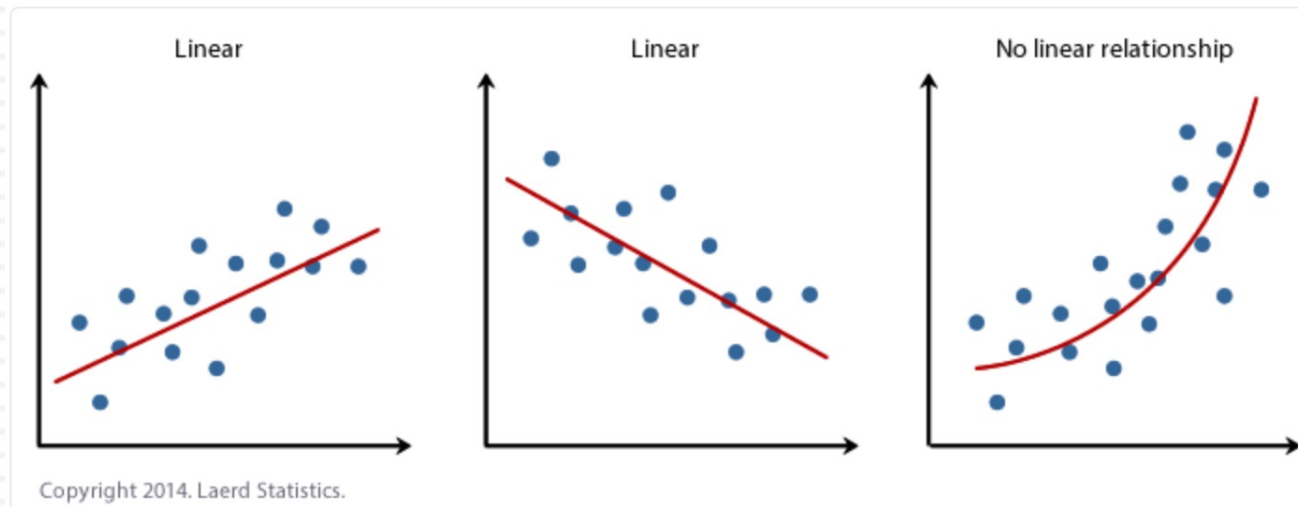
$$PE = a + b \text{ Growth Rate} + c \text{ Mkt Cap} + d (\text{Growth Rate} * \text{Mkt Cap})$$

Regression Diagnostics

- *Non-linearity*: In a regression, we are assuming that the relationship between the dependent and independent variables is linear. If it is not:
 - ▣ You can run a non-linear regression
 - ▣ or use mathematical transformation (square, natural log etc.) on either the dependent or independent variables to create a more linear relationship.
- *Multicollinearity*: In a multiple regression, the independent variables should, in a perfect world, be uncorrelated with each other. But we don't live in a perfect world...
- *Homoskedasticity*: The residuals (or prediction errors) should not reveal any patterns (get larger or smaller), as the independent variable(s) increase and decrease.
- *Normality*: The residuals should be normally distributed.

Linear versus Non-linear Regressions

- The easiest way to detect multicollinearity is to run scatter plots of the dependent variable against each independent variable.

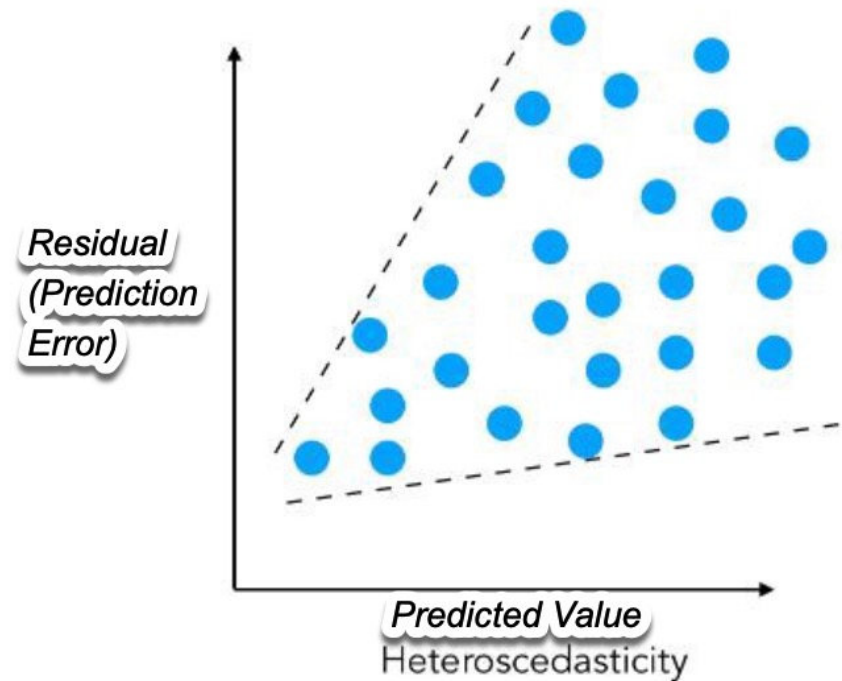
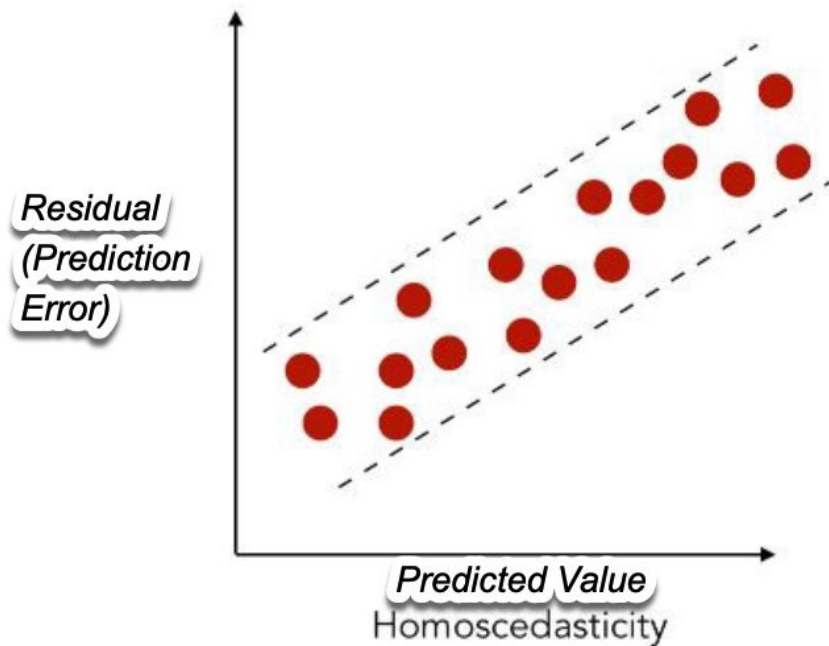


- A non-linear regression tries to find a best fit line through the non-linear plot. The process of fitting a line, though, is more complicated, and often involves trial and error (in the background, if you are using a statistics package).
- If you choose to transform a variable (dependent or independent), you may have to use trial and error (using the scatter plot) to find your best transformation.

Collinearity Diagnostics

- Multicollinearity can cause coefficient estimates on the independent variables to become unreliable and volatile.
 - ▣ It reduces the precision of these coefficients and the statistical power of the regression.
 - ▣ It generally does not influence the predictions or precision of the predictions.
- The multicollinearity in a multiple regression can be measured with a *variance inflation factor (VIF)*.
 - ▣ A VIF greater than 5 indicates severe multicollinearity
 - ▣ A VIF between 1 and 5 indicates moderate multicollinearity
- If there is multicollinearity, you can
 - ▣ Try *replacing or removing one of a pair of independent variables* that are correlated.
 - ▣ Run *individual simple regressions* against the independent variables
 - ▣ Do nothing, *use the multiple regression for predictions*, and don't over read the coefficients on variables.

Prediction Errors from Regressions: Homoskedasticity and Heteroskedasticity...



Regression Variants

- Generalized Least Squares (GLS): When the residuals from your OLS regression are heteroskedastic or autocorrelated, one alternative that will yield more unbiased estimates is a *Generalized Least Squares* (GLS) regression.
 - ▣ To run a GLS regression, you usually begin with an OLS regression and use the information in the prediction error terms to build a GLS regression.
 - ▣ The output from a GLS regression resembles that from an OLS regression and can be used similarly to make predictions and test hypotheses.
- Weighted Least Squares: In a weighted least squares regression, not all observations in a sample are weighted equally. Instead, a weighting mechanism is created that weights some observations more than others.
 - ▣ In some cases, the weighting may reflect your belief that some of your observations convey more information about the population than others.
 - ▣ In others, it can be to counter the problem of standard errors of predictions varying across different values for the explanatory variable.

Regression Predictions

- Precision: In almost every regression, your predictions from a regression will be imprecise, with a standard error and a range around the prediction. If the R-squared is low, and the range is large, a regression is less actionable.
- Stationarity: Regressions are run with observed data from the past. To the extent that the process that you are observing is non-stationary, with changes in population and structure, the predictions will not hold up in future periods.
- Observability: In a regression, you need to be able to observe the dependent variable (or variables) to be able to generate predictions of the independent variables. Regressions that use independent variables that don't lend themselves to timely observation cannot be acted upon.
- Actionability: If the predicted value for Y from a regression deviates from the actual value, being able to act on the divergence makes the reliable more useful (in many cases).



SESSION 5A: DATA RELATIONSHIPS IN FINANCE

Session 5

Correlations, Covariances and Regressions

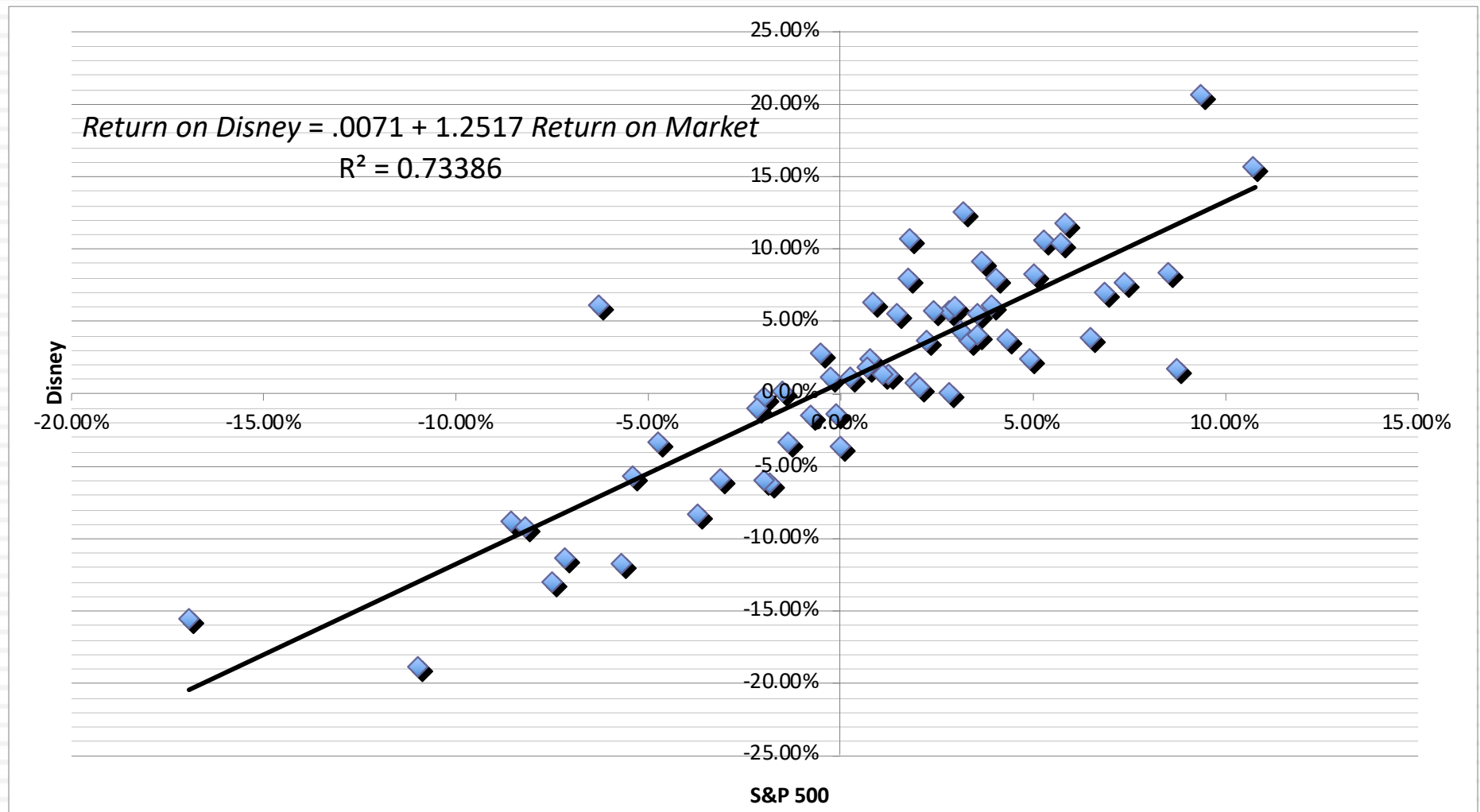
Macro and Micro Questions

- Micro versus Macro: In finance, we look for relationships between both macro variables (interest rates and inflation, for exam), as well as micro variables (a company's profit margins and its revenue growth).
- Motives: You can look at the relationship between variables both for understanding and for forecasting.
 - ▣ For analysis: We look at the relationship between two or more variables to get an understanding of why they move (or don't move) together, with the hope that this understanding will lead to better policy:
 - ▣ For prediction: We also look at relationships between two or more variables to make better forecasts of one of the variables for the future, in the hope of acting on those forecasts.

How risky is a stock?

- The Capital Asset Pricing Model (CAPM) is one of finance's most common used and misused models to estimate expected return on risky investments. In the CAPM,
 - ▣ The risk of an investment is the risk added to the market portfolio, which is a diversified portfolio of all risky, traded assets.
 - ▣ To measure that risk, you have to look at how the investment moves with the market portfolio, with investments that move more being more risky.
- The tool that is used to measure this co-movement is a regression, with past returns on the investment regressed against past returns on a market index.

1. A Scatter Plot & Best Fit Line: Disney versus S&P 500



Correlation and Covariance

- The correlation coefficient between Disney and S&P 500 returns is captured (in a correlation matrix):

R	Disney	S&P 500
Disney	1	
R Std Err		
t		
S&P 500	0.85666	1
R Std Err	0.00459	
t	12.64646	

- The covariance is the non-standardized measure of this co-movement, and it is captured in the table below:

	Disney	S&P 500
Disney	0.00574	
S&P 500	0.00337	0.00269

Reading a Regression

Linear Regression							
Dependent variable		Return(Disney)					
Independent variables		Return(S&P 500)					
Return(Disney) = 0.00712 + 1.25173 * Return(S&P 500)							
Regression Statistics							
R		0.85666	R-Squared		0.73386		
			Adjusted R-Squared		0.72927		
Durbin-Watson (DW)		2.20472	Log likelihood		109.38276		
ANOVA							
	d.f.	SS	MS	F	p-value		
Regression	1	0.25276	0.25276	159.93306	0.		
Residual	58	0.09166	0.00158				
Total	59	0.34442					
			95% Confidence interval				
	Coefficients	Std Err	LCL	UCL	t Stat	p-value	HO (5%)
Intercept	0.00712	0.00519	-0.00326	0.0175	1.37264	0.17515	Accepted
Return(S&P 500)	1.25173	0.09898	1.0536	1.44986	12.64646	0.00000	Rejected

A test of autocorrelation in residuals
<2 Positive
2-4 Negative

A function of sample size & number of independent variables.

SS is the squared error explained (regression) and unexplained (residual). MS is SS dividend by the degrees of freedom.

Measures whether independent variables are adding predictive value

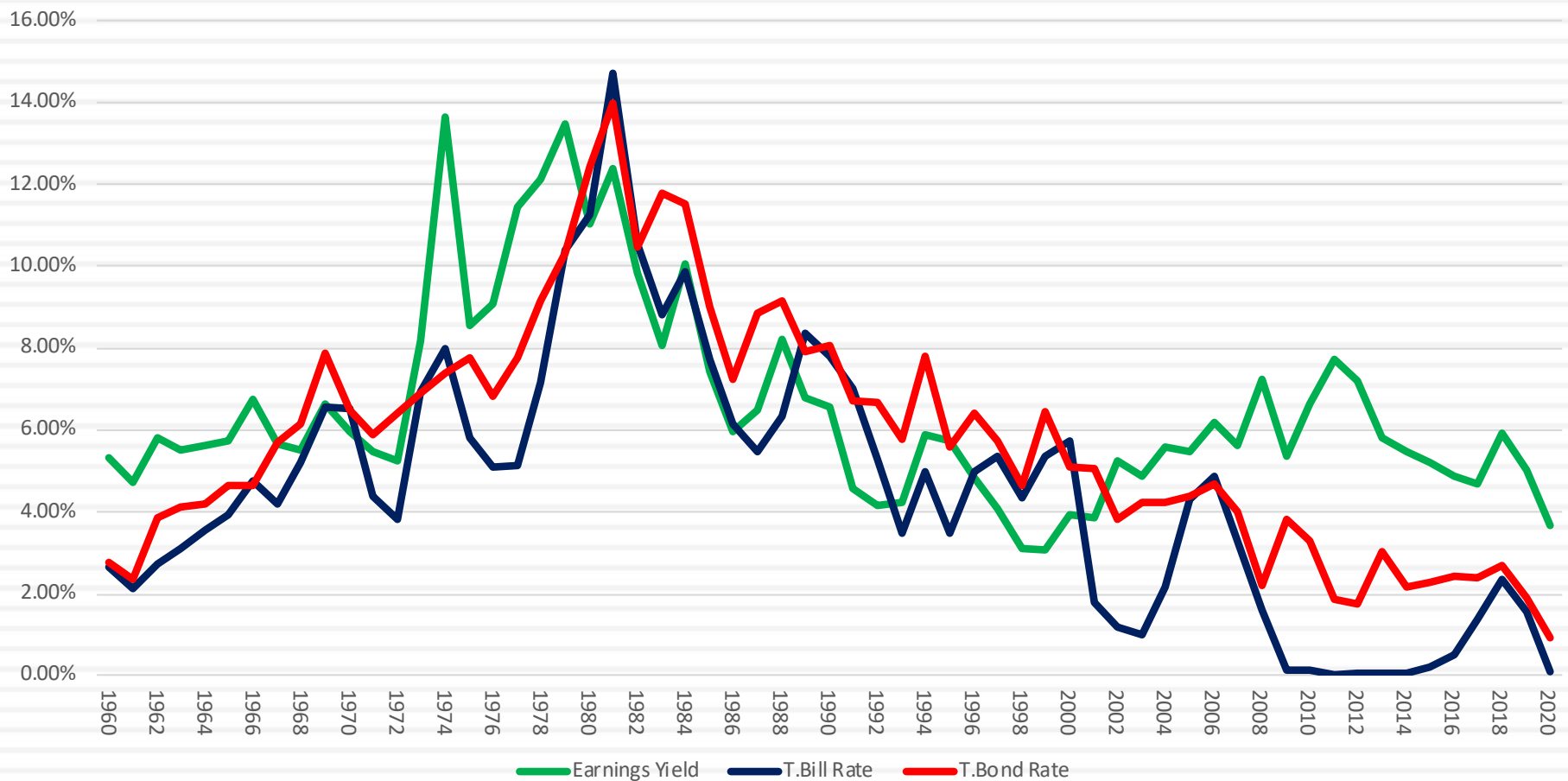
The lower (LCL) and upper (UCL) ends of the range are computed by subtracting or adding two standard errors to coefficient.

The t statistic and p value measure whether the specific independent variable adds predictive value in regression.

2. A Time Series Example: Earnings to Price ratios and Interest Rates

7

Earnings to Price Ratio and Interest Rates



Covariances and Correlations

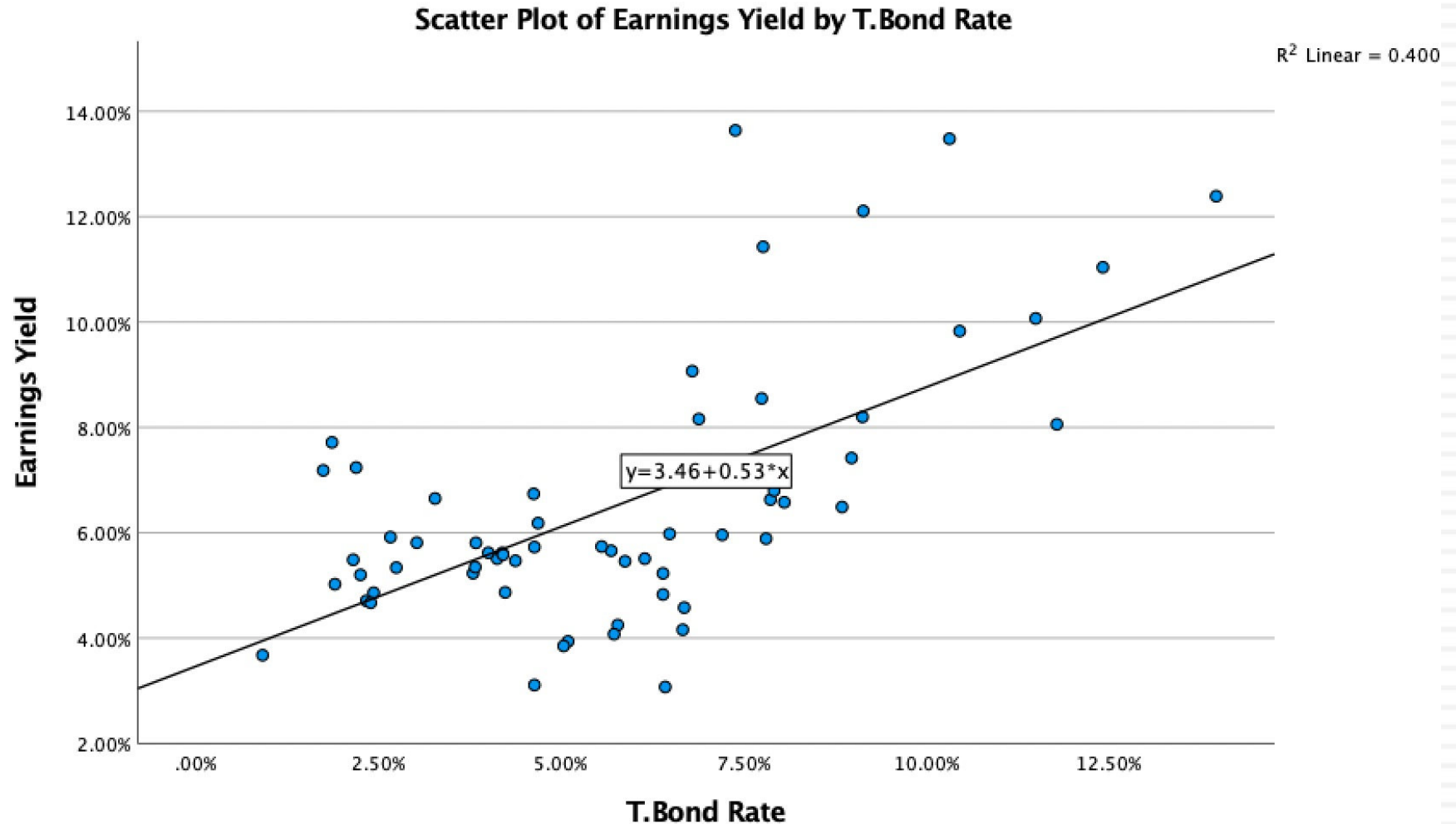
Covariances

	<i>Earnings Yield</i>	<i>T.Bill Rate</i>	<i>T.Bond Rate</i>
<i>Earnings Yield</i>	0.00059		
<i>T.Bill Rate</i>	0.00048	0.00101	
<i>T.Bond Rate</i>	0.00044	0.00086	0.00084

Correlations (Pearson)

R	Earnings Yield	T.Bill Rate	T.Bond Rate
Earnings Yield	1		
<i>R Std Err</i>			
<i>t</i>			
<i>p-value (2-tailed)</i>			
T.Bill Rate	0.61862	1	
<i>R Std Err</i>	0.01046		
<i>t</i>	6.04777		
<i>p-value (2-tailed)</i>	1.07728E-7		
T.Bond Rate	0.63253	0.93175	1
<i>R Std Err</i>	0.01017	0.00223	
<i>t</i>	6.27285	19.71042	
<i>p-value (2-tailed)</i>	4.53895E-8	0	
<i>Correlations in bold are significant at the 5% level (2-tailed).</i>			
<i>N of valid cases = 61.</i>			
R			
VAR vs. VAR	R	N	p-value
T.Bond Rate vs. T.Bill Rate	0.93175	61	0
T.Bond Rate vs. Earnings Yield	0.63253	61	4.53895E-8
T.Bill Rate vs. Earnings Yield	0.61862	61	1.07728E-7

A Scatter Plot of earnings yield on T.Bond rate, with a best-fit line..



Regression and Prediction

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.633 ^a	.400	.390	1.90780%

a. Predictors: (Constant), T.Bond Rate

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	143.218	1	143.218	39.349	<.001 ^b
	Residual	214.743	59	3.640		
	Total	357.961	60			

a. Dependent Variable: Earnings Yield

b. Predictors: (Constant), T.Bond Rate

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	3.457	.548		6.310	<.001
	T.Bond Rate	.530	.085	.633	6.273	<.001

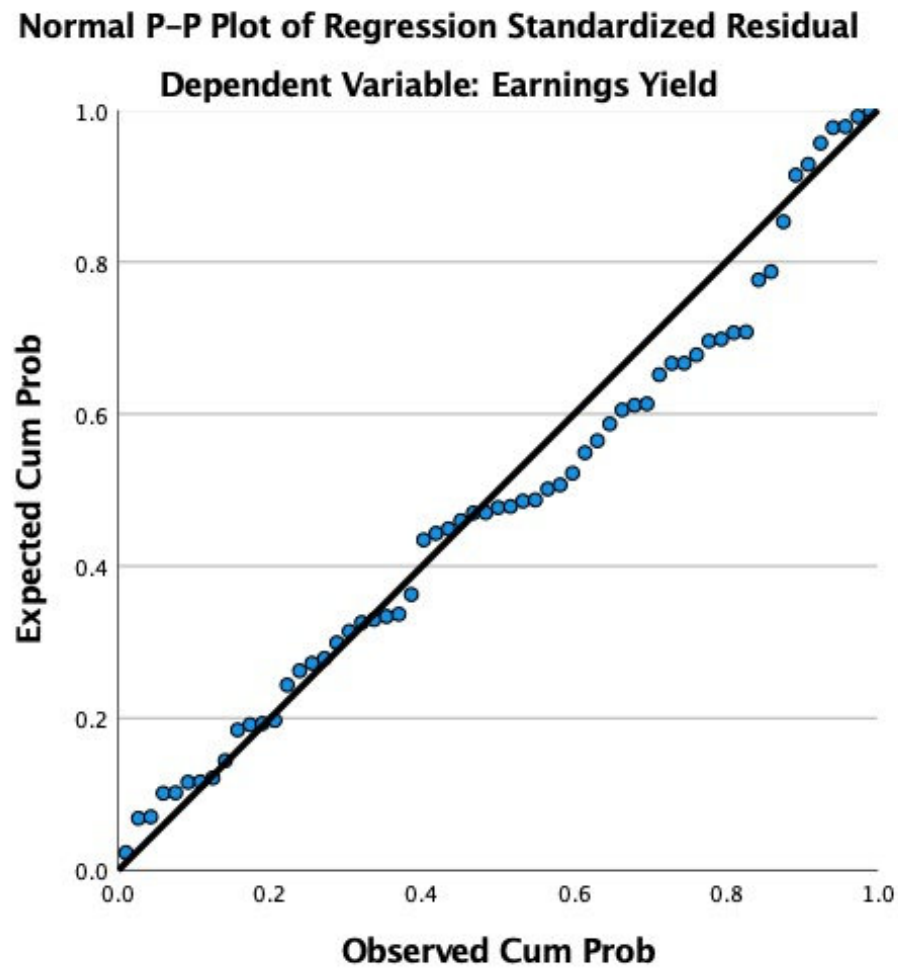
a. Dependent Variable: Earnings Yield

If the T.Bond rate right now is 2%, the predicted Earnings yield is:

Predicted E/P = 3.457% + .530 (2%) = 4.517%

The actual E/P ratio on January 1, 2021 was 3.68%, making stocks over valued.

Regression Diagnostics



3. A Cross-sectional Example: European Banks in 2013

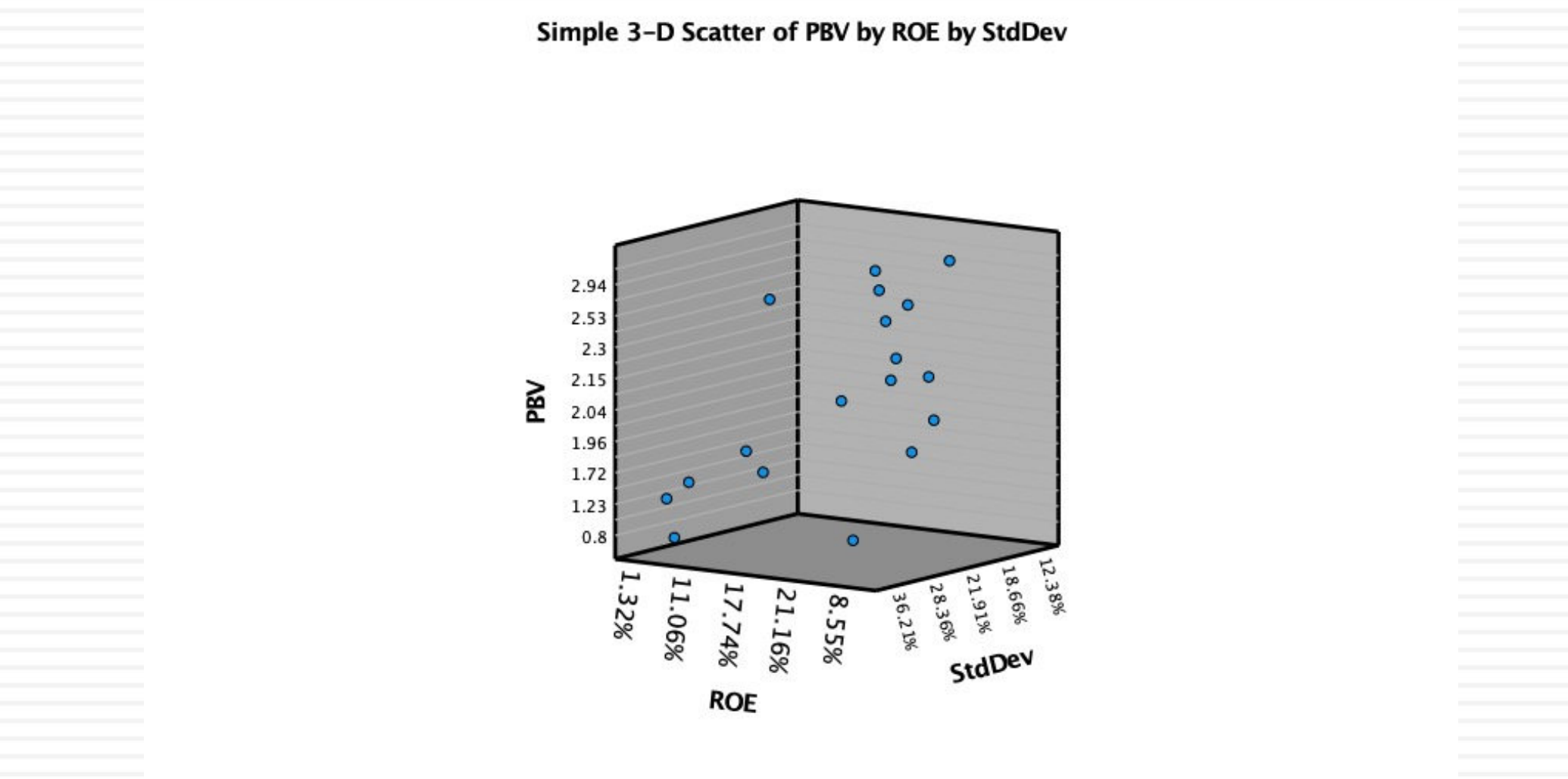
Name	Bloomberg symbol	PBV Ratio	Return on Equity	Standard Deviation
BAYERISCHE HYPO-UND VEREINSB	HVM GR Equity	0.80	-1.66%	49.06%
COMMERZBANK AG	CBK GR Equity	1.09	-6.72%	36.21%
DEUTSCHE BANK AG -REG	DBK GR Equity	1.23	1.32%	35.79%
BANCA INTESA SPA	BIN IM Equity	1.66	1.56%	34.14%
BNP PARIBAS	BNP FP Equity	1.72	12.46%	31.03%
BANCO SANTANDER CENTRAL HISP	SAN SM Equity	1.86	11.06%	28.36%
SANPAOLO IMI SPA	SPI IM Equity	1.96	8.55%	26.64%
BANCO BILBAO VIZCAYA ARGENTA	BBVA SM Equity	1.98	11.17%	18.62%
SOCIETE GENERALE	GLE FP Equity	2.04	9.71%	22.55%
ROYAL BANK OF SCOTLAND GROUP	RBS LN Equity	2.09	20.22%	18.35%
HBOS PLC	HBOS LN Equity	2.15	22.45%	21.95%
BARCLAYS PLC	BARC LN Equity	2.23	21.16%	20.73%
UNICREDITO ITALIANO SPA	UC IM Equity	2.30	14.86%	13.79%
KREDIETBANK SA LUXEMBOURGEOI	KBL LX Equity	2.46	17.74%	12.38%
ERSTE BANK DER OESTER SPARK	EBS AV Equity	2.53	10.28%	21.91%
STANDARD CHARTERED PLC	STAN LN Equity	2.59	20.18%	19.93%
HSBC HOLDINGS PLC	HSBA LN Equity	2.94	18.50%	19.66%
LLOYDS TSB GROUP PLC	LLOY LN Equity	3.33	32.84%	18.66%

Government	Percentage
Current government	85%
Previous government	15%

Simple 3-D Scatter of PBV by ROE by StdDev

The 3D scatter plot displays the relationship between PBV (vertical axis), ROE (horizontal axis), and StdDev (depth axis). The vertical axis (PBV) ranges from 0.8 to 2.94. The horizontal axis (ROE) ranges from 1.32% to 21.16%. The depth axis (StdDev) ranges from 8.55% to 36.21%. The data points are clustered in two main groups: one group at higher PBV and ROE values (around 2.53 to 2.94 PBV and 17.74% to 21.16% ROE), and another group at lower PBV and ROE values (around 1.23 to 1.96 PBV and 1.32% to 8.55% ROE).

ROE (%)	StdDev (%)	PBV
1.32	8.55	0.8
1.32	12.38	1.23
1.32	18.66	1.72
1.32	21.91	1.96
1.32	28.36	2.04
1.32	36.21	2.15
1.32	12.38	2.3
1.32	18.66	2.53
1.32	21.91	2.94
1.32	28.36	2.94
1.32	36.21	2.94
1.32	12.38	2.94
1.32	18.66	2.94
1.32	21.91	2.94
1.32	28.36	2.94
1.32	36.21	2.94
1.32	12.38	2.94
1.32	18.66	2.94
1.32	21.91	2.94
1.32	28.36	2.94
1.32	36.21	2.94
1.32	12.38	2.94
1.32	18.66	2.94
1.32	21.91	2.94
1.32	28.36	2.94
1.32	36.21	2.94
1.32	12.38	2.94
1.32	18.66	2.94
1.32	21.91	2.94
1.32	28.36	2.94
1.32	36.21	2.94
1.32	12.38	2.94
1.32	18.66	2.94
1.32	21.91	2.94
1.32	28.36	2.94
1.32	36.21	2.94
1.32	12.38	2.94
1.32	18.66	2.94
1.32	21.91	2.94
1.32	28.36	2.94
1.32	36.21	2.94
1.32	12.38	2.94
1.32	18.66	2.94
1.32	21.91	2.94
1.32	28.36	2.94
1.32	36.21	2.94
1.32	12.38	2.94
1.32	18.66	2.94
1.32	21.91	2.94
1.32	28.36	2.94
1.32	36.21	2.94
1.32	12.38	2.94
1.32	18.66	2.94
1.32	21.91	2.94
1.32	28.36	2.94
1.32	36.21	2.94
1.32	12.38	2.94
1.32	18.66	2.94
1.32	21.91	2.94
1.32	28.36	2.94
1.32	36.21	2.94
1.32	12.38	2.94
1.32	18.66	2.94
1.32	21.91	2.94
1.32	28.36	2.94
1.32	36.21	2.94
1.32	12.38	2.94
1.32	18.66	2.94
1.32	21.91	2.94
1.32	28.36	2.94
1.32	36.21	2.94
1.32	12.38	2.94
1.32	18.66	2.94
1.32	21.91	2.94
1.32	28.36	2.94
1.32	36.21	2.94
1.32	12.38	2.94
1.32	18.66	2.94
1.32	21.91	2.94
1.32	28.36	2.94
1.32	36.21	2.94
1.32	12.38	2.94
1.32	18.66	2.94
1.32	21.91	2.94
1.32	28.36	2.94
1.32	36.21	2.94
1.32	12.38	2.94
1.32	18.66	2.94
1.32	21.91	2.94
1.32	28.36	2.94
1.32	36.21	2.94
1.32	12.38	2.94
1.32	18.66	2.94
1.32	21.91	2.94
1.32	28.36	2.94
1.32	36.21	2.94
1.32	12.38	2.94
1.32	18.66	2.94
1.32	21.91	2.94
1.32		



Correlation Matrix

- You can compute correlations across pairs of variables for as many variables as you want, with the caveat that the number of correlations you have to compute will increase at a much faster rate than the number of independent variables.
- The pair-wise correlations computed can be read the same way as simple correlations, ranging between + 1 and -1.

R	PBV Ratio	Return on Equity	Standard Deviation
PBV Ratio	1.0000		
<i>R Std Err</i>			
<i>t</i>			
Return on Equity	0.8682	1.0000	
<i>R Std Err</i>	0.0154		
<i>t</i>	6.9977		
Standard Deviation	-0.8283	-0.7644	1.0000
<i>R Std Err</i>	0.0196	0.0260	
<i>t</i>	-5.9126	-4.7424	

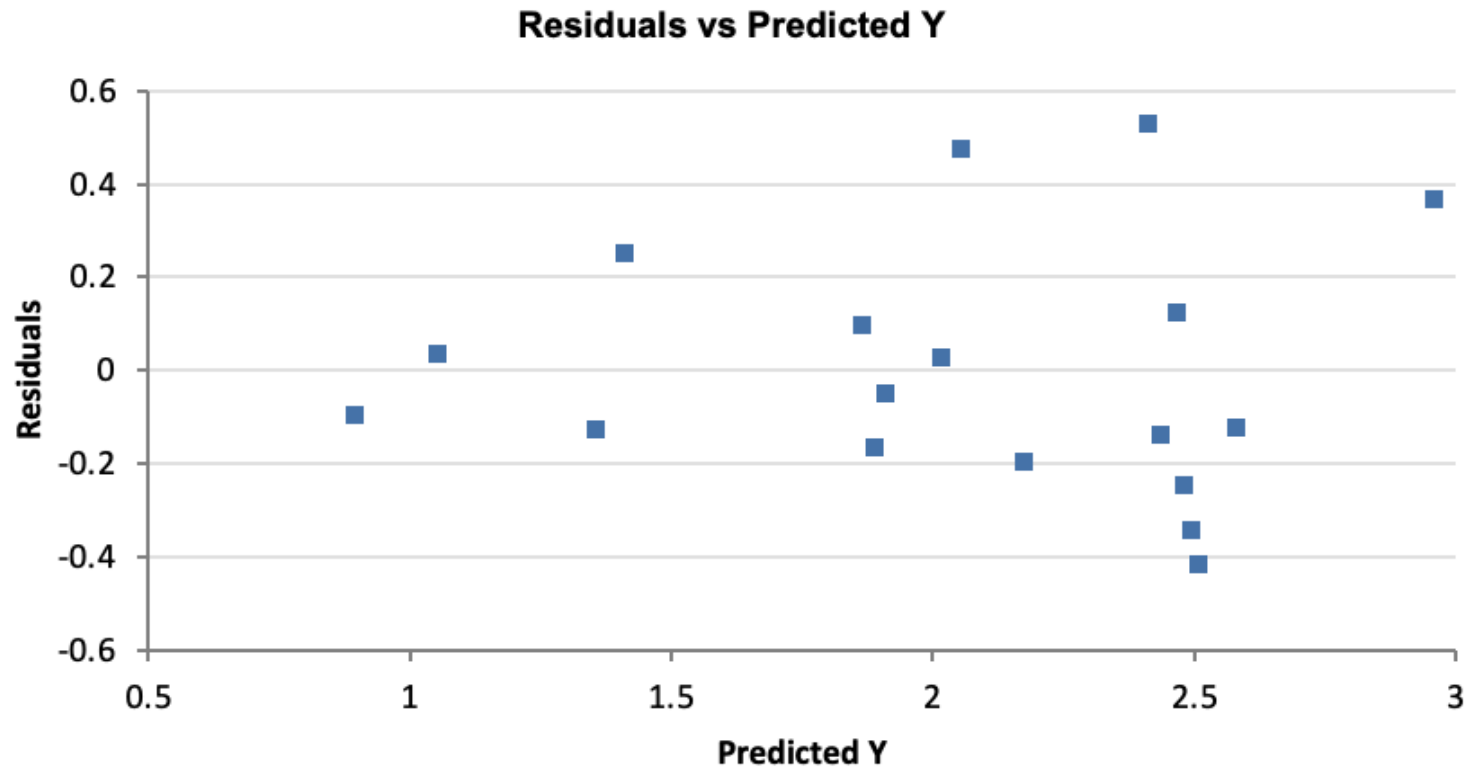
Multiple Regressions

Linear Regression							
Dependent variable		PBV Ratio					
Independent variables		Return on Equity, Standard Deviation					
Regression Statistics							
R		0.90494	R-Squared		0.81892		
MSE		0.08066	Adjusted R-Squared		0.79478		
PBV Ratio = 2.26677 + 3.62891 * Return on Equity - 2.67729 * Standard Deviation							
ANOVA							
	d.f.	SS	MS	F	p-value		
Regression	2	5.47185	2.73592	33.91886	2.71627E-6		
Residual	15	1.20991	0.08066				
Total	17	6.68176					
			95% Confidence Interval				
	Coefficients	Std Err	LCL	UCL	t Stat	p-value	HO (5%)
Intercept	2.26677	0.40802	1.39708	3.13645	5.55546	0.00006	Rejected
Return on Equity	3.62891	1.09369	1.29777	5.96005	3.31805	0.00468	Rejected
Standard Deviation	-2.67729	1.15203	-5.13278	-0.2218	-2.32398	0.03458	Rejected

Predictions and Residuals

	PBV Ratio	Predicted PBV	Residual
HVM GR Equity	0.79859	0.89324	-0.09465
CBK GR Equity	1.08757	1.05332	0.03425
DBK GR Equity	1.231	1.3566	-0.1256
BIN IM Equity	1.6605	1.40925	0.25125
BNP FP Equity	1.72301	1.88816	-0.16515
SAN SM Equity	1.85809	1.9088	-0.05071
SPI IM Equity	1.9612	1.86377	0.09743
BBVA SM Equity	1.97547	2.17367	-0.1982
GLE FP Equity	2.04314	2.01556	0.02758
RBS LN Equity	2.09153	2.50928	-0.41775
HBOS LN Equity	2.15222	2.4941	-0.34188
BARC LN Equity	2.23284	2.47984	-0.247
UC IM Equity	2.29741	2.43677	-0.13936
KBL LX Equity	2.45553	2.57913	-0.12361
EBS AV Equity	2.52774	2.05321	0.47453
STAN LN Equity	2.5889	2.46551	0.12339
HSBA LN Equity	2.94003	2.41157	0.52846
LLOY LN Equity	3.32586	2.95884	0.36703

Residual Chart for European Banks



Breusch-Pagan-Godfrey (BPG) test				
Test Statistic	3.34529	p-value	0.18775	H0 (5%)
F	1.71206	p-value	0.21394	Cannot reject
White test (All cross-terms)				
Test Statistic	4.45339	p-value	0.48614	H0 (5%)
F	0.78899	p-value	0.57729	Cannot reject



SESSION 5B: MORE ON REGRESSION APPLICATIONS

Session 5
Looking for links!

1. Building a Regression

- While the mechanics of a simple or multiple regression always involve finding independent variables to explain/predict a dependent variable, the process by which you find those variables is important.
- In general, you can find the independent variables statistically by scanning the data and finding variables purely based upon correlations with the dependent variable.
 - While this approach works statistically, it has key problems. If you have enough data and variables to work with, data mining can lead you to variables that provide high R-squared and statistical significance, but there is little you are learning about the past or gaining in terms of predictive value.
- A better approach is to start with intuition and/or a model that links the independent variables to the dependent variable, and then running regressions to see if the model holds. The plus is that if you find statistical significance, you have a much better basis for analysis and prediction. The minus is that the data might not back up your model, either because the model does not hold or because the independent variables that you picked at not good proxies.

An Example: PE Ratio Determinants

3

- To understand the fundamentals, start with a basic equity discounted cash flow model.

- ▣ With the dividend discount model,

$$P_0 = \frac{DPS_1}{r - g_n}$$

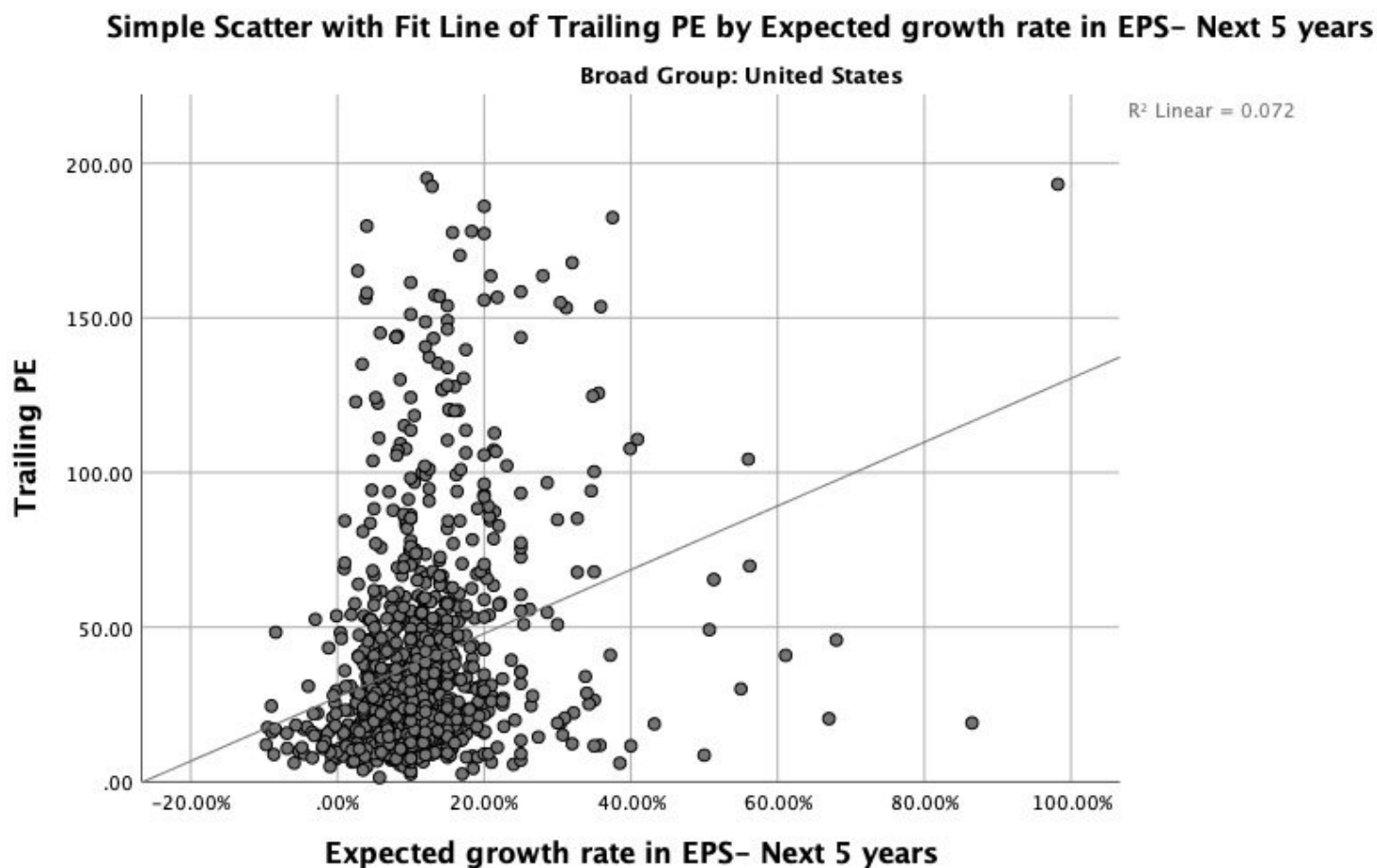
- ▣ Dividing both sides by the current earnings per share,

$$\frac{P_0}{EPS_0} = PE = \frac{\text{Payout Ratio} \cdot (1 + g_n)}{r - g_n}$$

- If this model is right, the PE ratio for a company should be determined by three variables:
 - ▣ Payout ratio, with higher payout ratios leading to higher PE
 - ▣ Growth rate, with higher growth rates leading to higher PE
 - ▣ Risk, with higher discount rates (r) leading to lower PE

Checking Linearity: PE versus Expected EPS Growth in January 2021

4



PE Ratio: Standard Regression for US stocks - January 2021

5

Model Summary^a

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.629 ^b	.396	.394	4035.87822

a. Broad Group = United States

b. Predictors: (Constant), Expected growth rate in EPS– Next 5 years, Beta, Payout ratio

The regression is run with growth and payout entered as absolute, i.e., 25% is entered as 25)

Coefficients^{a,b,c}

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	4.104	2.828		1.451	.147
	Payout ratio	.174	.017	.259	10.087	.000
	Beta	1.714	2.709	.015	.633	.527
	Expected growth rate in EPS– Next 5 years	2.304	.087	.681	26.512	.000

a. Broad Group = United States

b. Dependent Variable: Trailing PE

c. Weighted Least Squares Regression – Weighted by Market Cap (in US \$)

Don't fight the data: If a coefficient is not significant, take it out...

6

Model Summary^a

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.623 ^b	.389	.388	4049.88731

a. Broad Group = United States

b. Predictors: (Constant), Expected growth rate in EPS- Next 5 years, Payout ratio

Coefficients^{a,b,c}

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	5.913	1.650		3.584	.000
	Payout ratio	.171	.017	.254	9.921	.000
	Expected growth rate in EPS- Next 5 years	2.284	.087	.674	26.336	.000

a. Broad Group = United States

b. Dependent Variable: Trailing PE

c. Weighted Least Squares Regression – Weighted by Market Cap (in US \$)

If a coefficient has the wrong sign: The Multicollinearity Problem

7

Correlations^a

		Trailing PE	Payout ratio	Expected growth rate in EPS– Next 5 years	Beta
Trailing PE	Pearson Correlation	1	.144**	.270**	.071**
	Sig. (2-tailed)		.000	.000	.001
	N	2348	2320	1109	2293
Payout ratio	Pearson Correlation	.144**	1	-.220**	.080**
	Sig. (2-tailed)	.000		.000	.000
	N	2320	2434	1138	2364
Expected growth rate in EPS– Next 5 years	Pearson Correlation	.270**	-.220**	1	-.093**
	Sig. (2-tailed)	.000	.000		.000
	N	1109	1138	1649	1591
Beta	Pearson Correlation	.071**	.080**	-.093**	1
	Sig. (2-tailed)	.001	.000	.000	
	N	2293	2364	1591	6338

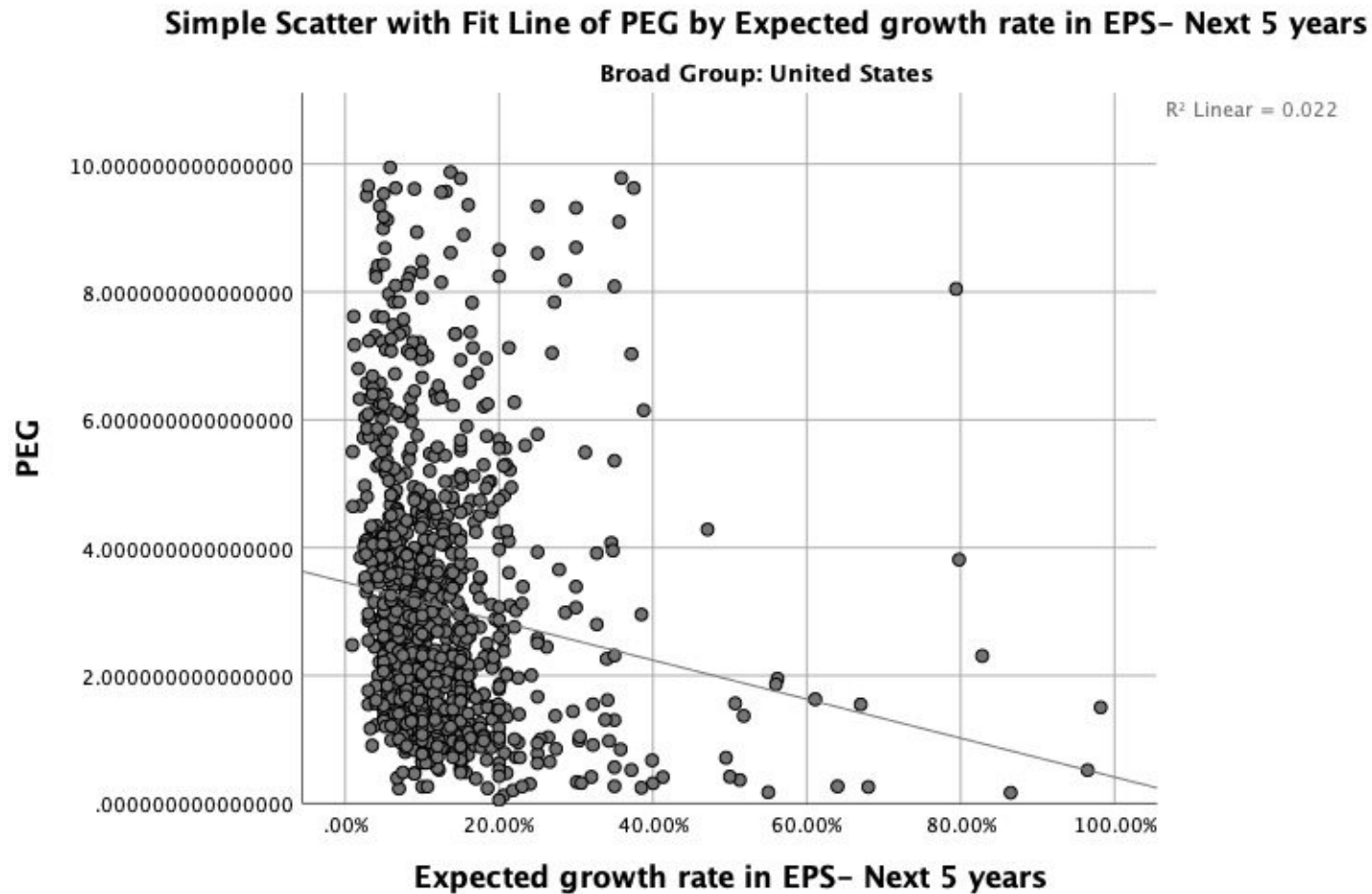
Improving R-Squared?

- Look for better proxies: In the regression, the variables used for growth, risk and payout (the expected growth rate in earnings per share, 2-year regression beta and a payout ratio from the most recent year) may not be the best proxies for future values of each.
- Add more independent variables: You can add more independent variables, but in doing so, you should start with a common sense model of why. Otherwise, you run the risk of kitchen sink regressions.
- Review outliers: In some cases, a few extreme outliers can alter your R-squared, even with large samples. Capping the values or removing the observations can help, with the caveat that you are “messing” with the data.
- Change to a weighted least square regression: If there are some observations that you believe should count for more than others in your regression, you can try a weighted least square regression.

Data Mining, P-hacking and Other Practices...

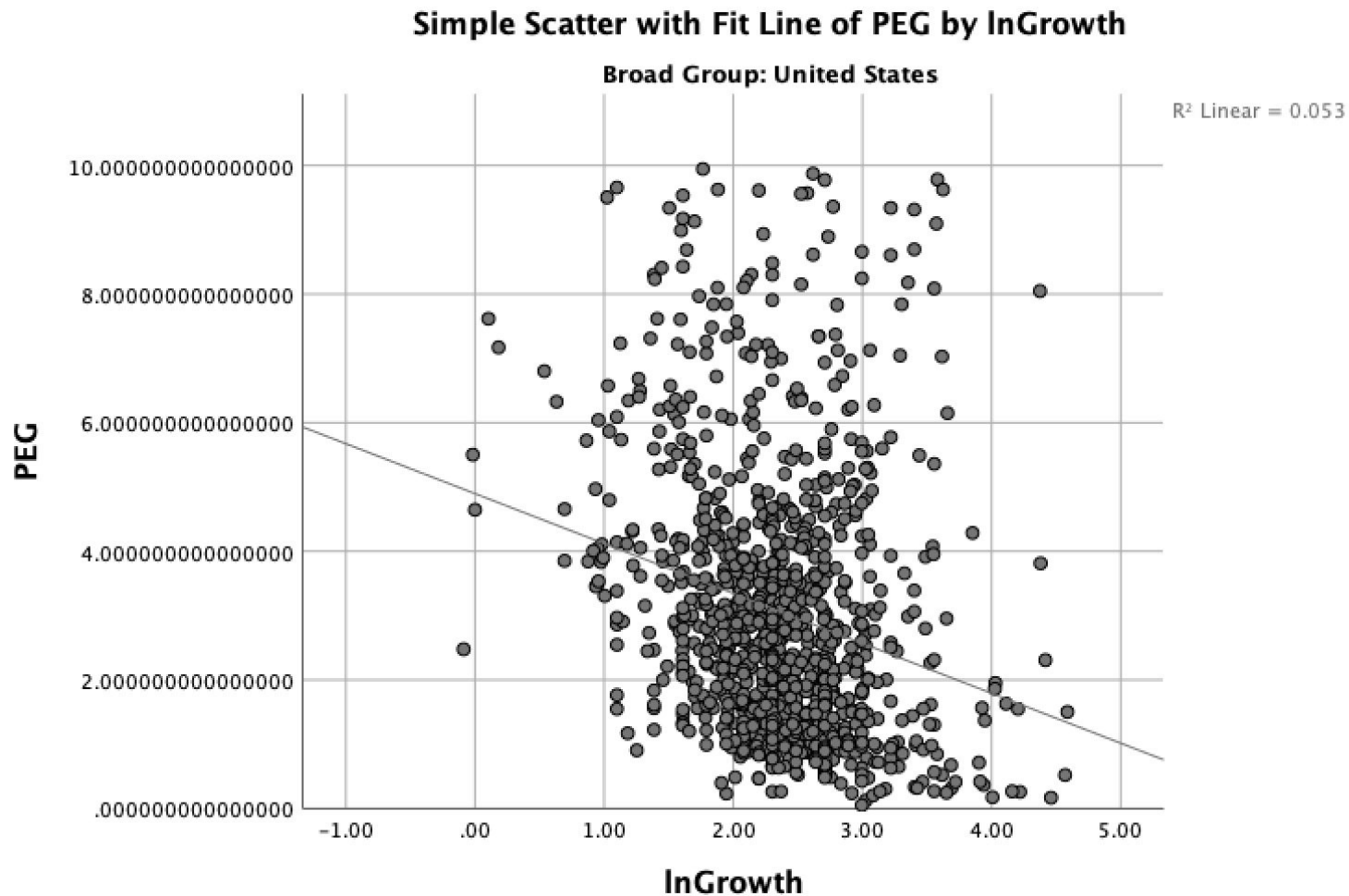
- **Data mining** refers to the practice of going through data looking for variables that have high explanatory power and presenting this as evidence of co-movement. While there is nothing statistically wrong with this practice, researchers should not attach economic or investing significance to nonsense variables.
- **P-hacking** refers to game playing with the data, changing time periods or variable measurement modes, with the intent of delivering p values and t statistics that are significant.
- Finally, with large samples, **statistical significance does not always translate into economic significance**. There are a lot more ways of making money on paper (based on research and statistical studies) than there are in practice.

2. A Non-linear Relationship: PEG Ratios versus Growth



PEG versus ln(Expected Growth)

11



PEG Ratio Regression - US stocks

January 2020

12

Model Summary^a

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.341 ^b	.116	.113	1.91045878

a. Broad Group = United States

b. Predictors: (Constant), Beta, Payout ratio, InGrowth

Coefficients^{a,b}

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	5.626	.321		17.521	.000
	Payout ratio	.004	.001	.107	3.294	.001
	InGrowth	-.660	.114	-.190	-5.799	.000
	Beta	-1.138	.170	-.210	-6.696	.000

a. Broad Group = United States

b. Dependent Variable: PEG

3. Sample sizes and Regressions

- In general, larger sample sizes are better than smaller ones, if you want statistical significance and explanatory power.
- In practice, there are times when you will have only small samples, no matter how hard you try. If you try to run a multiple regression, and use "too many" independent variables, you may get a regression that looks good (on R-squared) but are useless in either explaining or predicting the dependent variable.
- As a rule of thumb,
 - If your sample size is 10-15 observations, the best you can do is run a simple regression
 - For every additional 10-15 observations, you can add one more independent variable.

An Example

<i>Regression Statistics</i>	
Multiple R	0.867415154
R Square	0.752409049
Adjusted R Square	0.504818099
Standard Error	27.78683087
Observations	9

<i>ANOVA</i>					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	4	9385.496876	2346.374219	3.038919828	0.153548553
Residual	4	3088.431879	772.1079698		
Total	8	12473.92876			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	-13.47974744	22.33172411	-0.603614274	0.578658552	-75.48255353	48.5230587	-75.482554	48.5230587
EV/Revenues	20.68638521	40.16380231	0.515050469	0.633672769	-90.82620713	132.198978	-90.826207	132.198978
EV/EBITDA	-0.181441851	1.690097701	-0.107355836	0.919675869	-4.873905339	4.51102164	-4.8739053	4.51102164
Price/sales	-11.36003758	31.9251504	-0.355833487	0.739938381	-99.99846515	77.27839	-99.998465	77.27839
Operating Margin	0.55671716	0.734609703	0.757840739	0.490740927	-1.482886354	2.59632067	-1.4828864	2.59632067

4. Dummy Variable Regressions: Telecom ADRs in 1997

<i>Company Name</i>	<i>PE</i>	<i>Growth</i>
<i>PT Indosat ADR</i>	7.8	0.06
<i>Telebras ADR</i>	8.9	0.075
<i>Telecom Corporation of New Zealand ADR</i>	11.2	0.11
<i>Telecom Argentina Stet - France Telecom SA ADR B</i>	12.5	0.08
<i>Hellenic Telecommunication Organization SA ADR</i>	12.8	0.12
<i>Telecomunicaciones de Chile ADR</i>	16.6	0.08
<i>Swisscom AG ADR</i>	18.3	0.11
<i>Asia Satellite Telecom Holdings ADR</i>	19.6	0.16
<i>Portugal Telecom SA ADR</i>	20.8	0.13
<i>Telefonos de Mexico ADR L</i>	21.1	0.14
<i>Matav RT ADR</i>	21.5	0.22
<i>Telstra ADR</i>	21.7	0.12
<i>Gilat Communications</i>	22.7	0.31
<i>Deutsche Telekom AG ADR</i>	24.6	0.11
<i>British Telecommunications PLC ADR</i>	25.7	0.07
<i>Tele Danmark AS ADR</i>	27	0.09
<i>Telekomunikasi Indonesia ADR</i>	28.4	0.32
<i>Cable & Wireless PLC ADR</i>	29.8	0.14
<i>APT Satellite Holdings ADR</i>	31	0.33
<i>Telefonica SA ADR</i>	32.5	0.18
<i>Royal KPN NV ADR</i>	35.7	0.13
<i>Telecom Italia SPA ADR</i>	42.2	0.14
<i>Nippon Telegraph & Telephone ADR</i>	44.3	0.2
<i>France Telecom SA ADR</i>	45.2	0.19
<i>Korea Telecom ADR</i>	71.3	0.44

The Lead In: Data Hypotheses

- Growth and PE: Companies with higher expected growth should have higher PE ratios. In the table, the companies with lower expected growth rates are trading at lower PE ratios.
- Risk and PE: Companies that are riskier should trade at lower PE ratios. While these are all telecom firms, some are in developed markets and others are in emerging markets, and in 1997, emerging market companies were viewed as much riskier than developed market companies.
 - I used a dummy variable to capture this, with emerging market companies getting a value of one and developed market companies getting a value of zero.
 - Note that there are richer measures of country risk, like country risk scores and sovereign ratings, which allows for more variation, but they were not available in 1997.

PE, Growth and Risk

- Dependent variable is: PE
- R squared = **66.2%** R squared (adjusted) = **63.1%**

<i>Variable</i>	<i>Coefficient</i>	<i>SE</i>	<i>t-ratio</i>	<i>Probability</i>
Constant	13.1151	3.471	3.78	0.0010
Growth rate	121.223	19.27	6.29	≤ 0.0001
Emerging Market Dummy	-13.853	3.606	-3.84	0.0009

- You can use this regression to get predicted values for individual stocks. For instance, Indosat has an expected growth rate of 6% and is an emerging market company:
$$PE = 13.13 + 121.22 (.06) - 13.85 (1) = 6.55$$

At 7.8 times earnings, Indosat is expensive.

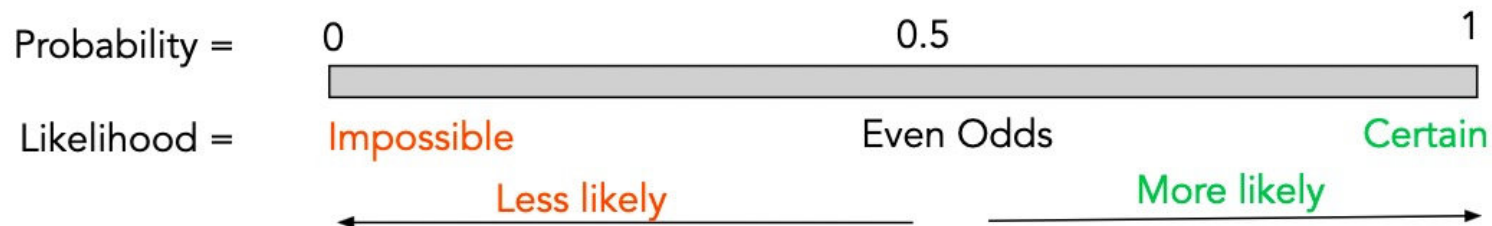


SESSION 6: PROBABILITIES & PROBABILISTIC TOOLS

Session 6
There is a chance!

What is a probability?

- When you cannot predict outcomes with certainty, probabilities allow you to measure the likelihood of an outcome occurring.
- The probability of an outcome will range from zero to one, with zero representing no chance and one representing total certainty.



- You can measure the probabilities of discrete events (an acquisition, bankruptcy etc.) as well as the probabilities of a continuous variable taking a value greater or less than a specified value (earnings will exceed zero, market cap will exceed a billion).

Two views of probability

- Frequentist view: In the frequentist view, probability measures the frequency with which an event will occur, if it is repeated many times.
 - ▣ In theory, you want to measure the frequency of an outcome if an experiment is repeated an infinite number of times.
 - ▣ While it is objective, it requires that an event be repeatable, and is thus narrow.
- Subjectivist (Bayesian) view: In the subjectivist view, probability is the value that a rational agent would assign to the likelihood of an event happening.
 - ▣ You can assign probabilities to events, even if they are not repeatable.
 - ▣ Unlike the frequentist view, rational individuals can disagree on the probability of an event occurring.

Cumulative and Conditional Probabilities

- When you have sequential events, the *cumulative probability* measures the likelihood of a specific series of outcomes. To compute cumulative probabilities, each event in the sequence has to be independent of the others in that sequence.
 - ▣ Thus, you can estimate the probability that an investor will beat the market ten years in a row, even if he or she is picking stocks randomly.
- A conditional probability measures the *likelihood of an event or outcome*, based upon the occurrence of a *prior event or outcome*.
 - ▣ For instance, you can estimate the probability that the market will be up tomorrow, given that it was up today.

Probability Rules

1. *It is bounded:* For any event X , $0 \leq P(X) \leq 1$
2. *It covers the universe of outcomes:* The sum of the probabilities of all possible outcomes is one.
3. *The Complement Rule* states that $P(\text{not } X) = 1 - P(X)$
4. *The General Addition Rule* states that for any two events, $P(X \text{ or } Y) = P(X) + P(Y) - P(X \text{ and } Y)$, where $P(X \text{ and } Y)$ is the probability of both X and Y occurring at the same time. If they are mutually exclusive, the rule simplifies to $P(X \text{ or } Y) = P(X) + P(Y)$.
5. *The Multiplication Rule* states that for two independent events to occur together, $P(X \text{ and } Y) = P(X) * P(Y)$. If they are not independent, $P(X \text{ and } Y) = P(X) * P(Y/X)$, where $P(Y/X)$ is the conditional probability of Y happening, given that X has happened.

Bayesian Statistics

- **Bayes' theorem** describes the probability of an event occurring, based upon prior knowledge of other variables related to that event.
 - In effect, it is a conditional probability, with the probability of an event conditioned on the information/knowledge that you have.
 - Since the information/knowledge that different individuals can have about an event can vary, Bayes' theorem allows for differences in probability estimates for the same event across individuals.
- In **Bayesian Inference**, you update the probability of an event happening as you receive new evidence or information.
 - The probability that you assign to an event before you receive the new information represent your priors.
 - The probability that you assign to that same event after receiving and processing new information represent your *posterior* estimate.

1. Probit & Logit Models: Techniques for estimating probabilities

- A *probit model* is a variant on the standard regression approach, with the key difference being that the dependent variable can take on only one of two values (binary) and a secondary one being a transformation of the linear model to ensure that the predicted value from the model is a number between 0 and 1 (limits for a probability estimate).
 - Example: You can estimate the probability of a hostile acquisition of a company based upon its size, past stock price performance and insider holdings.
- A *logit model* is a close relative to a probit, with the difference revolving around the function that they use to transform the standard linear model, with the logit model using a logistic distribution and the probit a normal distribution.

A Logit Model: Predicting Bankruptcy

- In investing and valuation, one of the concerns that you face is that the business that you are investing in or valuing may go out of business. Predicting the probability of bankruptcy becomes a part of the challenge.
- There are logit models that have been used to predict bankruptcy, where
 - ▣ You start with all firms at the start of the period
 - ▣ The dependent variable becomes the stand-in for whether a firm survives or goes bankrupt during the study period
 - ▣ The independent variables reflect what you believe are key drivers of bankruptcy (earnings level and volatility, debt level and payments due, market access to capital).
 - ▣ You build a logit model that will yield as output an equation that resembles a regression, but will yield a probability of bankruptcy.

A Probit Model: Hostile Acquisitions

- While there are no easy pathways to making money, it seems clear that investors in companies that are targeted in acquisitions (especially hostile ones) earn high returns, but only if they invest before the event.
- There are probit models for predicting companies that will be targeted, and they involve:
 - You start with all firms that publicly traded at the start of a period
 - The dependent variable becomes the stand-in for whether a firm is targeted in a hostile acquisition
 - The independent variables reflect what you believe are key drivers of hostile acquisitions, including poor stock price performance, lagging accounting returns and managers with little or no shareholdings.
 - You build a probit model that will yield as output an equation that resembles a regression, but will yield a probability of a hostile acquisition.

2. Decision Trees

- A decision tree is a flow chart used to capture the probabilities and outcomes of random events, where the likelihood of an event is affected by prior events occurring or not occurring.
- Within decision trees, you can build not only the outcomes of chance but also the decisions that follow from these outcomes and the end outcomes for each branch.
 - Event or Chance nodes represent when uncertainty is resolved, with event branches representing mutually exclusive and exhaustive, with probability estimates.
 - Decision nodes point where you decide to go on or stop. The branches represent mutually exclusive paths to follow.
 - ◀ Terminal or End node is the final outcome or payoff from each branch.

And follow through..

- *Folding (Rolling) back a decision tree:* If you have a complete decision tree, with probabilities attached to event branches, and outcomes following from the end of each decision branch, you can roll back the decision tree, by
 - ▣ Estimating the expected value of each branch, starting with the right and working towards the left, using the probabilities and payoffs.
 - ▣ Compute an expected value for the entire tree, across branches.
- If there is a time lag across the branches, and/or risk involved in the process, you can incorporate those elements into a decision tree as well in the form of a discount rate that you use discount payoffs at each leg.

3. Scenario Analysis

- Scenario analysis is best employed when the outcomes of a project are a function of the macro economic environment and/or competitive responses.
- There are a couple of ways in which you can structure scenario analysis
 - ▣ Best-case, Worst-case analyses, where you set all the inputs at their most optimistic and most pessimistic levels. In general, this leaves you with findings that are not of much practical use.
 - ▣ Plausible scenarios: Here, you define what you feel are the most plausible scenarios (allowing for the interaction across variables) and come up with estimates for each scenario.
- For scenario analysis to be most useful, you have to be able to
 - ▣ List scenarios to cover all possible outcomes
 - ▣ Estimate probabilities for the scenarios
 - ▣ Complete with an expected value.



SESSION 6A: PROBABILITIES & DECISION TREES IN FINANCE

Session 6
There is a chance!

1. Stock Market as a Random Walk

- One of the simplest, albeit weakest, tests of randomness is to test to see whether there is a 50:50 chance that markets will go up (or down). Thus, if you have a hundred days of market changes, you would expect to see 50 up and 50 down days.
- To test this proposition, you can look at price changes for n trading days, and compute the number of days that the market was up (and down). However, even if markets are random, the actual number that you find will deviate from 50:50. The standard error is a function of the number of trading days and can be computed as follows:
 - ▣ Std Error in probability = $\sqrt{\frac{0.5 \cdot 0.5}{n}}$ where n = Number of days in your sample.
 - ▣ This can be generalized more generally to any two-outcome experiment as
Std Error in probability = $\sqrt{\frac{p(1-p)}{n}}$ where p = Probability of one of the outcomes
 - ▣ Thus, if you observe a hundred trading days, the range, with 95% probability, on up (or down) days, even if markets have a 50:50 chance of going up and down, would be 45-55. (Std Error = 2.5%)

Probability that the market will go up (or down)...

- There were 1257 trading days between January 1, 2016 and December 31, 2020. Over that period, the S&P 500 was
 - ▣ Up on 700 trading days
 - ▣ Down on 557 trading days
- Based upon that data, the probability that the market was up during the period was 55.69% and the probability that the market was down was 44.31%.
- *Can you use this to accept or reject the hypothesis that there was a 50% chance of up/down days., at least during this period?*
 - ▣ *Std Error in probability, if random = $\sqrt{\frac{0.5*0.5}{1257}} = 0.0141$ or 1.41%*
 - ▣ *Range with 95% confidence, on up probability = $0.5569 \pm 2(.0141)$: Range: 52.87% to 58.51% probability of up (or down) days, during this period*
 - ▣ *At least during this period, you can reject the 50:50 up/down hypothesis.*
- *Can you extrapolate from this that there will be more up than down days in the market in the future?*

Conditional Probabilities...

- Using the same data (returns on the S&P 500 on a daily basis from January 1, 2016, to December 31, 2020), and breaking down into up and down days:

		Market tomorrow	
		Up	Down
Market today	Up	367	333
	Down	332	225

- Converting these numbers into probabilities, you get:

		Market tomorrow	
		Up	Down
Market today	Up	52.43%	47.57%
	Down	59.61%	40.39%

Cumulative Probabilities...

- If the probability that the market will go up (down) on a given day was 55.69% (44.31%), all through the time period, you can estimate the probabilities of the market going up or down two, three or even ten days in a row.
- Thus, to estimate the probability that the market was up three or five days in a row:
 - ▣ Probability of three up days in a row = $(0.5569)^3 = .1727$
 - ▣ Probability of five up days in a row = $(0.5569)^5 = .0536$
- Similarly, to estimate the probability that the market was down three or five days in a row:
 - ▣ Probability of three up days in a row = $(0.4431)^3 = .0870$
 - ▣ Probability of five up days in a row = $(0.4431)^5 = .0171$

2. Transition Probabilities

- In investing and finance, it is common to put portfolio managers and companies into groupings, reflecting their standing on a metric (returns, debt ratios, dividends) or rankings in performance.
- Those rankings are not only used to judge investors and companies, but are also used as predictors for the future. While every mutual or hedge fund touts the warning that past performance is not a predictor of the future, it is undeniable that funds that have performed/ranked well in the past market themselves on that basis, and that investors redirect their money to these funds.
- The test of whether chasing past performance is a good strategy can be converted into a test of whether there is stickiness in rankings. Put simply, if you ranked funds into groups based upon success, do these rankings persist?

Money Manager Performance

Report 5: Five-Year Transition Matrix – Performance over Two Non-Overlapping Five-Year Periods (Based on Quartile)

ALL DOMESTIC FUNDS	FUND COUNT AT START (SEPTEMBER 2014)	FIVE-YEAR PERCENTAGES AT END					
		1ST QUARTILE (%)	2ND QUARTILE (%)	3TH QUARTILE (%)	4TH QUARTILE (%)	MERGED/ LIQUIDATED (%)	STYLE CHANGED (%)
1st Quartile	485	31.75	21.44	18.14	20.21	8.04	0.41
2nd Quartile	484	23.76	21.69	21.07	18.18	14.88	0.41
3rd Quartile	485	14.02	22.47	23.30	17.53	21.86	0.82
4th Quartile	484	10.54	14.46	17.56	24.17	30.99	2.27

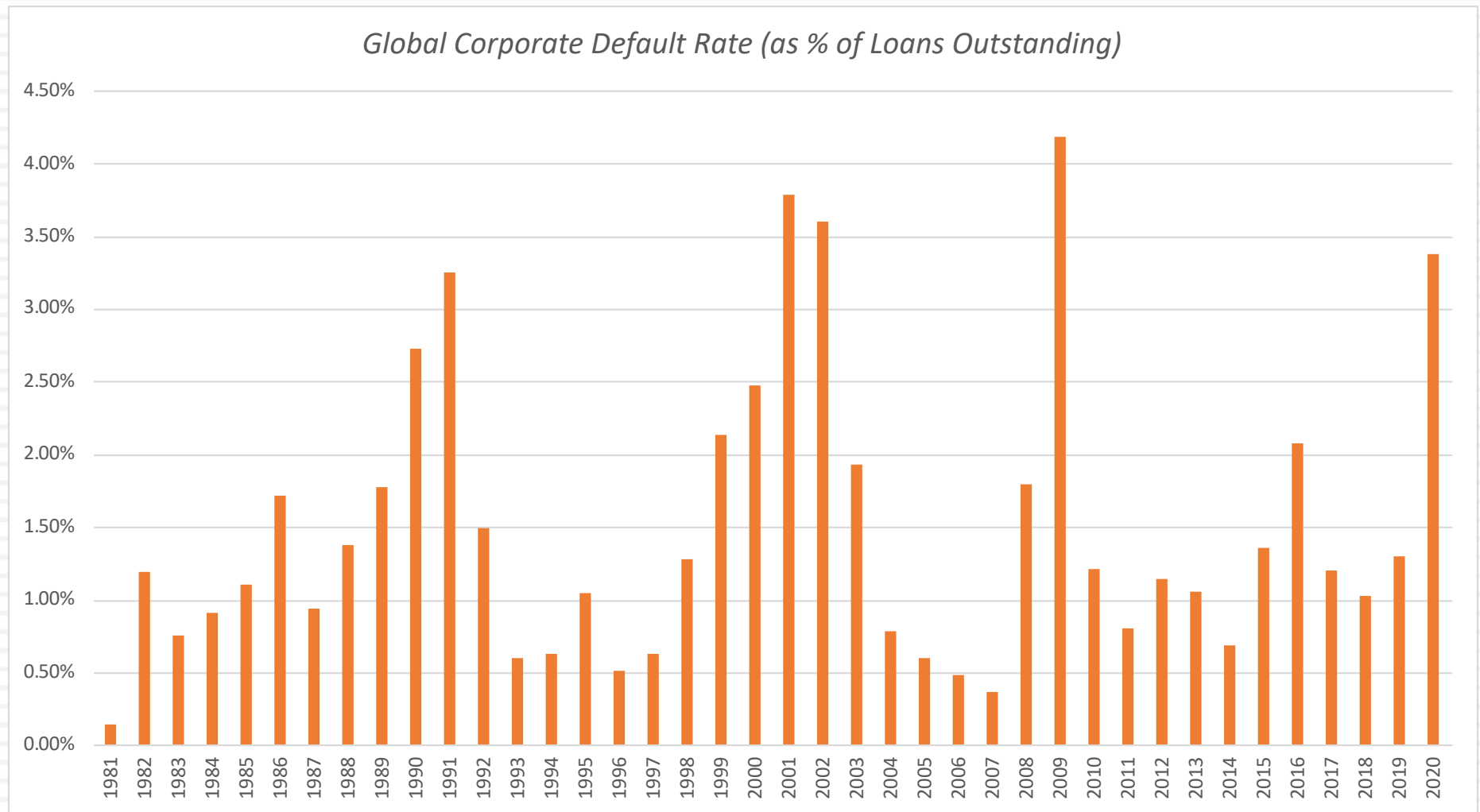
The performance persistence varies across

- Markets (Geographical, asset class)
- Time (Some periods have more persistence than others)
- Style classes (value versus growth, small vs large cap)

It is also difficult for investors to convert this statistical persistence into returns.

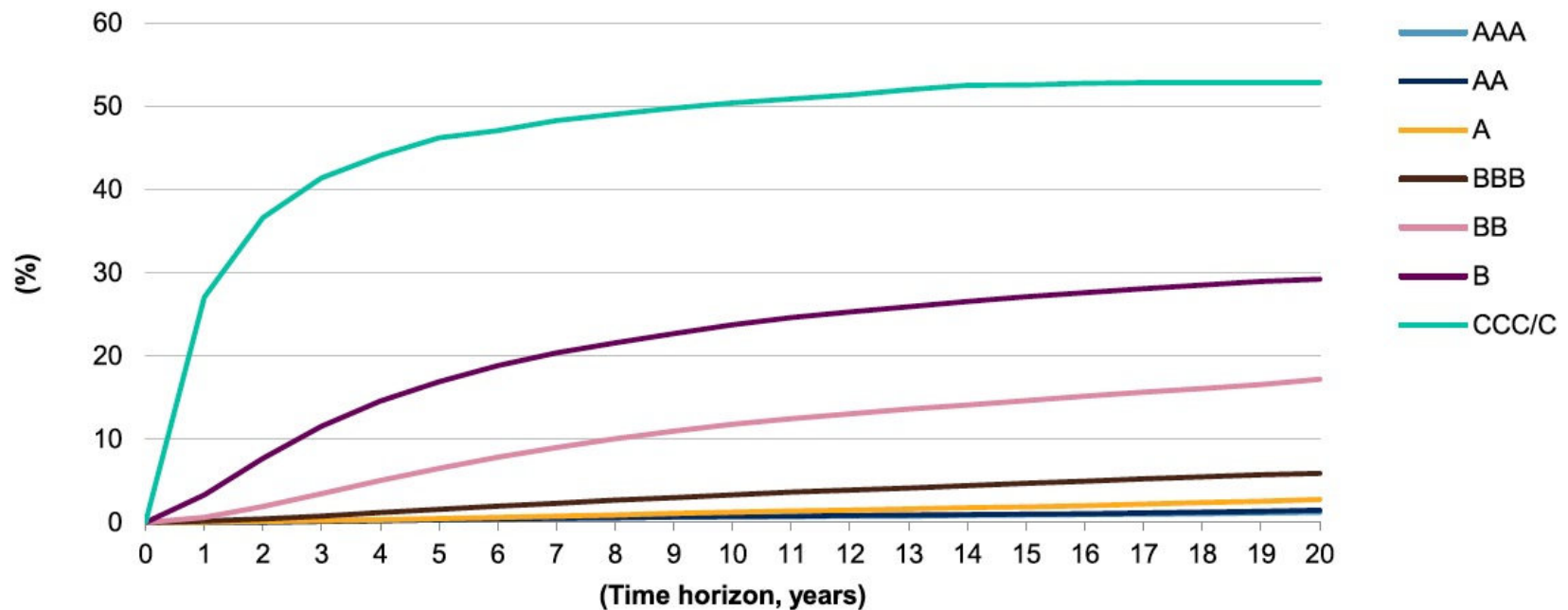
Each of these probability estimates has a standard error that will decrease as the sample size increases.

3. Probability of Corporate Default



Conditional Probabilities: Bond Ratings and Default Rates

Global Corporate Average Cumulative Default Rates By Rating (1981-2019)



Sources: S&P Global Ratings Research and S&P Global Market Intelligence's CreditPro®.

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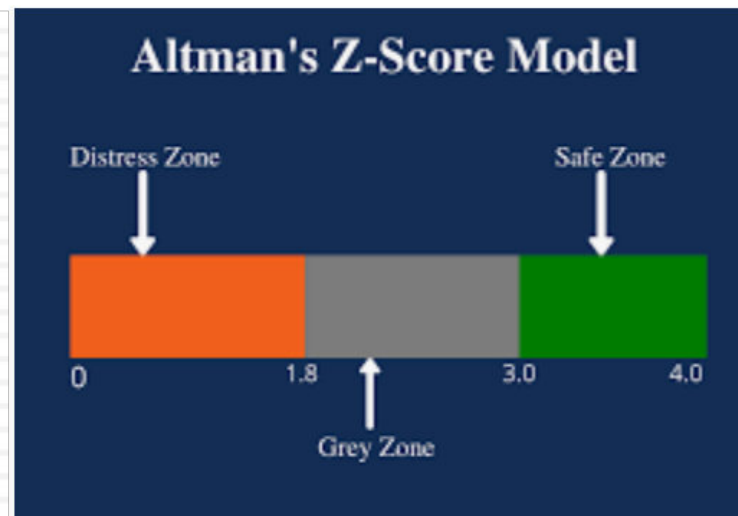
A Multiple Discriminant Model of Default: The Altman Z Score



Altman Z Score Formula

$$= \left(1.2 \times \left(\frac{\text{Working Capital}}{\text{Total Assets}} \right) + 1.4 \times \left(\frac{\text{Retained Earnings}}{\text{Total Assets}} \right) + 3.3 \times \left(\frac{\text{Earnings Before Interest and Task Payment}}{\text{Total Assets}} \right) \right. \\ \left. + 1.6 \times \left(\frac{\text{The equity's Market Value}}{\text{Total Assets}} \right) + 0.999 \times \left(\frac{\text{Total Sales}}{\text{Total Assets}} \right) \right)$$


Altman analyzed 66 manufacturing companies, of which 33 became bankrupt within the years 1946- 1965 and the other half were existing companies in 1966.



A Probit Model: Hostile Acquisitions

- While there are no easy pathways to making money, it seems clear that investors in companies that are targeted in acquisitions (especially hostile ones) earn high returns, but only if they invest before the event.
- There are probit models for predicting companies that will be targeted, and they involve:
 - You start with all firms that publicly traded at the start of a period
 - The dependent variable becomes the stand-in for whether a firm is targeted in a hostile acquisition
 - The independent variables reflect what you believe are key drivers of hostile acquisitions, including poor stock price performance, lagging accounting returns and managers with little or no shareholdings.
 - You build a probit model that will yield as output an equation that resembles a regression, but will yield a probability of a hostile acquisition.

Hostility Prediction Models, 1975–1996

Probit models predicting whether successful and unsuccessful takeover bids for exchange-listed target firms from 1975 to 1996 are hostile, using four measures of hostility. The dependent variables are dummy variables that equal one when a hostile bid is made for a target firm, and zero otherwise. Host(WSJ) is based on descriptions in the *Wall Street Journal Index* or *Dow Jones News Retrieval*, Host(SDC) is based on whether the target firm resisted an unsolicited offer as determined by the Securities Data Company (SDC), Host(Uns) is based on whether the initial or winning bid is unsolicited, and Host(Pre) is based on whether the target firm is in play (someone has filed a 13D form with the SEC showing an accumulation of shares within the past 12 months) or the subject of a takeover rumor reported in *DJNR*. Several variables measuring the performance of the target firm before the first bid are used in the model. ROE is earnings divided by average stockholder's (book) equity and Sales Growth is the growth in sales over the fiscal year before the first bid. Liquidity is the ratio of net liquid assets to total assets, D/E is the long-term debt to book equity, M/B is the ratio of market to book value of stockholder's equity, P/E is the ratio of stock price to the earnings for the last fiscal year, and Size is the logarithm of the market value of common stock, all measured at the end of the fiscal year before the first bid. Dummy variables are equal to one when the first bid occurs during 1980 to 1984, or 1985 to 1989, or 1990 to 1996, and zero otherwise, allowing for secular variation. The last two columns contain the coefficients and *t*-statistics for a regression of the hostility factor (principal component), Host(Factor), created from the three hostility variables with complete data (Host(WSJ), Host(Uns), and Host(Pre)) on the explanatory variables from the probit model. The marginal effect column transforms the probit coefficient into the marginal effect of the variable on the estimated probability, evaluated at the sample means of the explanatory variables.

Variable	Host(WSJ)			Host(SDC)			Host(Uns)			Host(Pre)			Host(Factor)	
	Coef.	<i>t</i> -statistic	Marginal Effect	Coef.	<i>t</i> -statistic	Marginal Effect	Coef.	<i>t</i> -statistic	Marginal Effect	Coef.	<i>t</i> -statistic	Marginal Effect	Coef.	<i>t</i> -statistic
Constant	-4.692	-7.66	-0.638	-3.145	-4.57	-0.882	-0.153	-0.43	-0.056	-1.660	-4.57	-0.603	-0.092	-1.36
ROE	-2.413	-2.79	-0.328	-0.104	-0.12	-0.029	-1.192	-2.11	-0.439	-0.483	-0.90	-0.175	-0.321	-3.65
Sales Growth	-0.715	-1.83	-0.097	-0.187	-0.54	-0.052	-0.433	-1.89	-0.159	-0.268	-1.19	-0.097	-0.121	-3.29
Liquidity	0.386	1.02	0.052	0.236	0.66	0.066	-0.027	-0.12	-0.010	-0.262	-1.13	-0.095	0.003	0.07
D/E	-0.243	-1.55	-0.033	-0.082	-0.72	-0.023	0.018	0.32	0.007	0.117	2.24	0.042	0.007	0.93
M/B	-0.068	-0.82	-0.009	-0.226	-2.83	-0.063	-0.151	-3.01	-0.055	-0.055	-1.48	-0.020	-0.018	-2.59
P/E	-0.020	-2.21	-0.003	-0.006	-0.89	-0.002	-0.003	-0.66	-0.001	-0.003	-0.65	-0.001	-0.002	-2.64
Size	0.359	7.49	0.049	0.256	5.68	0.072	0.032	1.11	0.012	0.124	4.29	0.045	0.038	6.73
1980–1984	-0.511	-2.85	-0.069	-0.508	-1.39	-0.142	-0.114	-1.05	-0.042	0.363	3.30	0.132	-0.018	-0.94
1985–1989	-0.188	-1.11	-0.026	-0.016	-0.04	-0.004	0.402	3.59	0.148	0.687	6.07	0.250	0.088	4.14
1990–1996	-0.675	-2.88	-0.092	-0.457	-1.24	-0.128	-0.356	-2.43	-0.131	-0.134	-0.93	-0.049	-0.096	-3.37
R^2		0.108			0.119			0.078			0.099			0.133
Log-likelihood		-275.6			-304.0			-704.5			-697.0			
Sample size, <i>N</i>		1,096			593			1,096			1,096			1,096

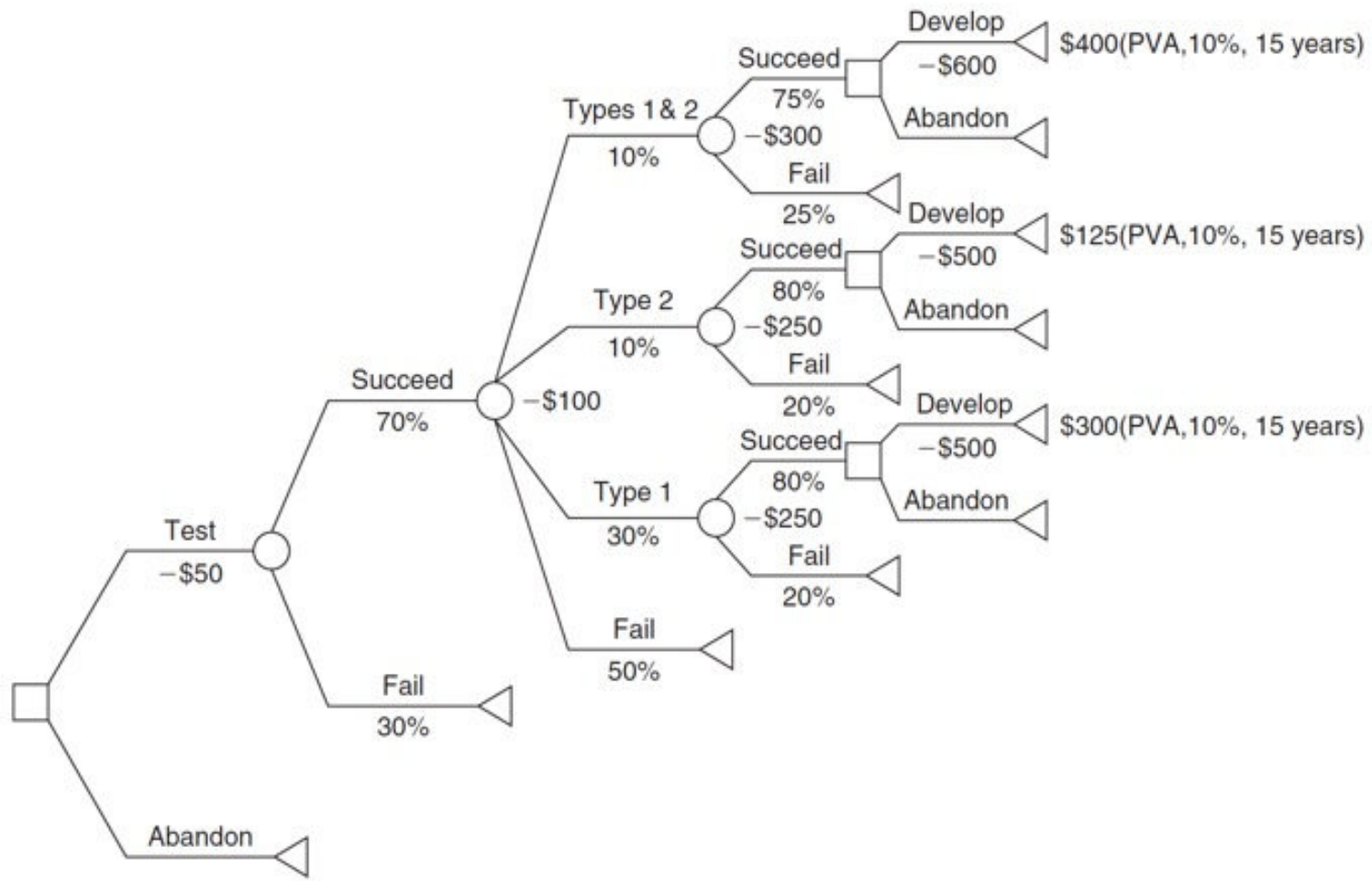
Source: *Hostility in Takeovers: In the Eyes of the Beholder*, G.W. Schwert, JF 2000

3. Decision Tree: An Example

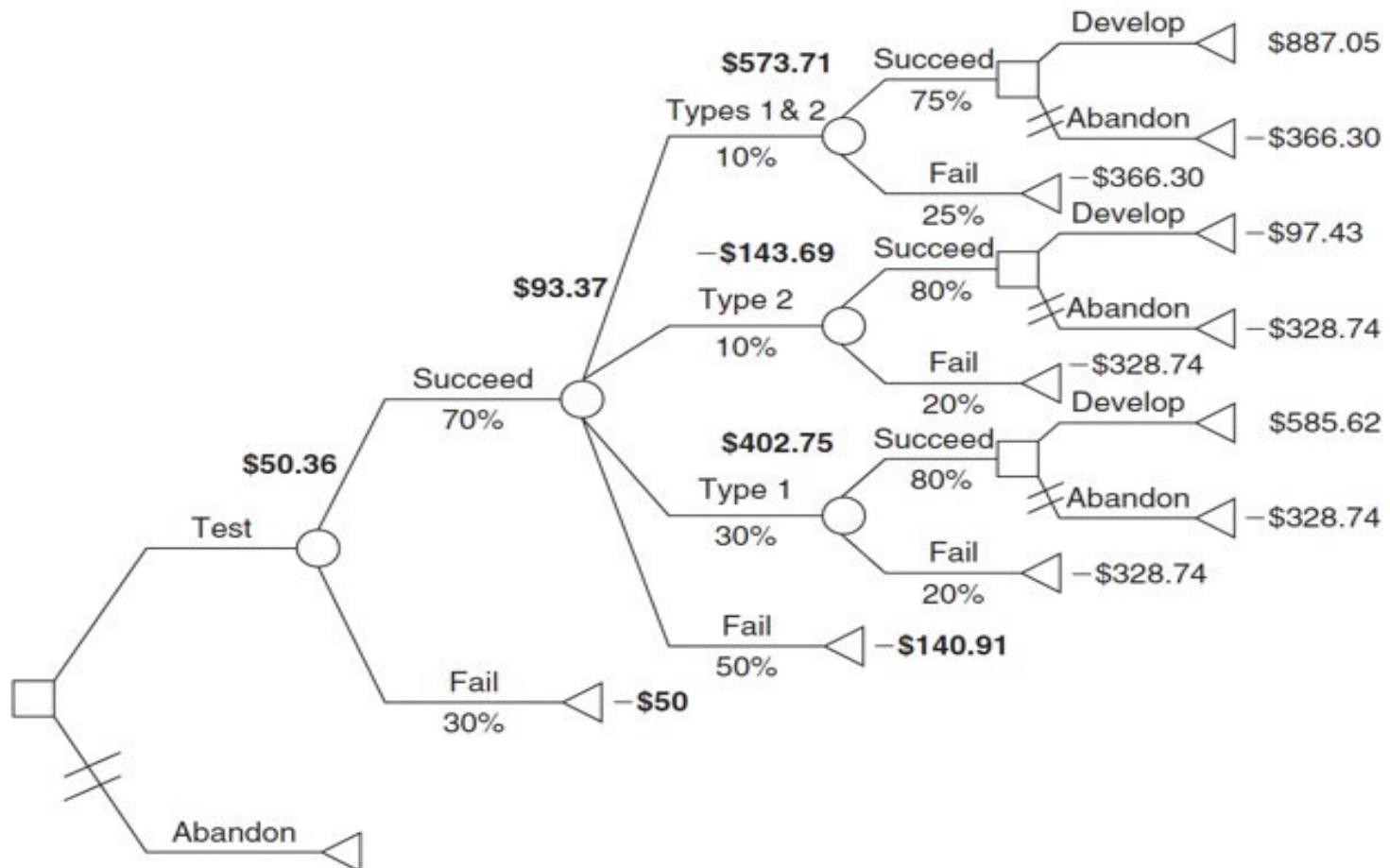
- Consider a pharmaceutical drug for treating Type 1 diabetes that has gone through preclinical testing and is about to enter phase 1 of the FDA approval process.
 - Phase 1 is expected to cost \$ 50 million and will involve 100 volunteers to determine safety and dosage; it is expected to last 1 year. There is a 70% chance that the drug will successfully complete the first phase.
 - In phase 2, the drug will be tested on 250 volunteers for effectiveness in treating diabetes over a two-year period. This phase will cost \$ 100 million and the drug will have to show a statistically significant impact on the disease to move on to the next phase. There is only a 30% chance that the drug will prove successful in treating type 1 diabetes but there is a 10% chance that it will be successful in treating both type 1 and type 2 diabetes and a 10% chance that it will succeed only in treating type 2 diabetes.
 - In phase 3, the testing will expand to 4,000 volunteers to determine the long-term consequences of taking the drug. If the drug is tested on only type 1 or type 2 diabetes patients, this phase will last 4 years and cost \$ 250 million; there is an 80% chance of success. If it is tested on both types, the phase will last 4 years and cost \$ 300 million; there is a 75% chance of success.
- If the drug passes through all 3 phases, the costs and annual cash flows are below:

Disease treatment	Cost of Development	Annual Cash Flow
Type 1 diabetes only	\$ 500 million	\$ 300 million for 15 years
Type 2 diabetes only	\$ 500 million	\$ 125 million for 15 years
Type 1 and 2 diabetes	\$ 600 million	\$ 400 million for 15 years

The Tree...



The Fold Back



4. Scenario Analysis: easyJet and Brexit in 2019

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	No Deal Brexit	Bad Deal Brexit	Soft or No Brexit
Restructuring cost (up front)	£500 million	£300 million	\$0
Revenue growth	3.00%	5.00%	5.00%
Operating Margin	6.00%	7.00%	8.00%
Sales to Capital Ratio	1.73	1.73	1.73

	No Deal Brexit	Delayed & Messy Brexit	Soft or No Brexit
Probability	25%	50%	25%
Value Per Share	£12.02	£15.70	£19.38

Expected Value per share = .25 (£12.02) + .50 (£15.70) + .25 (£19.38) = £15.70



SESSION 6B: SIMULATIONS

Session 6

Data, Distributions and Probabilities!

Point Estimates and Distributions...

- In both corporate finance and valuation, much of what we do is built around point estimates, made with the data that we have at the time of estimation.
- The reality is that what are estimating are distributions, with an expected value (that should be the point estimate) but also a substantial possibility of error.
- Our defense for using point estimates was that we lacked the data to estimate probability distributions and/or that doing valuations with distributions would require machine power that we did not have access to (at a reasonable price).

A Big Picture View of Simulations

- In a simulation, you estimate probability distributions for each variable that goes into an analysis.
- In each simulation, you draw an outcome from each of the distributions and estimate the end result with those outcomes. Since these outcomes can come from the low end or high end of the distributions, they will be different.
- You run as many simulations as you can and come up with a distribution of the outcomes, which you then use for decision making.

Start with a sound model, connecting inputs to your output variable

- In running a simulation, you are trying to estimate an output variable, using input variables that have probability distributions estimated for them.
- If the model you are using to connect the output variable to your input variables is flawed in terms of construct and connections, your output variable will reflect those flaws.
- In building this model, you are aiming for
 - ▣ Simple, over complex, since you have fewer variables to keep track off.
 - ▣ Transparent, rather than opaque, on the connections that underlie models. Models that have black boxes in them are dangerous.

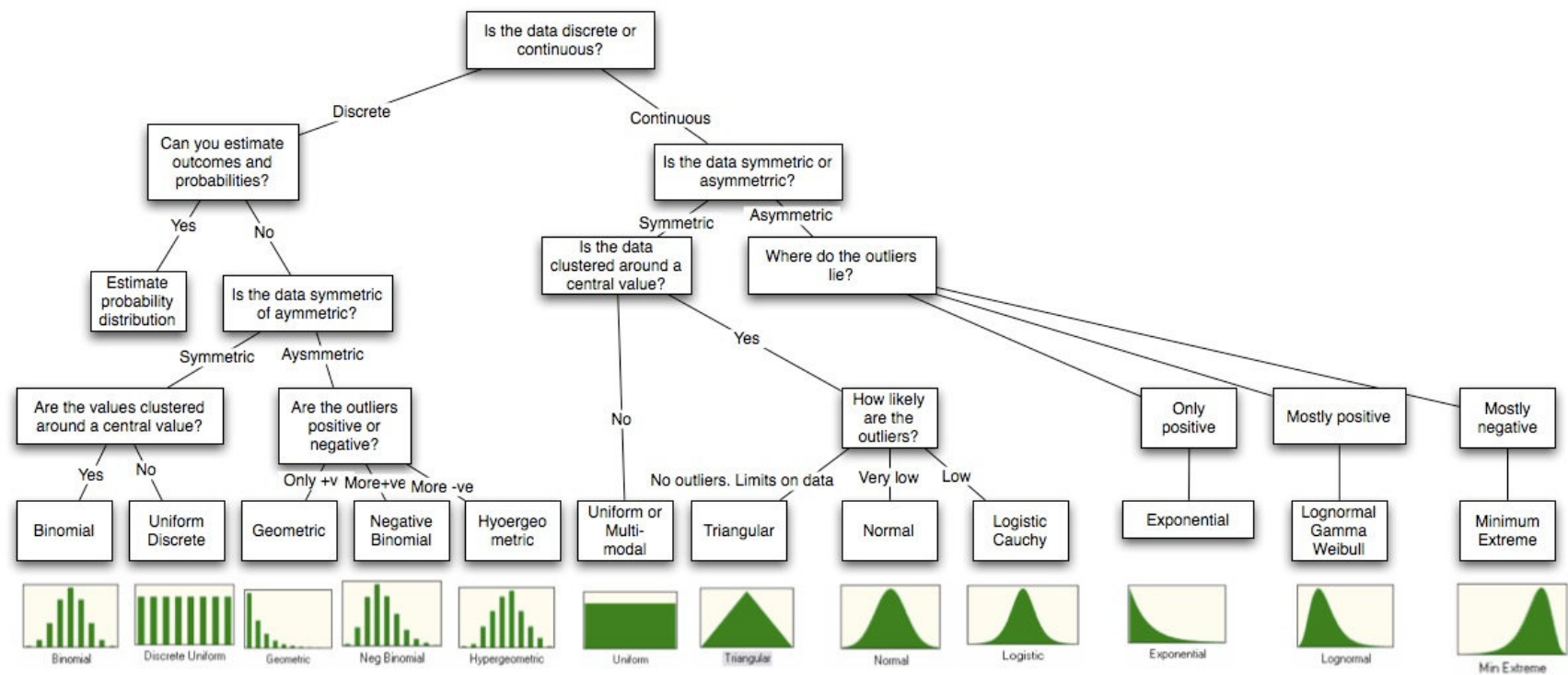
Which variables? Less is more!

- Almost every variable in an analysis has uncertainty attached to it, and the question you face in a simulation is whether you want to build a distribution for each.
- As a general rule, focus on
 - ▣ The variables that have the biggest effect on your decision or output variable.
 - ▣ The variables where you feel most uncertain and exposed.

Classifying uncertainties

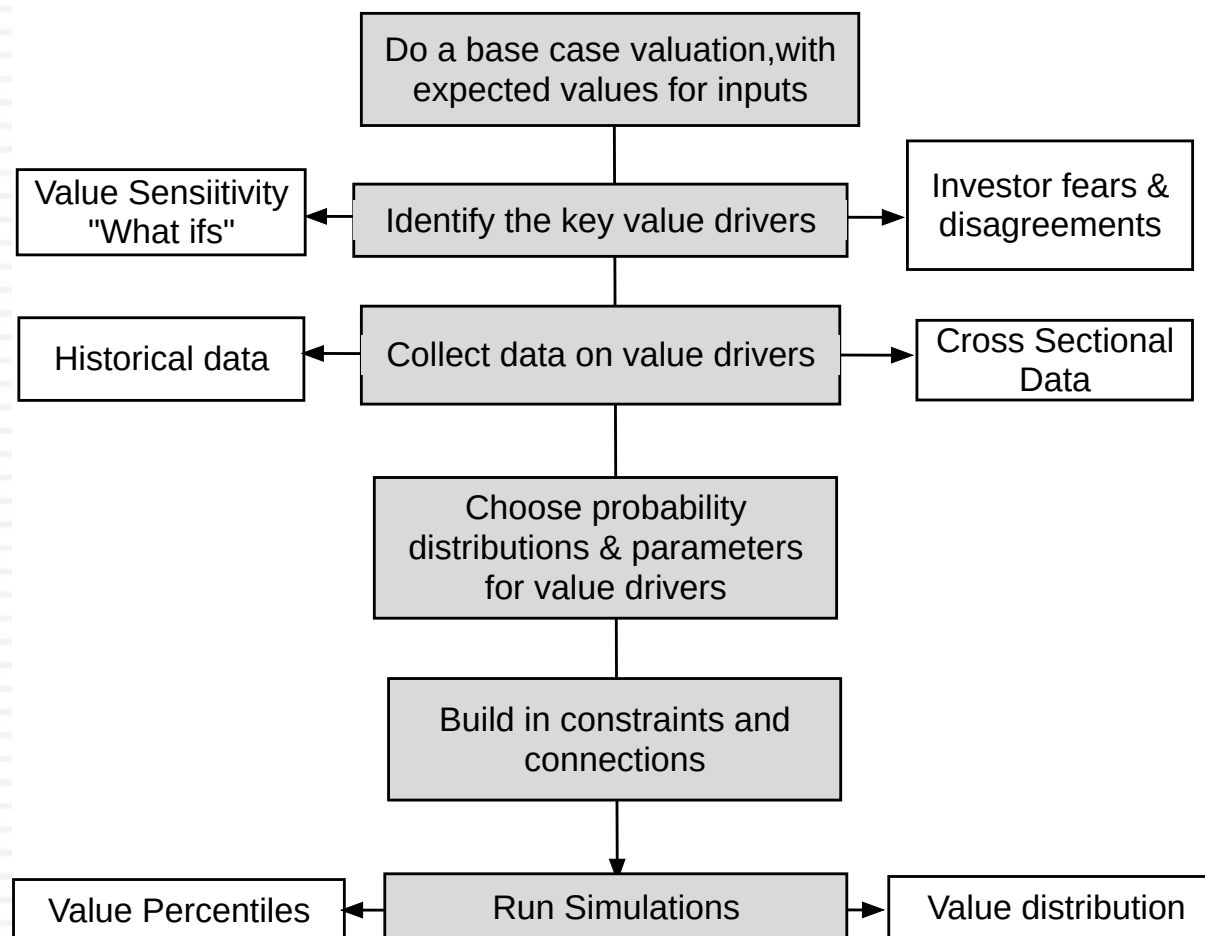
1. Discrete or Continuous? Risks that either occur or do not are discrete risks; you are not exposed to them much of the time, but when they do happen, they can be catastrophic. Risks that you are exposed to all of the time, albeit often in small doses, are continuous risks.
2. Symmetric or Asymmetric? If positive and negative outcomes are roughly equivalent in magnitude and probability, you have symmetric risks. If large positive (negative) outcomes are more likely, you have positively (negatively) skewed risks.
3. Extreme value likelihood, low or high? If outcomes that are very different from your expected value happen very infrequently, you have thin tailed distributions. If they occur often, you have fat tailed distributions.

Pick distributions



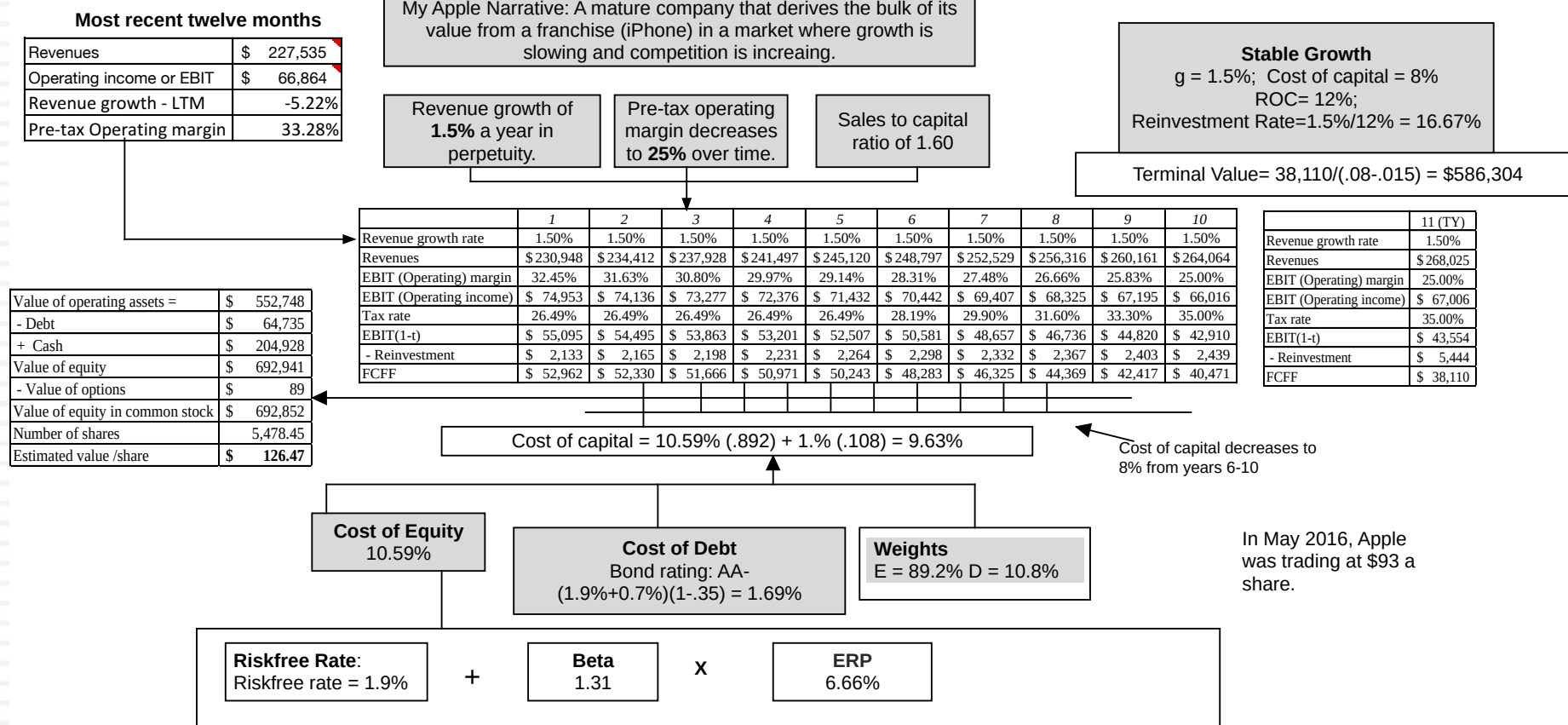
Simulation in Valuation

Value Simulation: The Steps

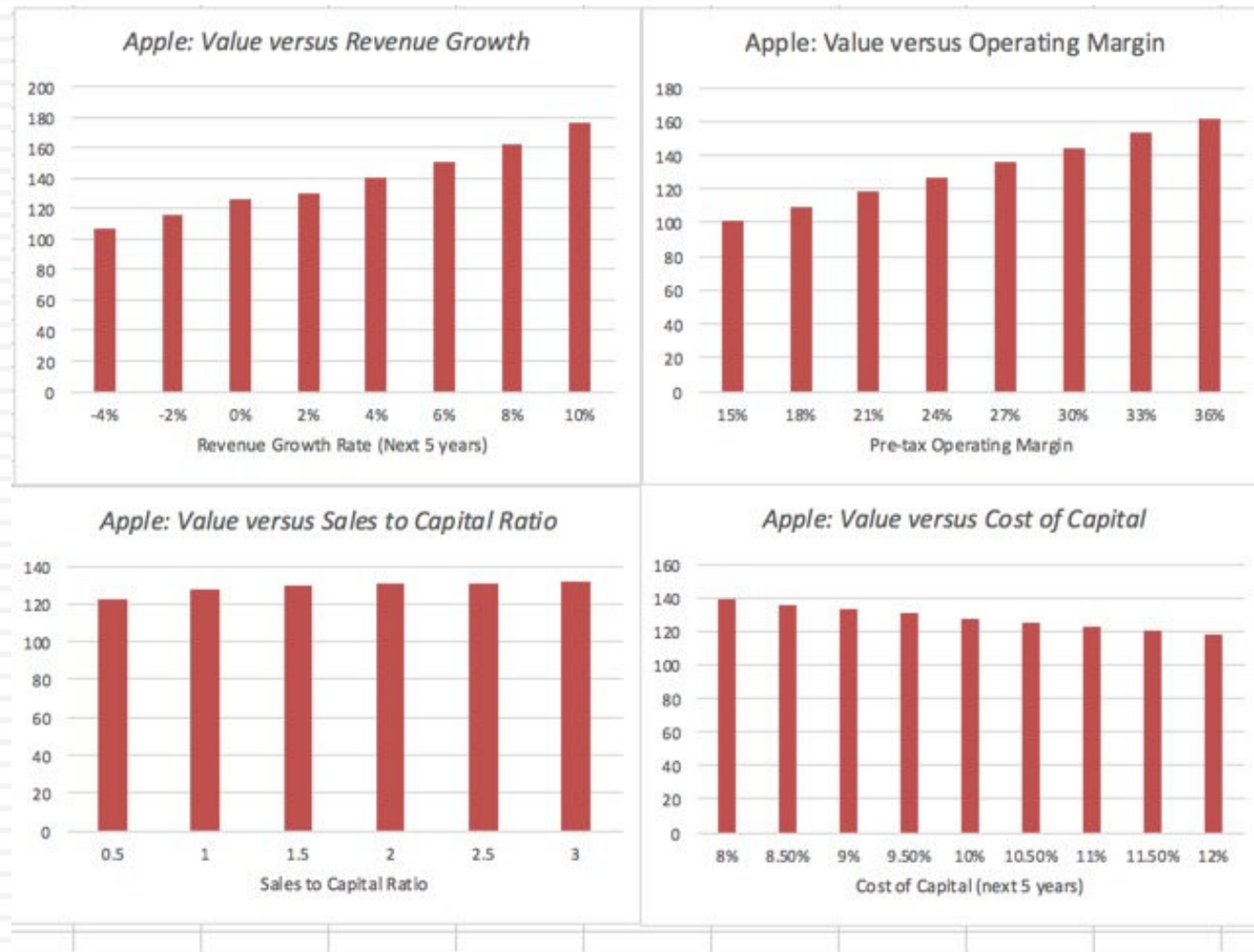


Step 1: Base Case Valuation

Apple: Base Case Valuation (May 2016)



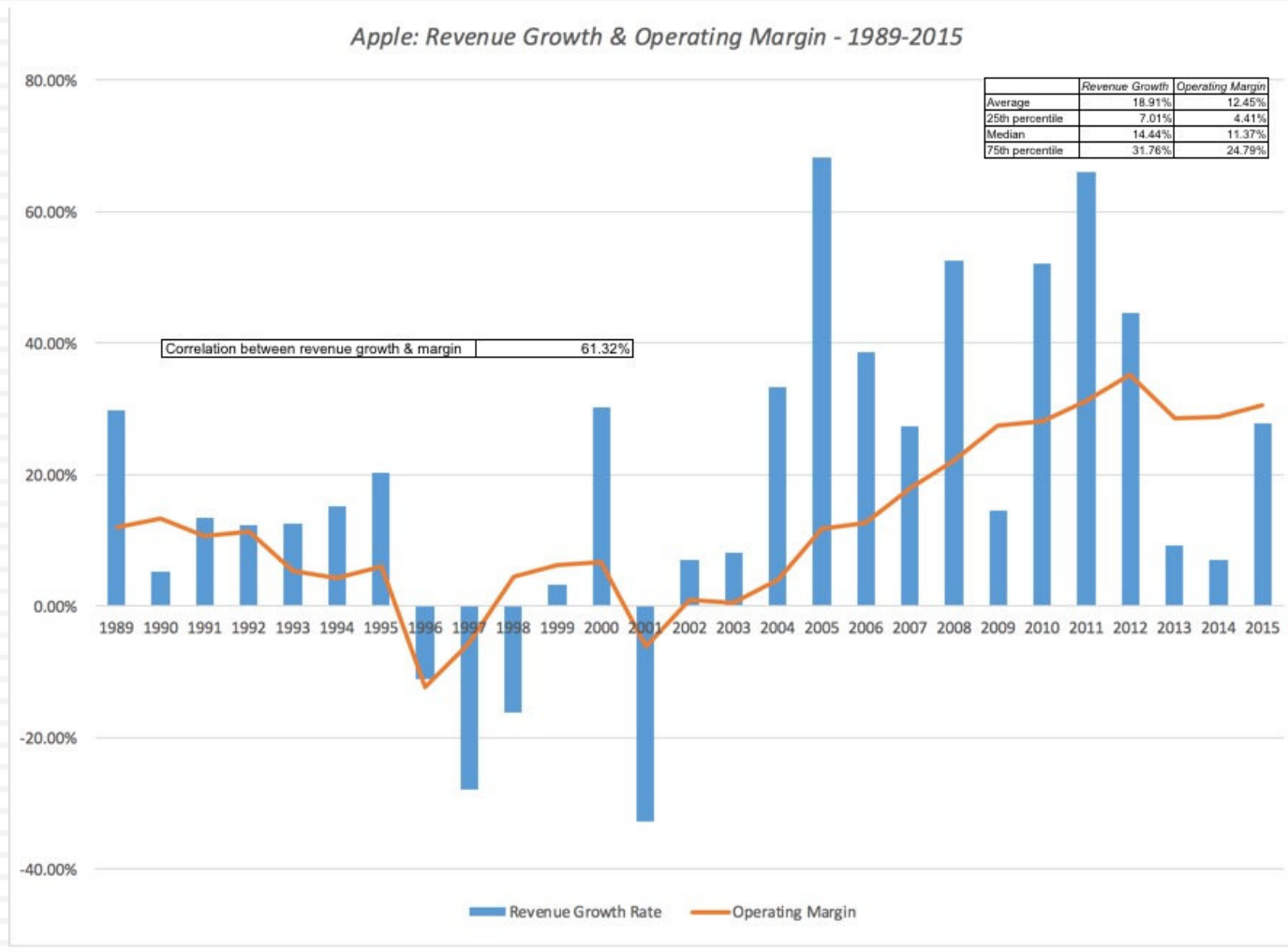
Step 2: Identify value drivers



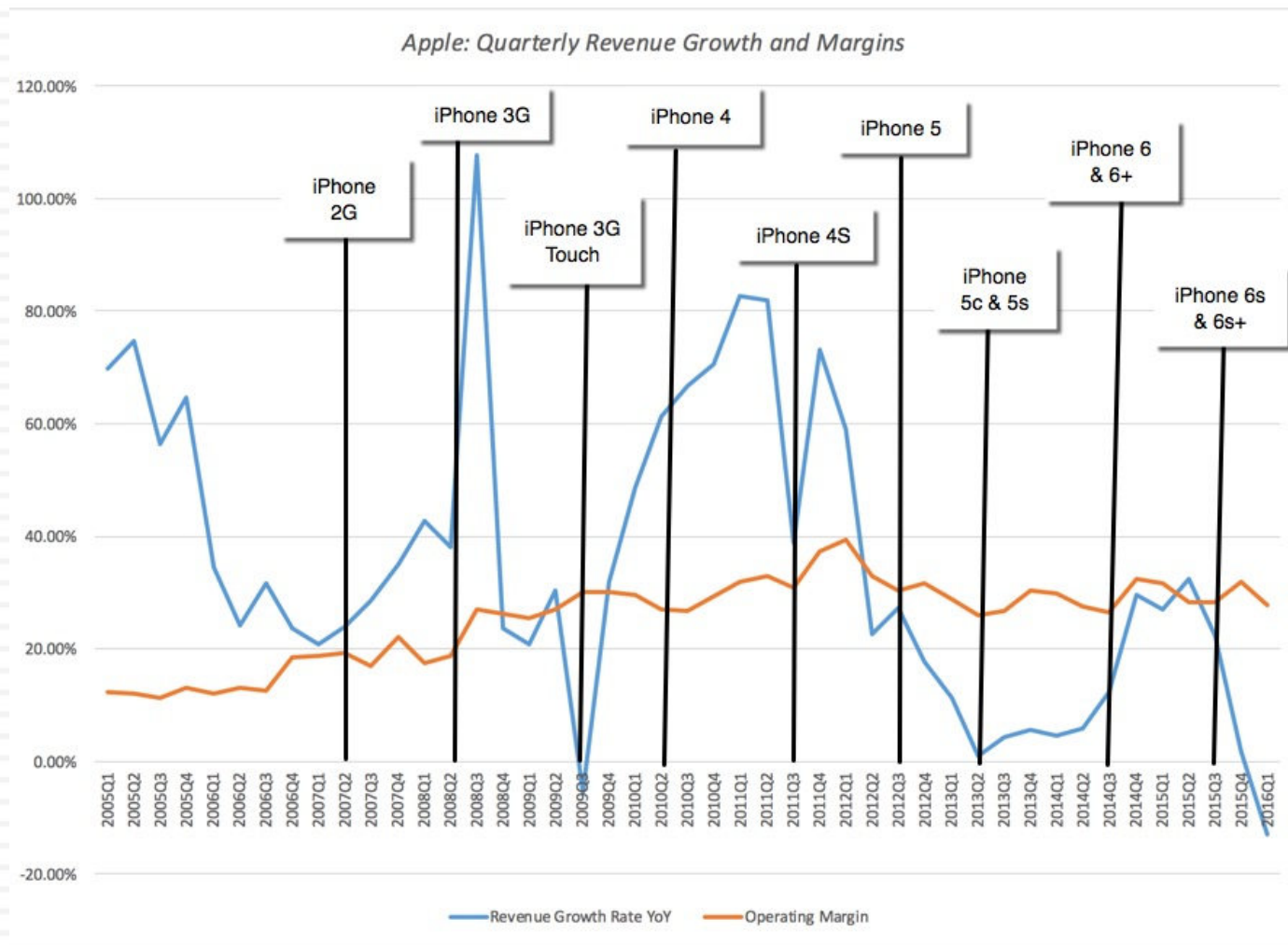
Step 3: Doing your homework

- Historical: The obvious place to start to get a sense of what uncertainties you face on a variable is to look at its historical behavior.
 - ▣ How much has it moved over time?
 - ▣ What factors seem to cause it to move?
- Cross-sectional: You can also look at differences on this variable across the sample today. Thus, in deciding how profit margins can vary for a software company, you can look at profit margin variations across software companies.
- Intuitive: You can keep your analysis grounded by bringing in common sense rules on the range for a variable.

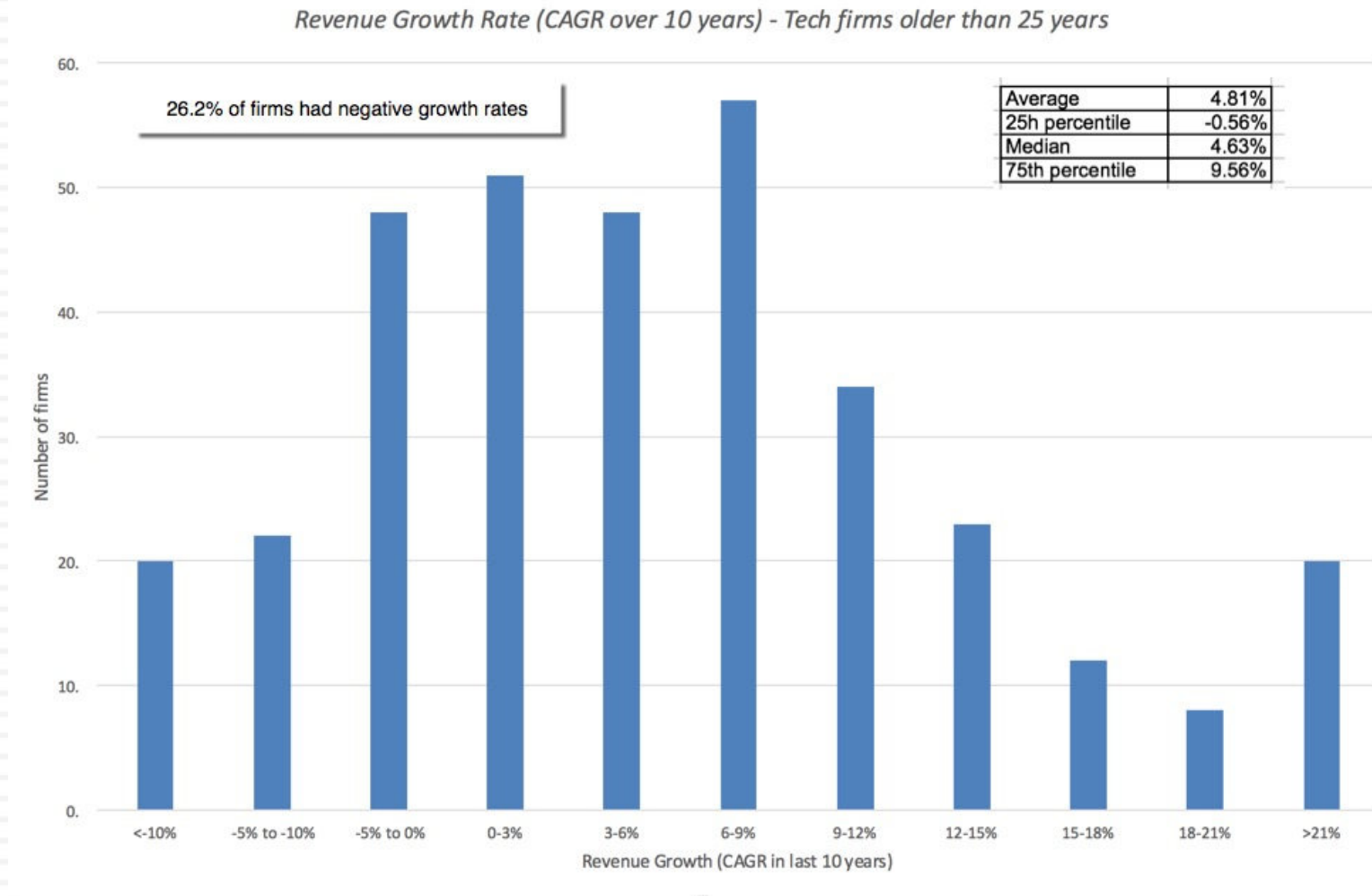
Apple's historical data



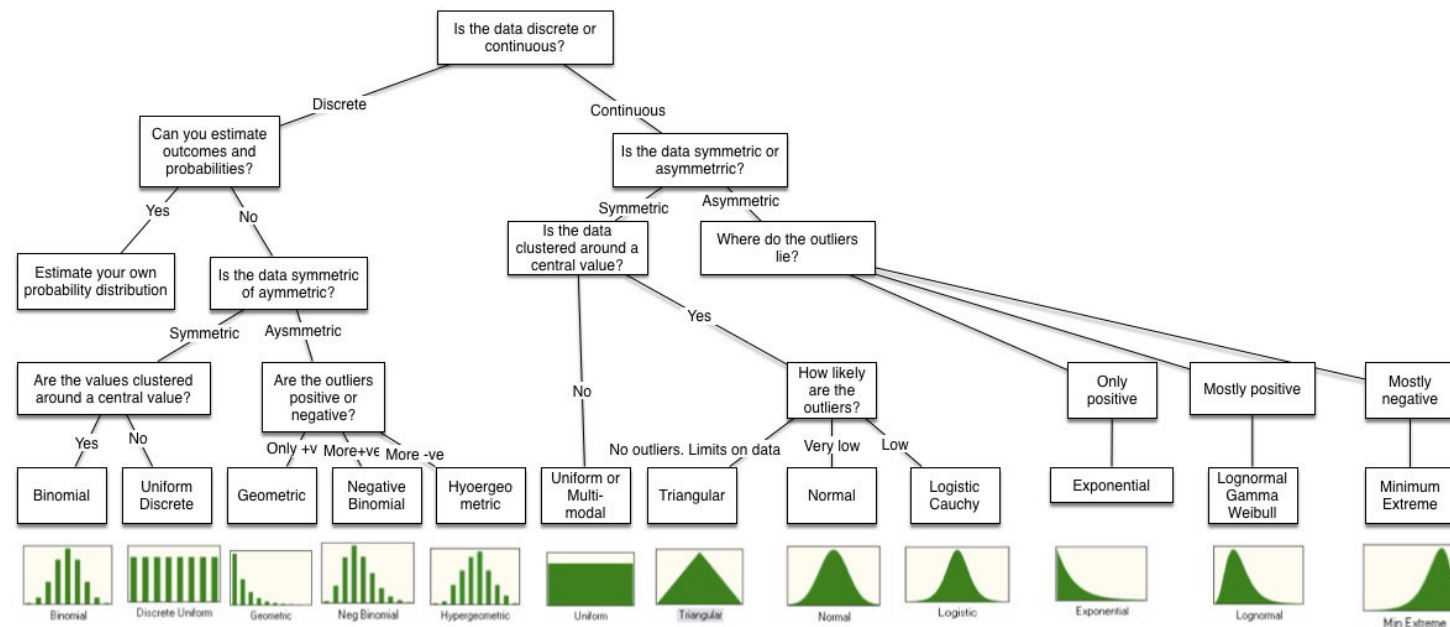
The iPhone Decade



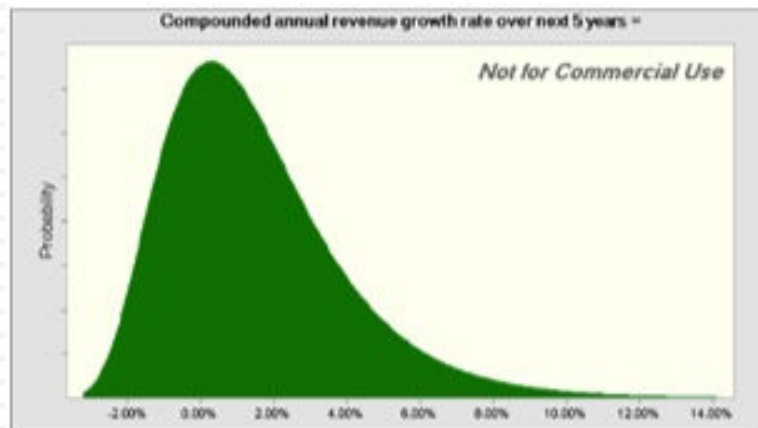
Revenue Growth at Aging Tech Firms



Step 4: Probability Distributions - Choices

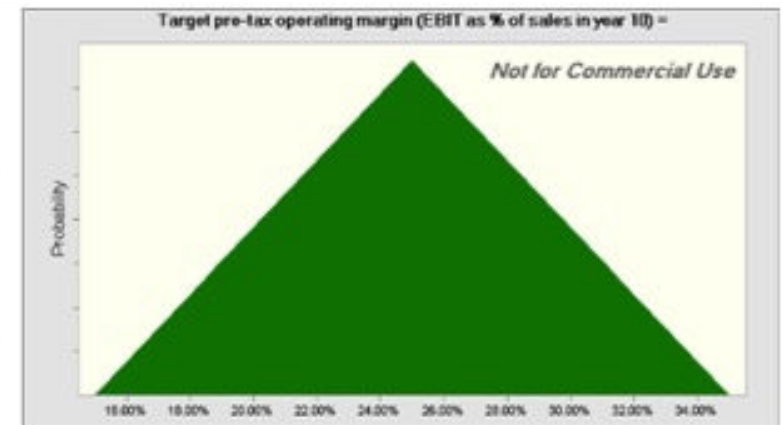


For Apple's revenue growth & margin



Distribution: Lognormal
Parameters: Location = -5%
Expected value = 1.50%
Std deviation = 2.5%

Correlation
between revenue
growth & margin =
0.50



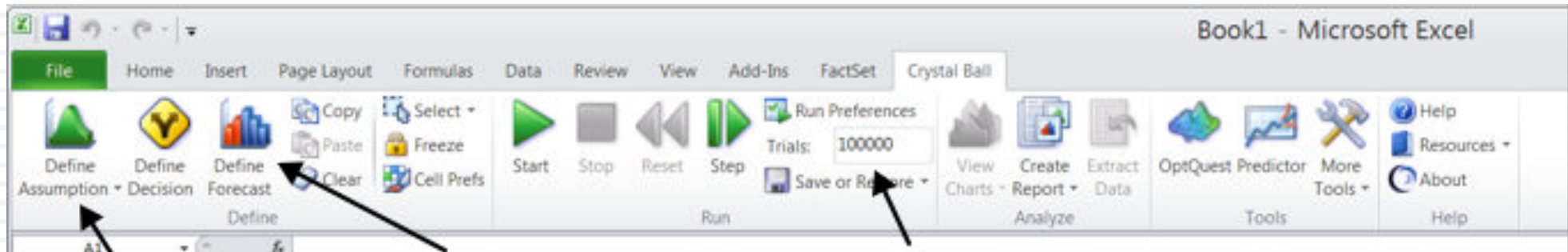
Distribution: Triangular
Parameters: Minimum = 15%
Expected value = 25%
Maximum = 35%

Step 5: Constraints, Correlations and Connections

- You can build in constraints that will affect the company's operations, and its value, that are either internally or externally imposed.
 - ▣ Internal constraints can include refusal to issue new stock, borrow money or pay dividends.
 - ▣ External constraints can include failure to make debt payments or meet regulatory capital requirements.
- You can also build in correlations between the variables that you are attaching probability distributions to.

Step 6: Run the Simulation

- ❑ **Crystal Ball:** <http://www.oracle.com/us/products/applications/crystalball/crystal-ball-product/overview/index.html>

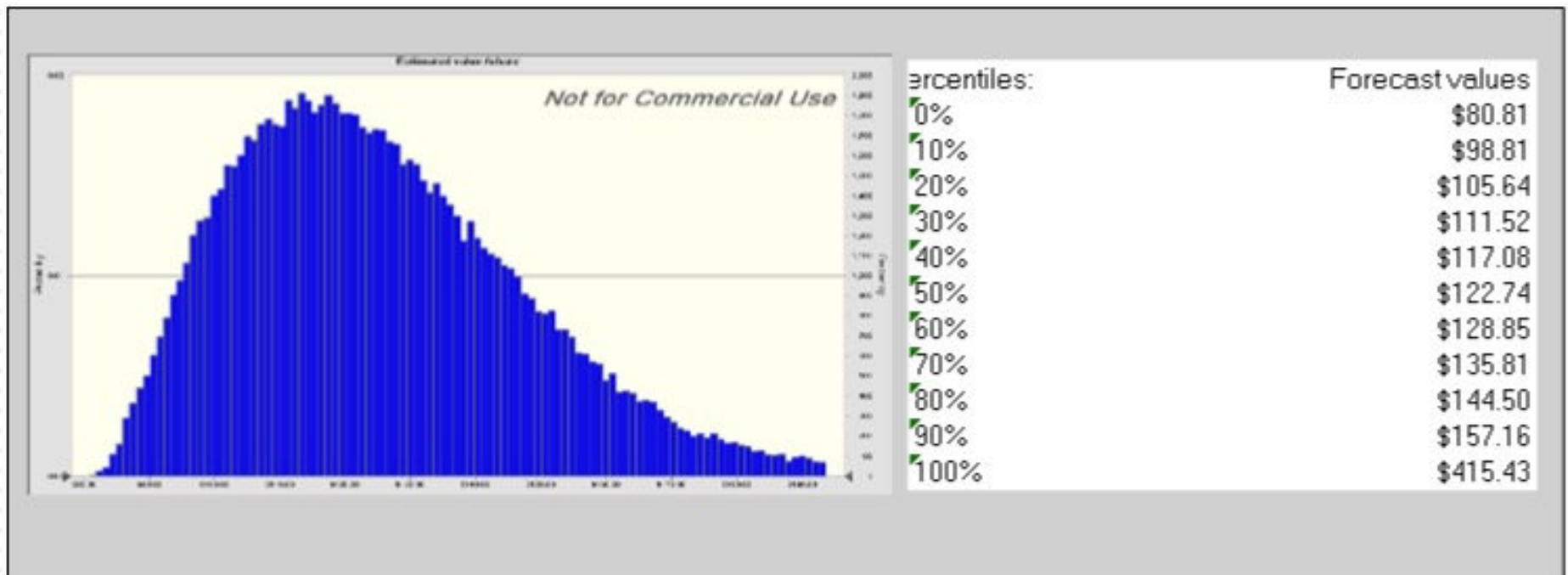


Define distributions
for variables

Specify output
variable

Number of
simulations

The Value Distribution



Conclusion



- ❑ Not looking at what you are uncertain about does not make it go away.
- ❑ Ironically, taking a closer look at what you fear (being wrong) can make you less fearful.
- ❑ Look into the (uncertainty) abyss. It might not be as dark and dangerous as you think it is.